INCH-POUND
MIL-C-52640B(ME)
7 August 1989
SUPERSEDING
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MILITARY SPECIFICATION

CRANE-SHOVEL, CRAWLER MOUNTED, 12-1/2-TON AT 12-FOOT RADIUS, 3/4 CUBIC YARD, DIESEL-ENGINE-DRIVEN, FOR MILITARY OPERATIONS

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel-powered, crawler-mounted crane-shovel suitable for use as a crane, clamshell, dragline, shovel, backhoe, wrecking ball, or piledriver. The crane-shovel shall be one size only with a rated lifting capacity of 12-1/2-tons at a 12-foot-load radius.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3810

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SPECIFICATIONS

FEDERAL

RR-C-271	- Chains and Attachments, Welded and Weldless.
RR-W-410	- Wire Rope and Strand.
VV-F-800	- Fuel Oil, Diesel.
ZZ-H-428	- Hose, Non-metallic and Hose; Preformed: (for the Coolant Systems of Automotive and Other Liquid-Cooled Engines).
GGG-B-485	- Block, Tackle, Wire Rope, with Swivel Hook; (for Crane Use).

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture- and Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-I-631	- Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-1283	- Can, Gasoline, Military, 5-Gallon.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-H-2514	- Hammers, Piledriver, Drop; 1,800 and 3,000 Pounds.
MIL-H-3199	- Heaters, Vehicular Compartment 28 Volt DC.
MIL-C-3432	- Cable (Power and Special Purpose) and Wire, Electrical (300 and 600 Volts).
MIL-C-3580	- Crane and Crane-Shovels, Truck, Crawler, and Wheel Mounted, Full-Revolving, and Their Attachments, Packaging of.
MIL-L-3661	- Lampholders, Indicator Lights, Indicator Light Housings, and Indicator Light Lenses, General Specification for.
MIL-L-3661/8	- Housings, Indicator-Light, Style LH76.
MIL-L-3661/14	- Lense, Indicator-Light, Dripproof, Style LC14.
MIL-L-3661/15	- Lenses, Indicator-Light, Dripproof, Style LC15.
MIL-S-3785	- Starter, Engine, Electrical: 24-Volt Direct Current.
MIL-G-3787	- Glass Laminated, Flat; (Except Aircraft).
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type, General Specification for.
MIL-W-5044	- Walkway Compound, Nonslip, and Walkway Matting, Nonslip.
MIL-W-5050	- Walkway, Coating and Matting, Nonslip, Aircraft, Application of.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-L-6085	- Lubricating Oil: Instrument, Aircraft, Low Volatility.
MIL-S-8660	- Silicone Compound, NATO Code Number S-736.
MIL-H-8796	- Hose, Air Duct, Flexible, Aircraft.

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MIL-E-10062 - Engines: Preparation for Shipment and Storage of.
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-L-11025 - Lead, Pile-Driver; with Catwalk.
MIL-B-11040 - Belt, V: Engine Accessory Drive.
MIL-A-11755 - Antifreeze, Arctic-Type.
MIL-H-13071 - Housing, Lamp, Resilient Mounting, for Construction Machinery.
MIL-C-13486 - Cable, Special Purpose, Electrical: Low-Tension, Heavy Duty, Single-Conductor and Multiconductor.
MIL-C-13516 - Circuit Breakers, Manual and Automatic (28-Volt, DC).
MIL-H-13531 - Hose, Rubber and Hose Assembly, Rubber (Hydraulic, Flexible).
MIL-C-14055 - Cup, Hydraulic Brake Actuating Cylinder: Synthetic Rubber.
MIL-R-14328 - Rubber Sheet: Synthetic, Medium Soft, General-Purpose Gasket Material (for Extreme Climatic Conditions).
MIL-I-15126 - Insulation Tape, Electrical Pressure Sensitive Adhesive and Pressure Sensitive Thermosetting Adhesive.
MIL-B-16028 - Buckets, Clamshell.
MIL-B-16029 - Buckets, Dragline.
MIL-H-21103 - Hammers, Piledriver, Self-Powered, Diesel.
MIL-B-23276 - Balls, Wrecking.
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw, NATO Code Number G-354, Metric.
MIL-B-43436 - Band Markers, Blank, General Specification for.
MIL-A-46153 - Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-L-46167 - Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176 - Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservation, Metric.
MIL-T-46786 - Tanks, Fuel, Engine: General Requirements for.
MIL-T-52176 - Teeth, Surface Ripping; (Points, Shanks, Keepers and Keeper Locks), Bucket and Dipper, Crane-Shovel.
MIL-T-52198 - Taglines, Crane and Crane-Shovel, Spring-Loaded Reel Type.
MIL-B-52203 - Boom Crane: 20-Ton, Rough-Terrain; 20-Ton, Truck-Mounted; 12-1/2-Ton, Crawler-Mounted.
MIL-B-52207 - Boom, Jib, Crane, 15-Foot-Long, for 3/4-Cubic-Yard (12-1/2-Ton Crawler Mounted, 20-Ton Wheel Mounted) Cranes.
MIL-A-52363 - Air Cleaner, Intake: Dry-Type (for Internal-Combustion Engine).
MIL-E-52649 - Engine Cold Starting Aids, Ether Fuel Primers.
MIL-C-62122 - Cable Assembly: Intervehicle Power: Plug and Receptacle.
MIL-T-83133 - Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-642 - Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-701 - Lists of Standard Semiconductor Devices.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-838 - Lubrication of Military Equipment.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1366 - Materiel Transportation System Dimensional and Weight Contraints, Definition of.
MIL-STD-1400 - Engines, Gasoline and Diesel, Methods of Test.
MIL-STD-1410 - Methods of Selection of Industrial Engines for End Item Application.
MIL-STD-1474 - Noise Limits for Army Materiel.
MIL-STD-1791 - Designing for Internal Aerial Delivery in Fixed Wing Aircraft.
MS15003 - Fittings, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Threads, Steel, Type III.
MS15795 - Washer, Flat-Metal, Round, General Purpose.
MS17165 - Lamp, Incandescent, PAR-46 24 Volt Floodlamp, 2 Contact Lugs.
MS21322 - Clamp, Loop - Aluminum, Plain and Cushioned.
MS21333 - Clamp, Loop - Steel, Plain and Cushioned.
MS21334 - Clamp, Loop - Double, Cable or Tube, Steel, Plain and Cushioned.
MS24207 - Coupling, Grease Gun; Hydraulic Type Nozzle.
MS24532 - Voltmeter, Battery-Generator-28 Volt DC.
MS24537 - Transmitter, Temperature Electrical Resistance - 24 Volt DC.
MS24538 - Transmitter, Pressure - 24 Volt DC 60 PSI.
MS24539 - Transmitter, Pressure - 24 Volt DC 120 PSI.
MS24540 - Indicator, Pressure - 24 Volt DC, Range 0 - 120 PSI.
MS24541 - Indicator, Pressure - 24 Volt DC, Range 0 - 60 PSI.
MS24542 - Indicator, Temperature, Electrical Resistance - 24 Volt DC, Range 160° - 320 °F.

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MS24543 - Indicator, Temperature, Electrical Resistance - 24 Volt DC, Range 120° - 240 °F.
MS24544 - Indicator, Liquid Quantity - 24 Volt DC, Engine Fuel.
MS27142 - Connector, Plug, Electrical-Pin Contact No. 12, 14, & 16 A.W.G., Waterproof.
MS27143 - Connector, Plug, Electrical: Socket Contact, No. 12 A.W.G., Waterproof.
MS27144 - Connector, Plug, Electrical: Socket Contact, No. 14 & 16 A.W.G., Waterproof.
MS27145 - Connector, Plug, Electrical: No. 14 A.W.G., Double Contact, Waterproof.
MS27146 - Connector, Receptable, Electrical: Double Contact, Waterproof.
MS27147 - Adapter, Connector - 3 Way No. 12, 14 & 16 A.W.G. Waterproof.
MS27148 - Contact; Pin, Electrical Connector: No. 12, 14 & 16 A.W.G.
MS33800 - Connector, Receptacle, Electrical-Pin Contact, No. 12, 14 & 16 A.W.G., Waterproof.
MS35000 - Battery, Storage, Lead-Acid, Waterproof.
MS35340 - Washer, Lock-Helical Spring, Extra Duty.
MS35425 - Nut, Plain, Wing, UNC-2B.
MS35647 - Padlocks, Key Operated.
MS35829 - Hinge, Butt, Continuous (Piano), 0.075 Thickness, 0.187 and 0.250 Pin Dia.
MS35844 - Plug, Machine Thread, Magnetic: Drain.
MS35916 - Tachometer, Mechanical, Fixed Mounting (4000 RPM).
MS39060 - Switch, Rotary, 28 Volt DC, Waterproof.
MS39061 - Switch, Toggle - SPST, SPDT, 24 Volt DC, 25 Amp (Waterproof).
MS51074 - Horn, Electrical - Automotive Type, 12 or 24 Volt DC.
MS51967 - Nut, Plain, Hexagon-Carbon Steel, Cadmium Plated, UNC-2B (IN./MM).
MS51968 - Nut, Plain, Hexagon - Carbon Steel, Cadmium Plated, UNF-2B (IN./MM).
MS53046 - Retainer, Battery - Automotive Tactical Vehicles.
MS53063 - Indicator, Air Cleaner, Intake, Restriction Mechanical Type.
MS90530 - Switch, Pressure-Warning, SPST, Waterproof, 24 Volt DC.
MS90725 - Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNC-2A.
MS90726 - Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNF-2A.
MS90727 - Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Alloy Steel, Grade 8, Cadmium Plated, UNF-2A, Plain and Self Locking.

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MS90728 - Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Alloy Steel, Grade 8, Cadmium Plated, Plain and Self-Locking, UNC-2A.

HANDBOOK

MILITARY

MIL-HDBK-113 - Guide for the Selection of Lubricants, Functional Fluids, Preservatives and Speciality Products for Use in Ground Equipment Systems.

MIL-HDBK-157 - Transportability Criteria.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government publication. The following other Government publication forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DEPARTMENT OF THE AIR FORCE

AFSC-HDBK DH-1-11 - Air Transportability.

DEPARTMENT OF THE ARMY

MTMCTEA Pamphlet 70-1 - Transportability for better strategic mobility.

(Application for copies should be addressed to the Military Traffic Management Command, Transportation Engineering Agency (MTMC/TEA), ATTN: MTT-TR, P.O. Box 6276, Newport News, VA 23606-0276.)

DEPARTMENT OF TRANSPORTATION

DOT Motor Carrier Safety Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI B29.1 - Precision Power Transmission Roller Chains, Attachments and Sprockets.
ANSI B29.2M - Inverted Tooth (Silent) Chains and Sprockets.
ANSI Z26.1 - Safety Code for Glazing Materials for Glazing Motor Vehicles Operating on Land Highways.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS, INC. (ASME)

ASME - Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications.

(Application for copies should be addressed to the Americal Society of Mechanical Engineers, Inc., 345 East 47th Street, New York, NY 10017).

THE AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING, INC. (ASNT)

Handbook Vol. II.

(Application for copies should be addressed to the American Society for Nondestructive Testing, Inc., 4153 Arlingate Plaza, Columbus, OH 43228.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 576 - Steel Bars, Carbon, Hot-Wrought, Special Quality.
ASTM D 1384 - Corrosion Test for Engine Coolants in Glassware.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC. (AWS)

AWS D1.1 - Structural Welding Code.

AWS D14.3 - Specification for Welding Earthmoving and Construction Equipment.

(Application for copies should be addressed to the American Welding Society, Inc., 550 NW Lejeune Road, Miami, FL 33126.)

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Commodities on Open-Top Cars.

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(Application for copies should be addressed to the Association of American Railroads, American Railroads Building, 50 F Street, NW, Washington, D.C. 20001.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTH COAST AIR QUALITY MANAGEMENT CONTROL DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management Control District, 9150 Flair Drive, El Monte, CA 91731.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The crane-shovel shall consist of a crawler-mounted, full revolving superstructure and front-end attachments. Each crane-shovel shall be capable of being converted for use in crane, clamshell, dragline, pile-driving, wrecking ball, shovel and backhoe operations. The crane-shovel shall be constructed so that conversion to any one of these uses can be effected in the field without the use of special tools or the necessity of installing additional shafts, drums, clutches, or other parts that should for reasons of practicability, be included at time of manufacture. The crane-shovel, equipped with boom, shovel front, or backhoe attachment, shall be capable of traveling over rough ground or under adverse or severe conditions. For maximum performance all operational functions for a given attachment shall operate independently and simultaneously where required.

3.1.1 Compatibility. The crane-shovel shall be compatible with front end attachments as specified in 3.9 and 3.20.1.

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3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2.1 Winterized first article model. When winterized crane-shovels are specified (see 3.10), in addition to nonwinterized crane-shovels, two first article models shall be furnished for examination and test. One of these first article models shall be nonwinterized and one winterized.

3.2.2 First article. When specified, (see 6.3) the contractor shall furnish to the Government three or more crane-shovels for inspection (see 4.3).

3.3 Material. Material shall be as specified herein and as shown on the contractor's drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Unless otherwise specified herein, rubber products furnished shall not exceed eight calendar quarters in age from date of cure of the rubber to date of acceptance of the crane-shovel by the Government.

3.3.1 Material deterioration prevention and control. The crane-shovel shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which it may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the crane-shovel may be newly fabricated from recovered materials to the maximum extent practicable, provided the crane-shovel produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the crane-shovel.

3.3.3 Common parts. Maximum practicable use shall be made of interchangeable hardware and fastening devices and of a minimum number of types and sizes of bolts, screws, nuts, washers, and similar common (standard) parts.

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3.3.4 Hose. Hose and hose assemblies shall be in accordance with the following:

a. Coolant hose shall be in accordance with SAE J20, heavy-duty type (SAE 20R1). Sizes of hose shall be as recommended by the engine and radiator manufacturers. Hose clamps shall be in accordance with SAE J536, type F, style 1, 2, or 4.

b. All other hose shall be in accordance with MIL-H-13531, and the fittings for this hose shall be in accordance with the following:

Hose diameter	Fitting
0.25 inch through 0.75 inch	SAE J516 female 37-degree flared.
1 inch through 2 inches	SAE J513 with 4-bolt clamp on hose ends.

Hoses furnished to the Government shall be not more than 12 months old from date of cure of the rubber to date of acceptance by the Government.

3.3.5 Breathers. All breathers for lubricant and fluid sumps on the crane-shovel shall preclude the entrance of water and airborne dust. Fuel tank breathers shall terminate at a level above the fill cap and shall be external to the superstructure.

3.3.6 V-belts. V-belts shall be in accordance with MIL-B-11040, and all belts for multiple belt drives shall be furnished as matched sets in accordance with MIL-B-11040. V-belts furnished to the Government shall be not more than 12 months old from date of cure of the rubber to date of acceptance by the Government.

3.3.7 Heater ducts. Heater air ducts shall be in accordance with MIL-H-8796.

3.3.8 Electrical cable. Low-voltage electrical cable not covered by component specifications shall be in accordance with MIL-C-13486.

3.3.9 Threaded parts. All capscrews 0.25-inch nominal size and larger shall be in accordance with MS90725, MS90726, MS90727, or MS90728. All nuts 0.25-inch nominal size and larger shall be in accordance with MS51967 or MS51968. Lockwashers shall be in accordance with MS35340. Grade B nuts shall be used with grade 5 capscrews and grade C nuts shall be used with grade 8 capscrews. All other threaded parts shall be in accordance with FED-STD-H28 except when otherwise specified herein. No self-tapping threaded parts shall be used.

3.3.10 Hydraulic cups. Synthetic rubber hydraulic brake cylinder cups for master and slave cylinders shall be in accordance with MIL-C-14055. Hydraulic

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cups furnished to the Government shall be not more than 12 months old from date of cure of the rubber to date of acceptance by the Government.

3.4 Design and construction. The design of the completely assembled unit shall be such that the maximum unit stress imposed by the required loading shall not exceed 60 percent of the yield point of any structural member when tested as specified in 4.5.2.3.

3.5 Superstructure. The superstructure shall consist essentially of a weather-tight cab enclosing the machinery and operator's position and shall be mounted on a fully rotating platform. The machinery in the enclosed cab shall be isolated or enclosed in the cab away from the operator of the crane. All operating machinery shall be mounted on and attached to the rotating platform. The rotating platform shall be a unit steel casting or heavy sections of structural steel welded together or a combination of cast steel and welded structural steel. The machinery shall be attached to the principal members of the rotating platform. The principal members shall be designed to receive the front-end attachments (see 3.9), and pins and pin retaining devices shall be provided. The pins shall be capable of being aligned with common handtools. The machinery deck shall be sealed with boots or grommets where controls pass through the deck. Two clips shall be furnished and fastened to the front edge of the operating frame to provide connection points for the inner end of the piledriver lead catwalk (see 3.9.2.8). Grab handles shall be provided alongside the doors, walks, and steps and shall be located to facilitate safe access to the operator's seat and to components requiring frequent attention. Grab handles shall provide the hand clearance shown on figure 1. To facilitate repair, maintenance, and replacement of equipment, and for access to the engine compartment, folding hinged panels with labyrinth-type weather seals (see 6.5 and figures 2 and 3) shall be provided. A ladder shall be attached to the front of the superstructure cab to facilitate accessibility to the gantry and A-frame sheaves. When the fuel tank filler neck is more than 36 inches from the ground, an expanded metal antiskid platform, not less than 15 inches square and 0.125-inch in thickness, shall be located adjacent to, and not more than 36 inches below the fuel tank filler neck, to enable the operator to fuel the crane tank with 5-gallon cans or to aid in bulk fueling. This platform shall be hinged to the bottom of the superstructure to fold up and latch against the superstructure cab wall when not in use. The standing area shall permit the operator the full use of both arms. A catwalk shall be installed all around the superstructure except for the boom-foot area. The catwalk shall be not less than 18 inches wide unless otherwise specified, and shall be surfaced with open, skidproof grating. The catwalk shall support a concentrated load of not less than 400 pounds (lb) per square foot. The catwalk shall be bolted to the superstructure and may be removable or hinged to meet the width requirement. Footholds or handholds to enable the operator to gain access to all platforms and catwalks shall be provided. All platforms and catwalks shall have guardrails installed. Removable cast-iron counterweights shall be provided. The complete counterweight shall be removable in not more than 30 minutes by two soldiers and set on a trailer or truck using only the lifting

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provisions on the crane-shovel. The complete counterweight shall be capable of replacement on the crane-shovel by two soldiers in not more than 10 minutes using only the lifting provisions of the crane-shovel. The counterweight may serve as the rear catwalk if the walking surface is treated with nonskid material in accordance with MIL-W-5044 and said nonskid material is applied in accordance with MIL-W-5050.

3.5.1 Functional requirements. With the engine operating at governed speed, the crane superstructure shall meet the following requirements:

a. A hoisting-drum line power of not less than 1,250,000 foot-pounds (ft-lb) per minute at minimum, bare hoisting-drum line speed.

b. A swing speed of not less than 3.5 revolutions per minute (rpm).

c. Bare hoisting-drum line speed between 160 and 180 feet per minute (fpm).

d. Simultaneously lift and rotate not less than a 7813-lb load using a single part line after establishing the maximum safe operating radius.

3.5.2 Forward deck machinery protection. A sheet-metal trough shall be provided to protect the forward machinery deck from mud and dirt scraped from the haul-in cable during dragline operations. The trough shall not interfere with any crane operations.

3.5.3 Cab. The sheet steel used for the cab shall be not less than 0.071 inch thick. The cab shall completely enclose the operator's seat, control levers and pedals, instrument panel, and all main-deck machinery. Hinged covers with slots for hoist cables shall be provided for weather protection of the cable drums. The arrangement of windows and windshield shall provide optimum vision for the operator. The seat, instruments, and controls shall be in accordance with SAE J899, SAE J209 and SAE J898. The cab shall have sufficient windows and glazed doors to permit 180-degree vision for the operator. The operator, varying from large to small, in accordance with SAE J833, wearing artic clothing and seated upright, shall have maximum unobstructed visibility of the work, including the point sheaves of the basic boom at a 10-foot operating radius. The seated operator shall also have 180-degree vision to all points projected on the ground at a horizontal radius of 12 feet and 20 degrees above horizontal for side window. All glass shall be replaceable and held in place by rubber channels. All glazing shall be 0.25-inch, AS2, tempered, safety plate glass in accordance with ANSI Z26.1. Not less than 75 percent of the front window shall be opened and shall slide into the roof of the cab for storage. The window shall latch in both the open and closed position. Both the side and rear doors shall be of the sliding type with accessible locks. The doors shall be constructed of metal and equipped with steel latches and locks. The seat shall be adjustable without tools. Vertical and horizontal seat adjustments shall be in maximum increments of 1 inch. The seat in the

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central position shall be located to enable a man ranging in size as indicated in SAE J833 and clothed in arctic clothing to reach the control levers and pedals. Cab doors shall be held closed with a positive locking device and shall be provided with steel latches to prevent accidental opening and closing. Walking surfaces and steps inside the cab shall be treated with non-skid material in accordance with MIL-W-5044, applied in accordance with MIL-W-5050. The seat base shall be provided with a quick release to allow the seat to tilt sideways toward the machinery or forward to permit operator to escape in case of emergency. Drainage provisions shall be made for wells or depressions that are subject to collection of water.

3.5.3.1 Cab heating. The cab shall be equipped with a heater in accordance with MIL-H-3199, type II, capacity A, 28-volt rating, with a three-position switch including two speeds and on-off. The heater shall be mounted and positioned to permit routine maintenance and inspection and to allow removal. The source of heat for the heater shall be from the crane-shovel engine. A ducting system shall be provided to direct heat toward the operator's feet and to the defroster. Valves to permit the operator to select air distribution between the defroster duct and for cab heat ducts shall be provided. A defroster shall be provided for defrosting the front windows. Ducting shall not interfere with operator performance or with maintenance, and shall be protected from damage. Controls for the heater shall be located within reach of the operator in his normal operating position. The area defrosted shall be at least that described by SAE J382, 3.2, Area A. Ducts shall be arranged to defrost not less than 75 percent of the front window area, and shall be mounted and positioned to permit routine maintenance and to allow quick removal.

3.5.4 Superstructure cab, disassembly. The superstructure cab shall be removable to reduce the size, weight, and shape of the cross-section of the crane-shovel for transportability (see 3.19). The framing shall tie the side panels and main portion of the roof together as one unit, and the top corners shall be provided with lifting eyes to accommodate slings. The supporting members of the cab shall be bolted to the machinery deck with fasteners in sufficient quantity and strength to hold the cab securely in place under all operating conditions. All free-standing edges of removable and fixed sections shall be provided with angle stiffeners and self-sealing, labyrinth-type weather seals as shown on figures 2 and 3 to keep water out. All electrical, air, and hydraulic lines shall be equipped with quick-disconnect, self-sealing couplings at the separation line of the cab and the machinery deck.

3.5.5 A-frame. An A-frame or gantry shall be provided that is capable of withstanding stresses induced by lifting from the ground a standard military boom 60 feet long. The principal vertical compression and tension members of the A-frame shall be connected so that the operating stresses are transferred to the main structural members of the rotating platform of the superstructure. When operating with booms from 30 to 60 feet in length and the A-frame in the raised position, the centerline of the A-frame top sheave shaft shall be not less than 5.6875 inches above the center of the boom foot pin when measured

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vertically, nor less than 79 inches to the rear of the boom foot pin when measured horizontally. The A-frame sheaves having their grooves in a vertical plane shall be located laterally from a vertical plane through the longitudinal centerline of the boom so the angle between the centerline of the reeved boom hoist cable and a plane through the centerline of the sheave groove shall be not more than 6 degrees with the boom at 82 degrees from the horizontal (see 4.5.2.11). The location of all sheaves in the A-frame shall permit reeving through a pendant equalizer in accordance with 3.9.1.2 and meeting all requirements specified herein. The gantry shall be of the folding or retractable type for overhead clearance and for reducing shipping space requirements during transportation.

3.5.6 Power load-lowering device. A power load-lowering device shall be provided on the primary drum to facilitate controlled load lowering. This device shall drive the hoist drum at a line speed 30 to 50 percent of the bare drum line hoisting speed as specified in 3.5.1. The device shall be controlled by a clutch and power the primary hoist drum in a direction to unspool the cable. The clutch shall be operated by a lever accessible to the operator in his seated position. The power-down clutch shall have sufficient capacity to hold on a single line, without slipping, a load in pounds equal to the maximum line pull developed by the hoist drum (see 4.5.2.13). The clutch shall have sufficient capacity to overdrive the primary hoist drum brake with hook loads on a four-part line, or simulated loading of both maximum capacity and one-half maximum capacity with the brakes adjusted for holding a maximum capacity load on a four-part line. The boom angle shall not change and the brake lining temperature shall be not greater than 600 ^{o}F (316 ^{o}C), or the maximum recommended by the manufacturer, whichever is less, when tested in accordance with 4.5.2.12. Provisions shall be made to prevent damage to the gear or chain reduction when free spooling the drum. The load hoist and load lowering controls shall be entirely independent of the boom hoist and boom lowering controls.

3.5.7 Crane-boom hoist. The crane-boom hoist shall be independently controlled. The boom hoist, equipped with a positive holding device, shall safely suspend the boom with recommended reeving without attention from the operator, shall provide for power lowering under the operator's control, and shall make free spooling of the boom cable impossible under any circumstance. The boom hoist shall raise and lower the 60-foot-long, standard military boom under loadings and conditions specified in 3.5.1. It shall have sufficient rope capacity to allow the boom tip to be lowered to ground level with the crane- shovel resting on level ground and three wraps of wire rope retained on the drum. The boom hoist shall be equipped with a positive lock to hold the boom in any set position.

3.5.8 Drums. The drums shall be independently driven and convertible for use with any of the specified front-end operating equipment. The drums shall be arranged to reduce eccentric pull to a minimum in order to avoid chafing of the rope and unequal wear on the drum shaft bearings. Drums shall have sufficient rope capacity to accommodate the specified rope lengths (see 3.9.1.6.2)

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and shall spool the rope in even layers. When clutches and brakes are not engaged, the complete drum assembly and mounting shall permit free rotation to allow gravity lowering of the hook block. The primary and secondary load hoist drums shall be operated by clutches controlled by levers. Once the clutch is engaged it shall be held in place by an overcenter lock (knuckle-in) on the control linkage. The clutches shall be unlocked by pushing or pulling the control lever. In addition, the machinery operating the primary and secondary drums shall be so arranged that the drum clutches may be simultaneously engaged and, with the main engine-clutch disengaged, shall reverse the operating machinery by lowering the load under control of both drum brakes simultaneously for precision load lowering without the boom raising or lowering. Pitch diameters of drum or drum laggings including boom hoist drum shall be in accordance with SAE J881. The drums shall be provided with a socket equipped with a wedge for dead-ending the wire rope. All drums shall be guarded to prevent the rope from leaving the drum during operation. Lubricating fittings or other forms of lubricating devices shall be provided and arranged to permit lubrication without unspooling the wire rope.

3.5.8.1 Hoist drum brakes. The drum brakes shall hold and slowly lower (without jerking or chattering) capacity loads supported by recommended reeving. Provision shall be made for locking the brakes in any position of the drum. Each drum brake (primary and secondary) shall hold the maximum load that the crane is capable of lifting on a single-part line at the minimum safe operating radius. The drum brakes shall be removable for lining replacement. All brake linings shall be riveted or bolted to the backing plate. The brake assembly shall be of a fail safe design.

3.5.8.2 Hydraulic lines. Hydraulic lines shall consist of tubing and hose. Hose shall be used where necessary to prevent failure due to vibration. Hydraulic lines shall be in accordance with applicable SAE handbook standards.

3.5.8.2.1 Hydraulic brake reservoir. A hydraulic reservoir shall be provided with sufficient capacity to supply the required reserve fluid to the master cylinder to prevent entry of air into the hydraulic system. The reservoir shall be located not less than 3 feet above the master cylinders. The reservoir shall be labeled "BRAKE FLUID RESERVOIR" and be labeled with the type and quantity of fluid required. Said labels shall be in accordance with 3.28.2.

3.5.8.3 Drum and swing clutches. Drum clutches shall be manually controlled by the operator and shall be arranged to engage and disengage instantly under full engine torque. Drum and boom hoist clutches shall be of the friction or electromagnetic type, accessible for adjustment and repair, and large enough to eliminate slipping under 150 percent of rated load. Drum clutches shall withstand high-speed, continuous operation without overheating or excessive wear. The hoist clutches and travel clutch shall each individually be capable of stalling the torque converter and shall be accessible for adjustment and

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repair. Swing clutches shall make not less than 3.50 cycles (a complete cycle of 90 degrees one way and 90 degrees back) per minute for not less than 1.50 hours without either fading or exceeding 500 °F (260 °C), with a 30-foot boom at an angle of 40 degrees above horizontal and with a load equal to 50 percent of maximum line pull. Clutches shall be controlled by mechanical linkage, air, or manual effort applied through hydraulic linkage or through variable-pressure, power hydraulic controls. When a power hydraulic control system is furnished, the force applied to control levers or pedals shall be in direct proportion to the force exerted on clutches. Regardless of the type of control mechanism or the constant of proportionality, the maximum force that must be exerted by the operator to move the control levers shall not exceed 10 pounds. All clutch facings shall be riveted or bolted to the backing plate. Hydraulic systems, if used to actuate the drum clutches, shall be capable of operation and storage without leaking or seepage of the hydraulic fluid. System component parts shall be compatible with one another and with the hydraulic fluid to prevent corrosion during storage or operation. The hydraulic master and slave cylinders shall have cylinder bodies and pistons of ferrous material. The cylinder cups shall be in accordance with MIL-C-14055.

3.5.9 Rollers. The rotating frame shall be supported by large fixed rollers or by a live ball or roller ring. Rollers, balls, roller rings, and ball races shall be heat-treated hardened steel and shall be accurately machined. Fixed rollers shall be fitted with lubricated bearings and shall be provided with grease fittings. The balls or rollers shall be removable. When the holding-down function is performed by hook rollers, they shall be either the dual-purpose type, in which each set of rollers performs both the supporting and holding-down functions, or the holding-down type only. Hook rollers shall operate either singly or in pairs on tandem bogies and shall be provided with grease fittings. Hook rollers shall be provided with means for vertical or horizontal adjustment as required. The hook rollers shall be removable without removal of the hook roller brackets from the revolving superstructure. The hook roller brackets shall not interfere with the removal of the superstructure from roller path and frame. The temperature of the supporting balls or rollers and hook rollers shall be not more than 175 °F (79.4 °C) under all operating conditions specified herein.

3.5.10 Swing-gear, lock and brake mechanisms. The revolving superstructure shall be actuated by a gear and pinion. The swing-gear pinion shall be of alloy steel and have accurately machined or ground teeth. The swing-gear pinion shaft shall operate in lubricated bronze or antifriction bearings. A swing lock shall be provided to lock the revolving superstructure within ±2.50 degrees of any desired position except the travel position. For travel, the crane boom shall be positioned to within 0.50 degree of the longitudinal centerline of the crawler frame. A manually-operated swing brake shall also be provided to control the swing of the revolving superstructure and to hold the superstructure from swinging when the crane-shovel is working on a slope.

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Both lock and swing-brake controls shall be located within reach of the operator while seated. The swing-brake shall be of a fail safe design. The rotation of the rotating frame shall be controlled through friction-type or electromagnetic clutches designed for fast cycling, craning, and other operations, and with a clutch capable of transmitting not less than 130 percent of the maximum engine torque. The temperature of the swing clutches shall be not greater than 500 °F (260 °C) and the clutch shall not fade when tested as specified in 4.5.2.14.b and c.

3.5.11 Power transmission. Unless otherwise specified (see 6.2), the engine shall be connected to the transmission through either friction clutch or friction disengaging clutch and fluid coupling. When specified (see 6.2), friction disengaging clutch and a hydraulic torque converter shall be furnished.

3.5.11.1 Clutch. The clutch shall be manually controlled from the operator's position and shall be arranged to engage and disengage instantly while the engine is running. The clutch shall be capable of transmitting not less than 150 percent of the maximum engine torque. The clutch shall be controlled by either mechanical linkage, air, or manual effort applied through hydraulic linkage. Regardless of the type of control mechanism, the maximum force that must be exerted by the operator to engage or disengage the clutch shall not exceed 10 pounds. Clutch facings shall be riveted or bolted to the backing plate.

3.5.11.2 Torque converter. The hydraulic torque converter shall be a standard commercial design that has proven satisfactory in extensive crane-shovel use. The torque converter shall have a torque input rating of not less than 150 percent of the maximum torque output of the engine. With an initial fluid temperture of not less than 180 °F (82 °C), in an ambient temperature of not less than 30 °F (-1 °C), the torque converter shall absorb maximum torque output for not less than 30 seconds without exceeding 250 °F (121 °C) and shall return to a stabilized temperature within 15 minutes when tested as specified in 4.5.2.15.

3.5.11.3 Drive chains. All drive chains shall be provided with a means of adjustment. This adjustment shall be made either by changing the center to center distance between the driving and driven shafts or by means of a roller or sprocket meshing with or riding on the chain. This roller or sprocket shall be located so the tensioning action shall be exerted on the exterior side of the slack side of the chain. The primary drive chain to the first sprocket shall be in accordance with a below, and shall be encased by an oil-tight steel chain case and run through an oil bath; or lubricant for the chain may be supplied by a circulating pump directing a spray of oil on top of the top chain span. The transmission chains shall be in accordance with ANSI B29.1 as follows:

a. Flat link-plate roller chain numbers: 50, 50-3, 60, 60-3, 60-4, 60-5, 80, 80-4, 80-6, 100, 100-3, 100-4, 120, 120-2, 140, 140-2, 160, and 200.

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b. Offset link-plate roller chains selected from table I shall be in accordance with ANSI B29.1.

c. 60-degree, inverted-tooth silent chains shall be in accordance with ANSI B29.2M and limited to chain numbers SC616, SC1020, and SC1024.

TABLE I. Chain, roller, offset link plates, power transmission, single strand.

Chain No.	1	2	3	4	5	6	7
Nominal pitch (in.)	6.000	5.000	4.500	4.073	3.500	3.067	2.000
Minimum ultimate strength (psi X 105).	3.6	2.5	1.93	1.41	1.22	0.90	0.5625

3.5.12 Electrical system. The crane-shovel shall be equipped with a 24-volt, negative-ground, splashproof, electrical system for lighting work areas at night, slave starting the engines of other vehicles, maintaining charge in the batteries, and starting the engine. The electrical system shall be in accordance with SAE Handbook, Electrical Equipment and Lighting section, except as specified herein. All components shall be provided with the maximum protection from damage, shock, and vibration through means of location and furnishing of resilient mounts and cushions. Sealed-beam units and bulbs shall be accessible for replacement without disturbing the permanently installed housings and guards. Components shall be in accordance with table II. Item quantities that are not specified shall be furnished as required for each crane-shovel.

TABLE II. Vehicle electrical system.

Description	Requirement	Quantity
Battery, storage, 12 V, 6TN	MS35000	-
Holddown frame	MS53046-3	1 ea per 2 batteries
Holddown bolts	J-Bolts, 10-in. long, 0.375-in. diameter	5 ea per frame
Washers and wingnuts	MS15795-815 and MS35425-74	5 ea per frame
Clamp assembly	Tension-type hook latches with strikers	-
Piano hinge	MS35829-11C	-

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TABLE II. Vehicle electrical system - Continued.

Description	Requirement	Quantity
Charging system, 60 amp at 28 volt	SAE J180 Commercial	1 ea
Low-tension cable	MIL-C-13486	-
Harness insulation	MIL-I-15126 and MIL-I-631	-
Loom	Polyethylene or SAE J562, oilproof loom	-
Cushioned support clamps	MS21322, MS21333 and MS21334	-
Circuit number bands	MIL-B-43436	-
Horn	MS51074	1 ea
Friction connectors	MS27142 through MS27148 and MS33800	-
Connector lubricant	MIL-S-8660	-
Connector holder clip	Plastic or metal spring clip	1 ea per friction connector
Threaded plugs and receptacles	Waterproof military type	-
Terminal connectors	SAE J561 straight or side type spade	-
Insulator sleeves	Heat shrinkable tubing	-
Circuit breaker	MIL-C-13516	-
Slave receptacle	MIL-C-62122	-
Slave cable w/plugs on each end	MIL-C-13486, 240-in. length, and MIL-C-62122	1 ea, assy.
Temperature sending unit	MS24537	-
Pressure sending unit, 60 psi	MS24538	-
Pressure sending unit, 120 psi	MS24539	-
Fuel tank sending unit	Compatible with MS24544	-
Dome light	Commercial, 4-inch dia., elastomer mount	3 ea
Panel light	Commercial with trade no. 313 bulb	3 ea, min.
Warning and indicator lights	MIL-L-3661/8 and MIL-L-3661/15	-
Battery-generator indicator	MS24532	1 ea
Pressure gage, 120 psi	MS24540	-
Pressure gage, 60 psi	MS24541	-
Temp. gage, 160 - 320 °F (71-160 °C)	MS24542	-
Temp. gage, 120 - 240 °F	MS24543	-

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TABLE II. Vehicle electrical system - Continued.

Description	Requirement	Quantity
(49-116 °C)		
Fuel gage	MS24544	1 ea
Tachometer, hourmeter	MS35916	1 ea
Rotary switch	MS39060	-
Toggle switch	MS39061	-
Pushbutton starter switch	Commercial waterproof, 15 amp. min	1 ea
Work lights	MIL-H-13071	4 ea
Sealed-beam units, flood	MS17165-4578	4 ea
Starter	MIL-S-3785	1 ea
Metal conduit	Commercial thinwall electrical type	-
Conduit clips	Commercial type	1 ea per 4-ft span

3.5.12.1 Fungus- and moisture-resistance. The electrical circuitry, including all components, and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish in accordance with MIL-V-173, composition as specified in 3.5.12.1.1, with 1-percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus- and moisture-resistant or which are hermetically sealed need not be treated.

b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter that would interfere with the adherence or function of the varnish.

3.5.12.1.1 Composition. Composition I shall be used unless composition II must be used in order to comply with local air pollution regulations in the application of the varnish (see 6.2). When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442 of the South Coast Air Quality Management Control District.

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3.5.12.2 Wiring. The wiring of the crane-shovel shall be fabricated and installed to provide a reliable electrical system. Rigid conduit shall be furnished for all circuits to within 12 inches of components, with the exception of the battery and slave cables. Loom or harness insulation shall be furnished from the end of the conduit to those components connecting two or more conduits. Conduit, clips, loom, grommets, and cushioned support clamps shall prevent whipping, chafing, or other mechanical damage to the wiring or harness. Wiring shall be routed within the framing members whenever possible. Unless protected by conduit or loom where two or more wires follow the same routing, the wires shall be formed into harnesses by binding them together with one-half overlapping turns of 0.010-inch-thick, 1-inch-wide tape or with insulation. The wiring shall facilitate defective wire replacement from bulk supply. Battery and slave cables shall be of adequate cross-section to satisfy the SAE starter voltage drop in SAE J541 for heavy-duty diesel use, 24-volt, and maximum current requirements. These cables shall not be spliced. The size of other wiring shall also be based on voltage drop and current required plus 15-percent allowance increase for deterioration and in no instance shall the cable be smaller than A.W.G. No. 14. All circuits shall be identified by number and color coding on circuit number bands furnished near each connector. Color coding shall be limited to these bands. Friction connectors (MS27142 through MS27148 for wire and MS33800 for components) shall be lubricated prior to assembly connection. Each friction connector assembly shall be secured to the crane-shovel with a connector clip. Terminal connectors shall be of the type equipped with either heat shrinkable or integral tubing sleeves. Removal of major assemblies such as engine, cab, and instrument panel cluster shall be accomplished by disconnecting one or more quick-disconnect connectors for all 24-volt circuits in each assembly with the exception of battery, starter, and slave cables. When circuit quantities exceed twelve in number, one plug and receptacle shall be furnished for each twelve circuits, or fraction thereof, and the multiple disconnect points shall be in the same location. All circuits shall be protected by circuit breakers.

3.5.12.3 Lights.

3.5.12.3.1 Floodlights. For illuminating the working area, the floodlights shall be in accordance with table II, size PAR-46. Each lamp shall be provided with a housing in accordance with MIL-H-13071 and the lamp housing shall be attached to the crane-shovel superstructure as specified herein by a mounting bracket in accordance with and having the adjusting features as specified in MIL-H-13071. The housings shall be mounted in a position to permit replacement of the lamp. Lamp housings shall not extend beyond the side of the revolving superstructure unless provision is made for folding the lamps clear of the sides of the superstructure for transport. The floodlights shall be located so lenses will not be seen by the operator in his normal operating position and shall illuminate the working area directly in front of the operator. One master light switch shall be furnished on the instrument or lighting panel to prevent operation of any light during blackout conditions, with the exception of engine indicator lights. The master switch shall be equipped with a red,

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spring-loaded, hinged cover to prevent inadvertent actuation by the operator. Four toggle switches, each with red indicator light, shall be furnished and banked near the master switch providing selective actuation of top-mounted floodlights, skirt-mounted floodlights, ceiling cab lights, and instrument panel lights, whenever the master light switch is in the "ON" position. Each switch shall be identified as to function and be labeled with the basic information needed for proper actuation.

3.5.12.3.1.1 Top-mounted floodlights. Two floodlights, one on each side of the boom hoist cables, shall be mounted at the top front edge of the cab.

3.5.12.3.1.2 Skirt-mounted floodlights. Two floodlights, one on each side of the superstructure skirt shall be mounted as near the side and bottom of the cab skirt as practicable but not higher than the mid-point between the bottom of the cab skirt and the bottom of the operator's window.

3.5.12.3.2 Interior cab lights. The cab shall have three interior ceiling lights effectively located with plastic lens not less than 4 inches in diameter. The light shall be of the flush-mount, waterproof type having either a flexible elastomer housing or an elastomer ring. Each light shall be equipped with a single contact, bayonet type bulb, Trade No. 1683. The overall depth of the light fixture from the face of the light to the ceiling shall be not more than 3 inches.

3.5.12.4 Warning horn. The revolving superstructure shall be equipped with a warning horn in accordance with MS51074. The horn shall be mounted in a protected location on the exterior of the revolving superstructure so that it does not extend beyond the sides of the cab. The horn shall be operated by a heavy-duty pushbutton switch (normally open) mounted on the swing lever of the revolving superstructure.

3.5.12.5 Trouble light. A handgrip type, portable trouble light shall be mounted in the cab adjacent to operator's door. The light shall be equipped with a Trade No. 4505, sealed-beam spotlight and furnished with a connecting, self-coiling cable not less than 25 feet long. The trouble light shall have a handgrip handle incorporating a thumb or finger on-off switch. The trouble-light mounting bracket shall be of the cushioned spring-over-center type holding and enclosing the sealed beam unit to protect it from damage during all operations.

3.5.12.6 Instrument panel. An instrument panel containing all switches, gages, and instruments, except hourmeter-tachometer, for complete control of the engine and lights shall be located within reach and visibility of the operator in his normal operating position. All switches shall be identified as to function and be labeled with the basic information needed to actuate the switch, and provided with universal symbols in accordance with SAE J1362 as a minimum requirement. Gages and instruments shall be in accordance with SAE J209, SAE J1362 and shall be labeled as to function in accordance with 3.28.2.

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Identification plates shall be securely fastened to the panel to prevent breakage, bending or other defacement and the material shall be in accordance with 3.28. All instruments and indicators shall be installed assuring a metal-to-metal return ground from the component to the battery. The instrument cluster group shall be removable from the front to permit access to all wiring and units. Instruments and gages required by the operator during night operations shall be illuminated. The display luminance shall be adjustable from 0.03 to 1.0 ft-L. The illumination control shall be provided with an off position labeled "OFF". Symbols used to indicate illumination intensity variation shall be in accordance with SAE J1362 for variable control. The instruments and gages shall be mounted within gage functioning tolerance. All warning and indicator lights shall be banked horizontally on the instrument panel. All primary instruments, indicators and controls for diesel engines as defined in SAE J209 shall be of the following items:

Item	Requirement
Battery-generator indicator	MS24532
Fuel gage	MS24544
Heat indicator (engine coolant)	
Indicator	MS24543
Transmitter	MS24537
High-temperature warning light	MIL-L-3661/14 red lens
Heat indicator (torque converter and engine)	
Indicator	MS24543
Transmitter	MS24537
High-temperature warning light	MIL-L-3661/14 red lens
Oil pressure gages	
Indicator	MS24540 or MS24541
Transmitter	MS24538 or MS24539
Low-pressure warning light	MIL-L-3661/14 red lens
Item	Requirement
Hourmeter-tachometer	MS35916

The hourmeter-tachometer may be mounted on the engine in a location that is visible to the operator in his normal operating position. The hourmeter shall have a totalizing mechanism to show not less than 9,999 hours to register the number of hours of engine operation. The tachometer shall be accurate within ±5 percent of actual engine speed. All panel wiring shall be routed to eliminate a tangle of crossed wiring behind the instrument panel. Electrical wires shall be bundled into a harness assembly routed along an edge of the instrument panel with component connection lines parallel to each other.

3.5.13 Operator controls. The operator's controls shall be located and installed within the maximum areas in accordance with SAE J898.

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3.5.13.1 Crane controls. The crane and other operating attachments shall be controlled by the least possible number of hand levers and foot pedals necessary for efficient operation. The levers and pedals shall be within reach of the operator when seated and shall not obstruct the operator's view of the work. All brakes shall permit locking and unlocking by foot action alone but shall not become locked without the operator's intent regardless of the operator's size, skill, or training. For ease of operation, the pedals shall rise from the depressed position in a backward as well as vertical movement. Hand- and foot-operated controls shall be in accordance with SAE J983. Latches on control levers shall not cause delay in operation. Information plates in accordance with 3.28.2 shall be located on or near each control lever or pedal to indicate function. The neutral and operating positions shall be shown by arrows and words. When the same lever controls more than one function, all functions shall be listed. The method of lettering and the method of fastening the plate to the panel board shall be in accordance with 3.28.

3.5.13.2 Engine hand controls. Mechanical push-pull remote controls shall be of the waterproof, antifriction type. As a minimum, all exposed parts shall be fabricated of corrosive-resistant steel. The cables shall be covered with a polyethylene or vinyl jacket. Cable ends shall be tailored to actuate all levers through their respective arcs without binding in the mounts or cable. Bend radii shall be as recommended by the cable manufacturers plus an increase in radii of 20 percent. The as-installed mechanical efficiency of each installation shall be not less than 71 percent and the no-load friction shall be not more than 5 ounces per foot. The continuous operation load rating, in both tension and compression, shall be not less than 150-pound rating for crane-shovel controls and 200-pound rating for engine controls and other controls.

3.5.13.3 Accumulator. The accumulator, when furnished in the hydraulic control system, shall be of the piston type and charged with nitrogen gas. The pressure in the accumulator after completion of the tests specified in 4.5.2.17 shall be not less than it was at the start of the test. The accumulator, when part of a hydraulic control system incorporating an engine driven pump, shall be of sufficient capacity to assure control of a clutch for not less than 15 full cycles of engagement and release without engine power.

3.6 Diesel engine. The diesel engine shall meet the requirements of MIL-STD-1410 for application in a class II end item. The engine shall be the heavy-duty diesel furnished commercially including all systems, components, accessories, and auxiliaries.

3.6.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the crane-shovel under the conditions specified herein shall not require horsepower or speed in excess of the intermittent brake horsepower rating of the engine at the applicable speed. The maximum governed speed shall be between 120 and 150 percent of the peak torque speed. Engine, with accessories installed, shall be capable of developing intermittent brake horsepower at full load governed speed of not less than 275 percent

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of the drum line power specified in 3.5.1.a. The engine shall have a maximum no-load governed speed appropriate for the torque converter used but not more than 2600 rpm.

3.6.2 Operation condition. All systems, including lubrication system, induction system, and coolant system, shall perform without leakage or malfunction when the crane-shovel is negotiating a 30-percent grade or a 15-percent side slope when tested as specified in 4.5.2.37.

3.6.3 Lubrication system. Oil consumption limits specified in MIL-STD-1400, test method 1500, table II, schedule III, shall not be exceeded. The oil shall be filtered by truck-type, diesel-engine oil filters. The lubricating oil filters shall be of the full flow type. The filters shall be located in an accessible location. The external oil lines leading to and from the filters shall be of flexible hose with reuseable fittings. The hose shall be constructed of corrosion resistant steel wire with a tube of tetraflouroethylene. The lines shall be compatible with lines and fittings in accordance with SAE J30 and SAE J1019. The lines shall be identified by markers in accordance with MIL-B-43436 labeled "ENGINE OIL" near each fitting.

3.6.4 Fuel system. A fuel tank of welded aluminum construction, 0.125 inch in thickness, in accordance with DOT Safety Regulations and with MIL-T-46786, type III, shall be furnished. The fuel tank shall have a capacity of not less than 50 gallons and shall be located under the machinery deck. The tank shall be equipped with a drain outlet and threaded plug. The drain outlet shall be located so that removal of the plug will result in complete drainage of the tank directly into a suitable container. The fill-pipe shall be marked "DIESEL FUEL ONLY" or "JP-8 FUEL" on a plate in accordance with 3.28.2. The tank filling components shall include fill-pipe, strainer, safety cap, and external vent. The fill-pipe shall extend through the outside of the cab wall in a protected and accessible location so that the tank can be filled from a 5-gallon container in accordance with MIL-C-1283. The fill-pipe shall be equipped with a safety cap which is captive chained with swivel to the strainer and fill-pipe. The strainer proper shall be not less than 4 inches ID and shall be made of corrosion-resistant steel screening not less than 80 mesh, protected in such a manner as to prevent contact with the filling nozzle. The strainer shall be externally protected from damage due to surging of fuel. The fill-pipe, strainer, and cap shall be in accordance with DOT regulations and shall permit filling the tank at the rate of 50 gallons per minute without spilling. The engine fuel intake line shall be in the center of the tank. The fuel pickup point shall be not less than 0.50 inch above the bottom of the tank and shall be located on the longitudinal center of the tank to prevent starving of the engine on grades and slopes. The tank shall be baffled to prevent surging of the fuel. The fuel lines from the tanks to the final filter on the engine shall be of flexible hose with reusable fittings, all in accordance with SAE J30 and SAE J1019. The lines shall be routed in protected locations not subject to chafing and shall be supported by cushioned support clamps. The lines shall be identified by markers in accordance with MIL-B-43436 labeled

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"ENGINE FUEL" near each fitting. There shall be no evidence of raw fuel leakage throughout the test specified in 4.5.2.37. The mounting of the fuel tank shall be as recommended by the fuel tank manufacturer to prevent shifting, damage, and fatigue as a result of travel and transport, as specified herein. Brackets and fuel tank straps shall be insulated from the tank by a non-fuel-absorbent cushioning material.

3.6.5 Cold starting aids. Unless otherwise specified (see 6.2), engines shall be provided with cold starting aids either in accordance with MIL-E-52649, type III, size as recommended by the engine manufacturer, or of the glow plug type, suitable for cold temperature engine starting for meeting the requirements specified in 3.24.1.

3.6.6 Cooling system. When the loaded crane is operated at 5,000 feet above sea level or in an ambient temperature of not less than 125 °F (52 °C), the cooling system shall maintain the coolant in the top tank at a temperature of not more than 210 °F (99 °C). The control of the engine operating temperature shall be accomplished by a bleed-hole-type engine thermostat or a full-blocking-type engine thermostat that prevents coolant flow through the radiator core when the thermostat is closed. The radiator shall be of the three-piece, bolted-construction type: top-header tank, core assembly, and bottom-header tank. The radiator shall incorporate a bleeding system that shall function to eliminate air from the engine coolant and shall permit operation of the crane-shovel on grades and side slopes, as specified herein, without sucking air into the tubes of the core. The cooling system shall include sectionalized, removable shrouds. The shrouds shall be fabricated of aluminum sheet metal and the baffle seals shall be of flexible, elastomer material. Fan guards of the sectionalized, removable type shall be required. The radiator shall be secured by two metal braces (one on each side). These braces shall be fastened to the radiator shell approximately halfway between the top and bottom tanks and shall extend approximately 45 degrees to a point where the other end shall be fastened to the machinery deck. These braces shall have sufficient rigidity to prevent vibration damage to the radiator or the radiator connections at all engine speeds. The cooling system shall be pressurized and shall be furnished with a large size filler neck and cap for either 7 or 13 psi systems in accordance with SAE J164. A cooling system filter shall be provided that is matched to the coolant capacity of the engine. The filter shall retard electrolytic corrosion by providing zinc anodes. The element shall contain ion exchange resins to treat the coolant and maintain a correct acid-alkali condition. All chemicals in the filter shall be compatible with all types of coolants and shall be noncorrosive to the metals used in the cooling system in accordance with ASTM D 1384. The mechanical filter of the element shall prevent passage of any particle larger than 20-micron size. The inlet and outlet of the filter shall include shutoff valves. The valves shall be provided with labels which describe valve function, double-ended arrows showing the direction of operation and labeled at each end to indicate the functional result (i.e. open, close). The filter shall be installed in accordance with the recommendations of the filter manufacturer.

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3.6.7 Air induction system. The engine shall be equipped with a dry-type air cleaner. The air cleaner shall be in accordance with MIL-A-52363, class III. An air cleaner restriction indicator, in accordance with MS53063, shall be installed inside the operator's compartment near the instrument panel and visible to the operator to inform the operator when the air cleaner requires servicing. The intake vacuum of the installed system shall not exceed maximum vacuum recommended by the engine manufacturer.

3.6.8 Engine exhaust system. The exhaust system shall expel gases to the rear of the crane-shovel at a height of not less than 10 feet above the ground. The exhaust pipe configuration shall prevent entrance of windblown precipitation into the cylinders of the engine. The exhaust muffler, if other than horizontal, shall terminate with a curved stack pipe. The curved stack pipe shall be of the truck-type having not less than a 14-inch radius of curvature and a beveled undercut to protect the system from the weather. The aiming of the exhaust pipe shall be angled between 0 to 90 degrees towards the side from the rear of the crane-shovel and shall direct the gases away from personnel standing on the ground. The exhaust pipes, muffler, and stack pipe, shall be inherently heat- and corrosion-resistant or shall be protected on both the internal and external surfaces to resist the effects of heat and corrosion. The exhaust pipes, muffler, and stack pipe shall be enclosed if they are on the exterior of the vehicle. The pipes shall be protected from heat, rust and discoloration by either a ceramic coating fused to both the interior and exterior of the pipe, or use of corrosion-resistant steel. The ceramic coating shall provide thermal shock resistance at all temperatures up to 1650 ^{o}F (899 ^{o}C). The mounting bracket assembly shall be installed to withstand a force of not less than 300 pounds imposed in any vertical and horizontal direction without exceeding 1.50-inch displacement and without permanent deformation, damage, or loss of sealing of any component. The method of mounting, bracing, and clamping of the exhaust components to the bracket assembly shall prevent damage from heat expansion and severe vibration from cross-country travel. The exhaust back pressure of the installed system shall not exceed the recommendation of the engine manufacturer. At least one flexible section of corrosion-resistant steel bellows shall be provided between the engine and the muffler. A muffler shall be furnished and shall have an internal volume of not less than four times the cubic inch displacement of the engine.

3.6.9 Governor. The engine shall be controlled by a tail-shaft governor in addition to the engine overspeed governor. Selected tail-shaft governor settings through the operating range shall be provided by a variable-speed governor control lever which shall be accessible to the operator.

3.6.10 Starting system. The engine starting system shall be as recommended by the engine and starter manufacturers for this application. The starting system shall include a starter switch, auxiliary control solenoids, engine speed disconnect, main starter solenoid, starter motor, and fly-wheel drive mechanism. The starter switch shall be of the waterproof type, rated at not less than 15-amp inductive load. The starter switch shall actuate a 24-volt

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auxiliary control relay which shall actuate the main starter solenoid and starter motor. The engine speed disconnect shall prevent starter engagement whenever the engine is running. The disconnect switch shall be in accordance with MS90530-1 and shall be operated by fuel or oil pressure. The starter motor shall be in accordance with MIL-S-3785, type II, grade A, class 1, heavy duty. Procedure for starter engagement periods shall be in accordance with the starter manufacturer's recommendations.

3.6.11 Charging system. An engine-driven 60-ampere, ac generator (alternator), in accordance with SAE J180 shall be furnished. The charging system shall be of the low cut-in type that provides not less than 25 amperes hot-charging current at engine idle speed with the lowest possible alternator speed. The alternator speed range shall be in accordance with the recommendations of the alternator manufacturer and shall not exceed 6,000 rpm maximum. The alternator shall be mounted on two fixed mounts and one adjustable mount in accordance with SAE J180. All alternator mounting bolts shall be not less than SAE grade No. 5 and not less than 0.50 inch in size. Hardened steel flat washers, not less than 0.125 inch thick, shall be furnished under all bolt heads, nuts, and between the adjusting lug and the adjusting strap. The alternator adjustable mounting bracket shall provide movement for adjusting the belt tension. The alternator fixed mount axis line shall be within a range of 20 to 60 degrees from the belt tension resultant load direction, as measured from the alternator armature shaft centerline. The fixed mounting bracket shall be in compression as a result of the load applied by the belts. A protective relay assembly, electrically matched to the ac generator, shall be provided to protect the crane-shovel electrical system from damage when (1) incorrect polarity is applied through the slave receptacle to the batteries, (2) the crane-shovel batteries are incorrectly installed, and (3) arc welding for any maintenance or repair is performed on the crane-shovel. Fuses are not acceptable. Protective diodes, if used within the system, shall be installed in a manner which permits removal and replacement without the use of tools. The diodes shall be in accordance with MIL-STD-701, with a minimum amperage of not less than 150 percent of the maximum current flow and a peak inverse voltage (PIV) rating of not less than 1200 volts.

3.6.12 Storage batteries. Storage batteries shall be in accordance with MS35000-3, except that electrolyte shall be furnished. The electrolyte for each crane-shovel shall be separately packaged in accordance with 5.1. The batteries shall be connected to provide 24 volts and shall be negative grounded.

3.6.12.1 Battery container. A battery container shall be provided for the batteries. The container shall be fabricated of noncorrosive sheet metal not less than 0.060-inch nominal thickness and the interior of the container shall have a finish in accordance with MS53046. The enclosure shall (1) hold the batteries level when the crane-shovel is on level ground, (2) include drains for water or electrolyte spilled within the enclosure, (3) provide gas venting to the outside, and (4) provide accessibility for servicing and removal of the

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batteries. Drainage shall be to the ground without draining over other parts or components. Clear vision shall be provided for checking of the electrolyte level in each cell of the batteries. Each pair of batteries shall be fastened to the enclosure to prevent damage from movement and vibration. Battery hold-down frames shall be in accordance with MS53046-3 and shall be installed with a total of five 0.375-inch, 10-inch-long "J"-bolts, with washers in accordance with MS15795-815, and wing nuts in accordance with MS35425-74. The size of the enclosure shall provide not less than 1-inch clearance on each side and top of a pair of 6TN battery cases, as shown on MS35000. The cover shall overlap the sides and ends of the enclosure, and the cover shall be either removable or hinged to open to an angle greater than 90 degrees to permit vertical removal of the batteries. A spring-loaded detent brace shall be furnished at each side to hold the cover open when hinged. Tension-type latches with strikers shall keep the cover closed. Battery containers that are located in a position providing a stepping surface shall support the maximum weight of personnel stated in SAE J833 without permanent deformation. A pictorial battery hookup plate, in accordance with 3.28.1, shall be furnished on or near the battery container. The positive and negative battery cables shall be red, labeled "+", and black, labeled "-", respectively.

3.6.13 Emergency starting and charging receptacle and cable. A slave receptacle in accordance with MIL-C-62122 shall be provided to facilitate electrical connections between the batteries and an external electrical power source for emergency starting of the engine and charging of the batteries. The receptacle shall be recessed in the wall of the superstructure so the plug-in face of the receptacle is flush with the outside face of the wall and shall be accessible to a man standing on the ground. The receptacle shall be located as near the batteries as possible and shall be connected into the battery circuit. A 20-foot-long, two-conductor shielded cable in accordance with MIL-C-13486, type II, class B, equipped with a slave plug in accordance with MIL-C-62122 on each end, shall be furnished with each crane-shovel. The emergency starting and charging receptacle and slave cable shall be in accordance with 3.5.12 and shall perform as specified herein when tested as specified in 4.5.2.18.1. Provision shall be made for carrying the cable within the superstructure cab. The receptacle shall be provided with a label which states "SLAVE RECEPTACLE, 24 VOLTS."

3.7 Contamination level. The particulate matter in hydraulic fluid removed from the system directly upstream of the filter shall not exceed the ISO Code level of 19/16 in accordance with SAE J1165 after completion of assembly of the crane-shovel.

3.8 Crawler mounting.

3.8.1 Crawler base. The crawler base shall be a rib steel casting or shall be rigidly constructed of heavy steel shapes, fitted and welded, and shall be constructed to prevent permanent distortion. The swing gear and turntable roller path shall be steel and shall be rigidly attached to or integral with

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the base. The swing gear shall have accurately machined or ground internal or external teeth. The turntable roller or shear ball path shall have an accurately machined or ground bearing surface, and shall be of sufficient size and strength to support the revolving superstructure under all conditions without distortion. Where the center pin performs the holddown function to prevent tipping, the pin shall be of high-strength steel, and shall be accurately machined or ground to size. The pin shall be provided with adjusting devices readily accessible from the rotating frame. The center pin may be of a separate flanged casting or forging and bolted to the underframe. Where hook rollers perform the holddown function, the lower member of the center pivot bearing shall be integral with the crawler base and shall be steel with smooth bearing surfaces. The center pin shall operate in bronze or antifriction bearings.

3.8.2 Track assembly. The crawler tracks shall be of a width and length suitable for the crane-shovel operation. Suitable adjustments shall be provided for maintaining proper track tension. The track shoes shall be of wear-resisting steel and of the flat self-cleaning design. The track connecting pins shall be of steel of sufficient size, hardness, strength, and shall be easily removed and replaced. The tracks shall be supported on a sufficient number of steel rollers in such a manner to distribute the weight of the entire crane-shovel uniformly over the ground for the entire length of the crawlers, without depending upon the tension in the track for support. All track assembly shafts and rollers shall be provided with guarded grease fittings. Where practicable, grease fittings shall be accessible from outside the crawlers and lower case.

3.8.3 Side frames. The crawler track side frames shall be steel castings or welded structural steel members. When side frames are not used, large track rollers shall be used, with not less than four axles supporting the lower base.

3.8.4 Crawler drive. The crawler shall be driven by suitable machine cut gears or roller chains. The travel mechanism shall have positive steering by means of independently operated friction brakes and adequate clutches or positive locking lugs engaged by sliding jaw clutch member and controlled from the operator's cab. The crawler clutches shall be of adequate size to permit constant operation without overheating. Brakes or other locking means shall be provided to hold the crane-shovel stationary during the working cycle while the crane-shovel is on level grade, or any grade at which it will operate satisfactorily, and to prevent the crane-shovel from accidentally rolling down a 40-percent grade. Brakes that can be applied at any time regardless of the position of the dogs on the propulsion drive shall be furnished on the travel mechanism. The brakes shall be of a fail safe design. It shall be possible to apply the brakes on both tracks at the same time, when required. Jaw clutches, when furnished, shall have a sufficient number of teeth to be readily engaged and to transmit maximum traction power without undue wear. One jaw clutch must be engaged at all times. The travel gears and bearings shall be

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protected against the entrance of dirt or other foreign matter. The propelling mechanism shall be provided with a positive traction lock or brake for locking the crawlers. The propelling shaft shall be arranged that it is not one or a part of the supporting axles. All shafts shall be properly bushed or provided with antifriction bearings and ball-check grease fittings.

3.8.5 Axle. Each axle or crossmember, when used, shall be either of cast, forged, or structural steel. When more than one-piece structural steel axles are furnished, the edges to be joined shall be aligned and welded as specified herein (see 3.35.5).

3.9 Front-end attachments. The revolving superstructure of each crane-shovel shall be equipped with all necessary equipment to permit operation of the crane-shovel as a crane, clamshell, dragline, piledriver, shovel, backhoe, and crane with wrecking ball. Tools, equipment, parts, and material that are not a component part of the crane-shovel, as specified herein, and are required for operation of front-end attachments, shall be furnished with each front-end attachment.

3.9.1 Primary front-end attachments. Unless otherwise specified (see 6.2), the following attachments shall be furnished with each crane-shovel. Such attachments shall be complete with all necessary parts, including wire rope required for satisfactory functioning.

3.9.1.1 Crane boom. The basic boom shall be 30 feet long and shall be in accordance with MIL-B-52203. Boom lengths up to 60 feet shall be accomplished by using 10-foot extensions (see 3.9.2.1).

3.9.1.2 Pendant equalizer assembly. A pendant equalizer assembly shall be furnished. Stresses in the pendant equalizer shall be not more than 60 percent of the yield point under all operating conditions specified herein and shall be tested as specified in 4.5.2.3.

3.9.1.2.1 Sheaves. The pitch diameter of load bearing sheaves shall be in accordance with SAE J881. Flanges shall extend beyond the rope grooves not less than 1.75 times the diameter of the rope and shall have a hardness of from 255 to 302 Brinell. All sheaves shall be guarded to prevent the rope from leaving the sheave groove during operation. The sheave pins shall be of medium carbon or alloy steel and shall be of sufficient size and strength to support the sheaves under maximum operating conditions.

3.9.1.2.2 Sheave shaft. Sheave shafts shall be steel ASTM A 576, grade 1040-1045, with a hardness of 229 to 277 Brinell. Means of retaining the sheave shaft in the assembly shall be provided.

3.9.1.2.3 Clevis pin. Clevis pins shall be furnished for attaching the pendant equalizer assembly to the open-swaged socket of the pendant cable assembly, as specified in MIL-B-52203. Clevis pins shall be provided with

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means of retaining the pin and shall be steel ASTM A 576, grade 1040-1045, with a hardness of 240 to 265 Brinell.

3.9.1.3 Boom safety stop. A telescopic boom stop shall be furnished in order to provide a safeguard against overtopping of the boom and shall be installed to limit the maximum boom angle to 82 degrees above the horizontal plane. The complete boom stop with supporting attachments shall be designed with sufficient strength to develop the ultimate strength of the boom in bending at the point of attachment of the boom stop to the boom. The boom stop shall provide a shock-absorbing action and shall remain assembled when the boom point is lowered to the ground. The boom stop shall be attached to the boom with the boom stop attachment and shall be removable with common tools. Under all operating conditions, as the boom is raised from the ground, the boom stop shall telescope freely without sticking, binding, or bending while the boom is raised to an angle of 82 degrees with the horizontal.

3.9.1.4 Boom angle indicator and chart. An indicator which will show the position of the boom at all times shall be provided. The indicator shall be of the heavy-pendulum type swinging on a double-sealed (sealed-for-life) anti-friction bearing. The pointer shall indicate the angle of the boom with the horizontal plane. The indicator shall be provided with means of attaching to the standard military boom without cutting, welding, or drilling the boom. The indicator shall be located on the boom, near and visible to the operator in his normal operating position. An angular radius relationship chart shall be furnished and mounted in the cab near the operator. The chart shall show angles and radius with boom length up to 60 feet and include a statement of additions to be made when using the boom jib. The chart shall be of the same material as the plate specified in 3.28.2. The chart may be combined with the crane capacity plate specified in 3.20.1.5.

3.9.1.4.1 Overload indicating system. When specified (see 6.2), an overload indicating system shall be furnished in accordance with SAE J159 and as specified herein. This system shall automatically indicate and warn the operator when an overload condition exists that exceeds the rated loads, through the use of lights, an audible signal, and dials mounted in the cab within the operator's view. The audible and light signals shall cease and dials shall indicate a safe loading condition when corrective action has been taken by the operator. Any components of the system mounted on the military standard boom shall be installed without cutting, welding, or drilling the boom. The indicating system may utilize the 24-volt electrical system of the superstructure. The electrical system shall be waterproof and shall operate under all temperatures and use conditions specified herein without any malfunction.

3.9.1.5 Crane hook block. The crane hook block shall be in accordance with GGG-B-485 except that the hook block shall have a safe working capacity required to meet the crane-shovel requirements. The hook block shall have a proof test load capacity of twice the safe working load capacity and shall contain two sheaves. The overall length shall be not more than that specified for the

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size 20; the gross weight and hook opening shall be not less than specified for the size 10 of GGG-B-485. Lubrication fittings shall conform to 3.11.2.

3.9.1.6 Rope. All rope, except tagline and dipper triplines, shall be wire rope in accordance with RR-W-410, type I, class 2, 6 by 19, construction 3, IWRC-improved plow steel, uncoated, right regular lay, and shall be preformed. The breaking strength of ropes shall be not less than the wire rope strength factors in accordance with SAE J959. The hook-block load rope shall be not less than 0.625-inch diameter, and the boom hoist shall be not less than 0.50-inch diameter.

3.9.1.6.1 Rope sockets. All rope, except taglines and dipper triplines, shall have ends attached to drum laggings or to dead-end connections by wedge-type sockets. Clamps, slips, and screw wedge connectors shall be permitted only on tagline and dipper triplines.

3.9.1.6.2 Wire rope lengths. Sufficient wire rope shall be furnished for the primary and secondary drums with the basic boom at minimum radius shown in table III, the hook block at ground level, and a four-part line. There shall be not less than three wraps of wire rope on the hoist drum. The boom hoist shall have sufficient rope to allow the tip of the boom to be lowered to rest on level ground plus three additional wraps on the drum. The primary and secondary hoist drum shall be equipped with drum-lagging with a rope capacity for a four-part hoist line and a boom length of not less than 60 feet.

TABLE III. Crane function to determine rope lengths.

Function	Position of boom	Position of attachment
Crane hook	At minimum radius shown in table V.	At ground level.
Clamshell	60 degrees from horizontal.	20 percent of boom length (in feet) below ground level.
Dragline	40 degrees from horizontal.	50 percent of boom length (in feet) below ground level and 30 percent of boom length (in feet) beyond boom tip.
Piledriving	At basic boom height.	Two basic boom lengths plus 50 feet below ground level.

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A chart shall be provided and mounted in the cab showing the correct length and size of wire rope required for operation of all front end attachments. The chart shall be of the same material as the plate specified in 3.28.2.

3.9.2 Secondary front-end attachments.

3.9.2.1 Boom extensions. When specified (see 6.2), boom extensions shall be furnished and shall be in accordance with MIL-B-52203.

3.9.2.2 Boom jib. When specified (see 6.2), the boom jib shall be furnished and shall be in accordance with MIL-B-52207. Each boom jib shall be provided with an overhaul block and hook, with a safety latch actuated by a stainless-steel spring.

3.9.2.3 Clamshell bucket. When specified (see 6.2), the clamshell bucket shall be furnished and shall be in accordance with MIL-B-16028, type II, size 75. Bucket teeth shall be in accordance with MIL-T-52176. When specified (see 6.2), side and corner teeth shall be furnished.

3.9.2.4 Tagline. When specified (see 6.2), the tagline shall be furnished, shall be of the reel type, with an adjustable tension mechanism, shall control a bucket, as specified in 3.9.2.3, and shall be in accordance with MIL-T-52198. The tagline shall be provided with means of attaching to the standard military boom without cutting, welding, or drilling the boom, and shall be removable with common tools.

3.9.2.5 Dragline bucket. When specified (see 6.2), the dragline bucket shall be furnished and shall be in accordance with MIL-B-16029, type II, class S, size 75. Bucket teeth shall be in accordance with MIL-T-52176.

3.9.2.6 Dragline fair-lead. When specified (see 6.2), the dragline fair-lead shall be furnished and shall be in accordance with MIL-B-52203.

3.9.2.7 Wrecking ball. When specified (see 6.2), the wrecking ball (skull cracker) shall be furnished and shall be in accordance with MIL-B-23276, size 1 or 2 as specified (see 6.2).

3.9.2.8 Piledriver leads, boom adapter, and catwalk. When specified (see 6.2), the piledriver leads, boom adapter, and catwalk shall be furnished and shall be in accordance with MIL-L-11025. Lead sections shall consist of one top section and four lower sections for each crane-shovel.

3.9.2.9 Piledriver hammer. When specified (see 6.2), the piledriver hammer shall be furnished. The piledriver hammer shall be a drop hammer in accordance with MIL-H-2514, or a diesel piledriver hammer in accordance with MIL-H-21103, size as specified.

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3.9.2.10 Shovel front. When specified (see 6.2), a shovel front shall be furnished.

3.9.2.10.1 Shovel boom. The shovel boom shall be fabricated from structural steel shapes and plates rigidly welded to form a box section up to the shipper shaft for an inside dipper stick, and to the boom tip for an outside dipper stick. Replaceable wearing shoes or other suitable fittings shall be provided where the dipper stick passes by or through the boom. Replaceable buffer rails shall be provided where the dipper strikes the boom.

3.9.2.10.2 Dipper stick. The dipper stick shall be rigidly constructed to withstand all working stresses and strains imposed by normal shovel operations.

3.9.2.10.3 Shovel dipper. The shovel dipper shall be of either heavy steel plate, cast steel, or a combination of both. The front lip shall be of abrasion-wear-resistant, hard-surfaced steel. The dipper lips shall be equipped with tooth sockets, shanks, and teeth in accordance with MIL-T-52176. The dipper shall have a discharge gate (greater in area than the dipper mouth) in the bottom that shall be securely hinged to the back of the dipper with large steel hinges. The dipper front-latch opening on the wearing side shall have a replaceable wear plate where the latch bar engages. The dipper angle shall be adjustable for different digging conditions. A power trip controlled from the operator's seat and a slack takeup to prevent fouling of the tripline shall be provided. The shovel dipper capacity shall be not less than that specified in 3.20.1.2.

3.9.2.10.4 Dipper crowd and retract. The dipper crowd shall be self-adjusting at all boom angles and shall be provided with a tension adjustment for maintaining the correct tension on the chain or rope. The dipper crowd and retract shall be operated by an independent positive chain or rope mechanism.

3.9.2.10.5 Sheave. The shovel boom point sheaves shall be in accordance with 3.9.1.2.1 except that the pitch diameter shall be not less than 15 times the diameter of the rope used. The sheave shaft shall be in accordance with 3.9.1.2.2.

3.9.2.11 Backhoe. When specified (see 6.2), a backhoe shall be furnished.

3.9.2.11.1 Boom, dipper arm, and gantry. The boom, dipper arm, and gantry shall be fabricated from structural steel shapes and plates rigidly welded to withstand all working stresses and strains imposed by backhoe operations.

3.9.2.11.2 Dipper. The backhoe dipper shall be of heavy steel plate or cast steel, shall have a cutting lip of abrasion-resistant, hard-surfaced steel, and shall have the bottom reinforced and protected by the extension of the tooth sockets or wearing bars. The dipper lips shall be equipped with not less than four tooth shanks and teeth in accordance with MIL-T-52176, size as

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applicable. The shank shall be pinned or bolted into the socket. Side cutters shall be replaceable cast alloy or drop-forged, abrasion-resistant steel, heat treated and tempered. The dipper angle shall be adjustable for different digging conditions. The dipper width measured over nominal-size side cutters shall be as specified in table VII.

3.10 Winterization. When specified (see 6.2), the crane-shovel shall be furnished with a winterization system that will permit efficient engine heating and starting, engine coolant heating, battery heating, personnel cab heating, window defrosting, and equipment operation at any ambient temperature from -50 to +105 °F (-46 to +41 °C). Cab doors, radiator winter-front and engine-housing side panels may be removed or opened for operation above -25 °F (-32 °C). Engine operation after starting shall not be dependent on the winterization system. Winterized crane- shovels shall start in 5 minutes in any ambient temperature warmer than -25 °F (-32 °C) and function exactly like nonwinterized crane-shovels without overheating or other malfunction at any temperature up to 105 °F (41 °C).

3.10.1 Performance (winterized crane-shovel). With the winterized crane-shovel stabilized at a temperature of not higher than -50 °F (-46 °C), the winterization system shall perform as follows:

a. Heat the engine, engine coolant, and batteries sufficiently to obtain a start within 5 minutes after 4 hours of standby 115-volt electric heat.

b. Heat the personnel cab from -50 °F (-46 °C) to a temperature not colder than +41 °F (+4 °C) in 1 hour after the engine starts, measured at 24 inches above the seat reference point at the foot pedals, and automatically maintain the temperature of +41 °F (+4 °C) or warmer.

c. Defrost not less than 75 percent of each front and side cab window in 1 hour after the engine starts, and continue to maintain not less than 75 percent of each window frost free.

3.10.2 Engine heater. An electrical-standby, 115-volt, oil-pan heater shall be provided with sufficient capacity to raise the engine oil temperature in the sump to an average of +10 °F (-12 °C) and assure a positive engine start after 4 hours of standby heat. The heater coil shall be installed so it is covered with oil when the oil level is at the low mark and as near the pump pickup as possible. The watt (W) density of the heater shall be not more than 18 Watts per square inch. The system shall be controlled by a 180 °F (82 °C) thermostat. The heater coil shall be covered with oil when the crane-shovel is on a 40-percent longitudinal and 30-percent side slope in all directions.

3.10.3 Coolant heater. An electrical-standby, 115-volt coolant heater with a thermosyphon system shall be provided to raise the engine coolant to +50 °F (10 °C), measured at the coolant inlet of the heater after 4 hours of standby

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heat. A thermostat shall be furnished and set to cut off the standby power when the coolant at the inlet of the heater reaches +100 °F (+38 °C). Shutoff valves shall be installed in the lines to enable removal of the heater without draining the system. The valves shall be provided with labels which describe valve function, double-ended arrows showing the direction of operation and labeled at each end to indicate the functional result (i.e. open, close).

3.10.4 Cab heater and defroster. The cab heater shall be a conventional coolant heat-transfer type, mounted in the cab not to interfere with the operator performing normal functions or obstruct entering or leaving the cab. A ducting system shall be provided to direct cab heat to the operator's feet and to defrost and maintain not less than 75 percent of each window (front and side) frost free. The area defrosted shall be at least that described by SAE J382, 3.2 Area A. The controls for the heater shall be accessible to the operator and the heater thermostatically controlled to maintain the temperature requirements specified in 3.10.1. The heater shall be provided with an air-intake proportioning device with connection and controls to draw any desired mixture of inside and outside air.

3.10.4.1 Coolant lines. The heater lines shall be in accordance with 3.10.11 and shall be of the size specified for the heater. Lines shall be connected to ensure maximum circulation of the heated coolant throughout the full length of the engine cylinder block and heads. The shortest possible coolant lines shall be used and shall be supported to minimize vibration or interference with engine maintenance and operation. Flexible metallic or flexible, steel-reinforced rubber hose and connections shall be used to prevent damage by line vibrations, expansion, or contraction.

3.10.5 Battery winterization.

3.10.5.1 Battery box. The battery box shall be made of corrosion-resistant steel not less than 0.060-inch thickness. THe box shall provide a minimum internal clearance of 1 inch around the perimeter and 2 inches over the top of each battery. Batteries shall be supported on nonconductive rails or multiple-posts at least 1 inch above a perforated plate over a bottom-plenum at least 1 inch deep. The box shall incorporate a gasketed cover, a drainhole not less than 0.50 inch in diameter, provision for firm acid-resistant battery holddown and spacing, a heating-air inlet to the bottom-plenum not less than 3 inches in diameter, at least 15 holes in the perforated-plate with a total cross sectional area not less than 10 square inches, and check-dampered, 3-inch-diameter air outlet vent. The battery supports, holddowns, and spacers shall not interfere significantly with the free circulation of heating air through the spaces around the batteries. Rubber grommets or boots shall be provided to cushion and seal the opening where battery cables enter the box. The cover of the battery box shall not be used as a holddown device, and shall be provided with latches and handles large enough to facilitate operation by arctic mittened hands. The box shall be faced with nonconductive material on the bottom side to provide a safety factor against accidental short circuiting of

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batteries and cables. The battery box shall be located and installed to allow full accessibility to the batteries for servicing and replacement. A connection for a duct from the cab-heating system shall be provided. If installed in the cab, the box shall be on or near the floor. An acid or alkali-resistant pipe or hose shall be provided to discharge any liquid on the ground. The outlet airvent shall be extended out of the cab to a downward discharging position. If not installed in the cab, the battery box and its cover shall be insulated externally with a 1-inch thickness of plastic foam, and the insulation shall be protected by a hard facing on the cover and on exposed sides or ends. Bottom support shall not be through the insulation unless it can be shown that its strength and rigidity are suitable for such application. A pictorial battery connection plate, in accordance with 3.28.1, shall be furnished on or near the battery container and shall be viewable while the batteries are being maintained. The positive and negative battery cables shall be colored red, labeled "+" and colored black, labeled "-", respectively.

3.10.5.2 Battery heating and temperature control. The batteries shall be heated during operations by fresh air from the cab heater, delivered through a duct of not less than 3 inches in diameter, through a check-damper, and through a thermostatically controlled damper, the latter governed by a temperature-sensing element in the battery box to maintain air-space temperature normally between 40 and 100 ^{o}F (4 and 38 ^{o}C). A bypass or override control shall be provided, operable from the panel containing other heater controls, by which air at maximum system temperature can be delivered continuously when preheating after "soakout" to extreme low temperatures. The positions of this control shall be clearly marked "BATTERY PREHEAT" and "BATTERY NORMAL HEAT".

3.10.5.3 Battery charger. A 24-volt, 15 amp-charger shall be mounted in the cab to maintain the batteries at the initial charge level during standby operations. The charger shall be equipped with a limiting device to reduce the charger output equal to the battery state of charge. A direct reading ammeter indicating the charger output shall be fitted to the charger case.

3.10.6 Winterized cab. The winterized cab shall be as specified in 3.5.3 and the additional requirements specified herein. The doors shall be of the swinging type and rubber sealing gaskets shall withstand temperatures of -50 to +105 ^{o}F (-46 to +41 ^{o}C). Doors shall open a minimum of 90 degrees. Doors shall be equipped with positive latches of a type that will keep the doors closed when subjected to vibration. Positive hold open devices to secure doors in the open position shall also be provided. Friction or coil-spring compression devices are not acceptable. All windows, except one front window, shall be tinted safety glass in accordance with MIL-G-3787, class 1, type II. All controls or open housings shall be covered with a boot to prevent cold air from entering the cab. A storage compartment shall be provided for the electrical 25-foot cable specified in 3.10.8.

3.10.6.1 Front window. The front window glass shall be laminated, tinted, safety-plate glass in accordance with MIL-G-3787, class 1, type II. The front

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window shall be equipped with one or more 24-volt electric window wipers. The wiper shall have not less than two speeds of operation, and shall have a park position for holding the blade when the glass is stored overhead. The wiper shall clear at least 75 percent of the front window area when in operation and shall be capable of operation from inside the cab by handcrank. The area cleared shall be at least that described by SAE J198, 3.1.1, Area A.

3.10.6.2 Cab insulation. The interior of the cab shall be insulated with mastic-type insulation not less than 0.0937 inch thick or polyvinylchloride sheet insulation not less than 0.25 inch thick. All insulation shall be non-hydroscopic, fire-retardant, resistant to attack by rodents, insects, and fungi, and shall maintain its form under all operating and environmental conditions specified herein.

3.10.6.3 Weatherstripping. Weatherstripping in accordance with MIL-R-14328 shall be used for sealing the doors and covers to the cab framework and shall not interfere with normal operation of the doors or covers. The type of weatherstripping and method of securing it shall prevent the entrance of blowing snow and shall reduce vibration and noise to a minimum. The weatherstripping used for sealing the bottom of the doors shall be so installed that it will not be damaged by personnel stepping on it while entering or leaving the cab.

3.10.6.4 Floor mat. The floor part of the cab occupied by the operator shall be fitted with a fiberglass or a with polyvinylchloride shell insulation not less than 0.375 inch thick with an integral antiscuff skin surface. The floor mat shall be in sections in accordance with the size of the floor plates to permit removal of floor plate sections without distrubing adjacent areas of the mat. Floor insulation shall be bonded to floor plates.

3.10.7 Radiator winterfront. The radiator shall be furnished with a two-piece, removable, manually-controlled metal winterfront of the sliding type. The winterfront shall not jam from blowing snow or become inoperative when covered with a sheet of ice up to and including 0.25 inch thick. The open area shall be not less than 30 percent of the frontal area of the radiator core. A locking device shall be provided on each section to positively lock the winterfront in the following positions: Full open, half open, and closed. Operations of the winterfront shall be accomplished without the use of tools or pry bars.

3.10.8 Wiring system. The 115-volt electrical heating elements for the engine sump, engine coolant, and battery box shall be connected through a distribution box with a separate circuit-breaker switch for each outgoing line. This box shall be weatherproof or shall be mounted in the cab and connected to an easily accessible, externally-mounted, waterproof male receptacle. A single three-wire power cable, 25 feet long, in accordance with MIL-C-3432, of adequate capacity to serve all heating elements, shall be furnished. Cable terminals

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shall be the three-prong type and shall ground the vehicle when connected. All winterized elements and the electrical system shall be radio suppressed.

3.10.9 Air cleaner. Warm air from the engine housing may be directed through the air cleaner if necessary for more efficient engine operation.

3.10.10 Cold starting aids. The cold starting aids specified in 3.6.5 shall be installed in the cab.

3.10.11 Hose and hose assemblies. Hose and hose assemblies shall be in accordance with the following specifications:

Hot air duct hose.	MIL-H-8796
Coolant hose (radiator and coolant heater).	ZZ-H-428, type I or II, grade A, class 2

3.10.12 Fuels, lubricants, hydraulic fluids, and antifreeze for winterized cranes. The crane- shovel shall be assembled, run-in, tested, and delivered with the products listed in table IV, except that fuels and oils used during run-in and testing shall be suitable for use in the prevailing ambient temperature range.

TABLE IV. Standard military fuels, lubricants, hydraulic fluids, and antifreeze, for winterized cranes.

Product	Specification No.	Symbol
Compound, antifreeze, arctic type.	MIL-A-11755	-
Fuel oil (an alternate fuel suitable for arctic temps: or turbine fuel, aviation, kerosene type, grade JP-8, diesel, arctic grade.	VV-F-800 MIL-T-83133	DF-A NATO F-34
Lubricating oil, internal-combustion engine, arctic.	MIL-L-46167	OEA
Lubricating oil, gear, multipurpose.	MIL-L-2105	GO-75
Greases, aircraft and instrument.	MIL-G-23827	NATO G-354
Greases, automotive and artillery.	MIL-G-10924	GAA
Hydraulic fluid, automotive, all-weather (brake system).	MIL-B-46176	BFS
Hydraulic fluid, petroleum base, for preservation and operation.	MIL-H-6083	OHT
Lubricating oil, instrument.	MIL-L-6085	OAI

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3.10.13 Fabrication. The completely winterized crane-shovel in accordance with all the requirements specified herein for nonwinterized crane-shovels including the weight and dimensional requirements. All latches, handles, grab-handles and access doors shall be proportioned to permit grasping and manipulating by personnel wearing arctic type mittens.

3.11 Lubrication. All surfaces requiring lubrication shall be provided with accessible lubrication fittings. A lubrication fitting shall be located at each area where two moving metal surfaces are in contact and under pressure.

3.11.1 Lubricants. The procedure for the selection of lubricants shall be in accordance with section 5 of MIL-STD-838. Lubricants selected shall be in accordance with chapter 2 of MIL-HDBK-113. When the specification of the lubricant selected includes a requirement for a Qualified Products List (QPL) the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.11.2 Lubrication fittings. Lubrication fittings shall be in accordance with MS15003. Fittings shall be located in a protected position and shall be accessible to a grease gun in accordance with MIL-G-3859. The gun shall be equipped with a hydraulic coupler in accordance with MS24207 and a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories and parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.11.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure-lubricating equipment will damage grease seals or other parts.

3.11.4 Enclosures with integral reservoir of lubricant. Enclosures such as gear cases and transmission housing that contain a reservoir of lubricants for the parts enclosed shall be equipped with dipsticks, check plugs not less than 0.50-inch pipe size, or sight gages to determine the level of the lubricant. If dipsticks are used, dipstick marking shall comply with SAE J614. Each enclosure shall be equipped with a means for filling the enclosure with a lubricant and for draining. The fill pipe shall be not less than 1-inch in diameter. The drain shall be fitted with a drain plug in accordance with MS35844. The drain shall be located to provide complete drainage of the lubricant from the enclosure. Drainage shall be into a suitable container when the crane-shovel is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain into suitable containers. The drain plugs, the filling means, and the lubricant-level dipsticks or holes shall be accessible without disturbance of accessories or parts.

3.11.5 Lubrication service.

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3.11.5.1 Grease lubrication. Bearing lubrication, including sealed bearings, shall be with greases as follows:

a. MIL-G-23827 - For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250 °F (121 °C).

b. MIL-G-10924 - For other bearings where operating temperatures do not exceed 175 °F (79 °C).

The crane-shovel shall be run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.11.5.2 Oil lubrication. Unless otherwise preserved for packaging (see 5.1), all components shall be filled to the operating level with winter grades of military oils designated for use in the temperature range of 0 to 32 °F (-18 to 0 °C). Each lubricated component shall be tagged in a conspicuous place to indicate the temperature ranges and grades of oils used.

3.12 Fuels and lubricants. Unless otherwise specified (see 6.2), and prior to delivery to the Government, the winterized crane-shovels shall be serviced with military fuels, lubricants, greases, hydraulic fluids, and antifreeze in accordance with table IV during manufacturing, assembly, run-in, and testing. The ambient temperatures at the time of manufacture may dictate that lubricating oil of heavier weight than winter weight is required for run-in and operation testing.

3.13 Lifting and tiedown attachments. The lifting and tiedown provisions on all major components (those components that will be shipped unboxed, uncrated and lifted as a separate unit) of the disassembled crane-shovel shall meet the criteria of MIL-STD-209. The tiedown provisions shall permit tiedown of all components of the disassembled crane-shovel to the floor (or deck) of the transport medium to restrict shifting or movement in any direction.

3.14 Antirotation devices for shipping. Steel rods shall be furnished for securing the revolving superstructure to the track frame to prevent rotation during shipment. The steel rods shall be of a size to restrain a movement equal to twice the gross weight of the revolving superstructure (without boom) multiplied by the distance from the center of gravity to the center of rotation. One end of each rod shall be permanently affixed in a single position on the track frame while in use as an antirotation device and for storage. Means shall be furnished to secure the loose end of the rod to the track frame for storage.

3.15 Toolbox. A toolbox shall be provided for all the maintenance and special tools. The toolbox shall be welded steel construction not less than

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0.071 inch thick. The cover shall be attached to the box on one edge by a continuous hinge in accordance with MS35829-7. To secure the cover in a closed position, an attachment to accommodate a lock, in accordance with MS35647, shall be located on the side opposite the hinge. The toolbox shall be located in an accessible, protected area. Means for drainage of the toolbox shall be provided.

3.16 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments. Covers, safety guards, and plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fastenings. All access opening dimensions shall conform to SAE J925. The configuration of the crane-shovel shall provide maximum accessibility for removal of engine and drive train components. Where incorrect installation of an item could cause malfunction of that item or of the system in which it is installed, an asymmetrical mounting means shall be provided or the mating parts shall be doweled or matchmarked. Ease of maintenance provisions shall incorporate features ensuring operating tool and lubricant component clearances, and shall facilitate maintenance and servicing operations at extremely low temperatures. Intricate locking devices and controls with threaded fastenings which can be overtorqued by personnel lacking feel becaue of thick gloves or because of numb hands shall be avoided. Each operation shown on the following maintenance operations list shall be accomplished by one man using common tools and special tools furnished on the crane-shovel within the time limit specified when tested as specified in 4.5.2.27. After completion of the maintenance operations, the crane-shovel shall operate as specified herein.

Maintenance operations list:	Time allowed
a. Remove, replace, and adjust fan and accessory drive belts.	30 minutes
b. Remove and replace alternator and readjust belt tension.	30 minutes
c. Remove and replace electric starter.	30 minutes
d. Remove, clean, and replace all screens, strainers, and filter elements in the hydraulic systems.	30 minutes
e. Drain engine cooling system, remove and replace engine cooling system hoses, and refill cooling system.	2 hours
f. Drain, change filters, and refill engine oil system.	1 hour

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g.	Remove and replace fuel filter elements.	30 minutes
h.	Disconnect battery cables, remove and replace batteries, and reconnect battery cable.	30 minutes
i.	Remove and replace engine. (A hoist and operator may be used for this operation.)	4 hours (each operation)

3.17 Maintenance ratio. The ratio of manhours of maintenance required to the hours of operation performed shall not exceed 0.25 when tested as specified in 4.5.2.28.

3.18 Filling, draining, and checking provisions. The hydraulic system reservoir and transmission housings shall be equipped with dipsticks to determine the fluid level. Other enclosures such as gear cases and reservoirs which contain lubricant or hydraulic oil shall be equipped with check plugs of not less than 0.50-inch pipe size. Dipsticks shall be etched, knurled, sandblasted, or phosphate coated and marked in accordance with SAE J614. Each enclosure shall be equipped with means for filling and draining. All fill ports shall have a label indicating the type and quantity of fluid for the reservoir the port accesses. Said label shall be in accordance with 3.28.2. All drains shall be labeled "(item) DRAIN" (i.e. TRANSMISSION DRAIN) and the labels shall be in accordance with 3.28.2. The drain shall be fitted with a magnetic drain plug having either tapered pipe threads or shoulder and gasket. In lieu of the magnetic drain plug, a separate magnetic device shall be fastened inside the enclosure at or near the bottom and shall be removable for cleaning without removal of other components (except access covers). Each drain plug or valve when removed shall result in complete drainage of the fluid from the enclosure (except torque converter). When the crane-shovel is in a level position, drainage of fluids shall be into a suitable container without draining on or over any part of the crane-shovel. Integral tubes, hose, or troughs may be used to convey the fluids from the drain outlet. Accessibility to the filling means, the fluid-level checking device, and the drain plug or valve shall be provided without removal or adjustment of accessories and parts, except plates equipped with hand- operable, quick-disconnect fasteners, or capscrews in applications where quick-disconnect fasteners are subject to damage. The drain outlets for the engine-cooling system shall provide complete drainage of the system. The radiator drain shall be such that coolant will not drain onto any part of the crane-shovel. Extension hoses are permitted.

3.19 Transportability. The crane-shovel shall be transportable worldwide by the highway, rail, water, and air modes without special (non-routine) permits. Transportability criteria are defined in MIL-STD-1366 and MIL-HDBK-157.

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3.19.1 Disassembly and reassembly. Front-end attachments and the counterweight shall be capable of removal from the crane-shovel and reassembled when tested as specified in 4.5.2.29. The crane-shovel shall have a maximum of two hours for disassembly and two hours for reassembly using 2 people. Only hand tools and the special tools as furnished with the crane-shovel are to be used. A crane may be used for component removal and replacement as required. Cutting and welding are not permitted.

3.19.2 Air transportability. All components of the disassembled crane-shovel shall be transportable by C-5 aircraft. Criteria for transport in U.S. Air Force cargo aircraft are defined in MIL-STD-1791 and Air Force Systems Command (AFSC) Handbook DH-1-11.

3.19.3 Rail transportability. All components of the disassembled crane-shovel, when loaded on a 50-inch high flatcar, shall meet the dimensional requirements of the Gabarit International de Chargement (GIC) outline diagram as shown in MTMCTEA PAMPHLET 70-1. All components of the disassembled crane-shovel shall sucessfully complete the modified MIL-STD-810 rail impact test (see appendix B).

3.19.4 Highway transportability. The disassembled crane-shovel and its prime mover combination shall be within the maximum highway permit limits established by the individual States, see appendix C.

3.19.5 Water transportability. All components of the disassembled crane-shovel, when loaded on a ship shall meet the criteria as defined in MIL-STD-1366 and MIL-HDBK-157.

3.20 Performance.

3.20.1 Capacities.

3.20.1.1 Crane capacity. Each crane-shovel, when used as a crane with a 30-foot boom, shall have rated lifting loads of not less than those specified in table V when tested in accordance with 4.5.2.31.

TABLE V. Rated crane-lifting capacities.

Radius of load (ft)	Rated load (lb)
12	25,000
30	6,900

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3.20.1.2 Shovel capacity. The shovel dipper capacity shall be not less than specified in table VI. The working heights, radii, and the digging depth of the shovel shall be not less than specified in table VI with the crane-shovel resting on firm level ground and the shovel boom positioned at a 45-degree angle.

TABLE VI. Shovel operating range.

Shovel dipper capacity (cu yd)	Dumping		Cutting		Digging depth below floor level (ft)(in)	Cleanup radius at floor level (ft)(in)
	Height door open (ft)(in)	Radius max height (ft)(in)	Height (ft)(in)	Radius (ft)(in)		
3/4	14 6	21 0	21 0	26 6	5 6	17 6

3.20.1.3 Backhoe capacity. The backhoe bucket capacity shall be not less than specified in table VII. The ground clearance at start of dump and the digging depth of the backhoe shall be not less than specified in table VII with the crane-shovel resting on firm, level ground.

TABLE VII. Backhoe working range and dipper width.

Minimum dipper capacity cu. yd.	Minimum dipper width in.	Minimum ground clearance start of dump ft. in.	Minimum digging depth ft. in.
3/4	36	9 6	17 0

3.20.1.4 Clamshell and dragline capacity. Equipped with the basic boom, the crane-shovel shall be adaptable to and operable as a 3/4-yard-capacity clamshell and 3/4-yard-capacity dragline.

3.20.1.5 Crane capacity plate. All crane-shovels shall be provided with a crane capacity plate mounted inside the cab in a location conspicuous to the operator. The plate shall be similar to and of the same material as specified in 3.28.2 and shall contain the following information (see 4.5.2.35):

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a. Rated loads (see 6.9.3) in 5-foot increments for the full range of operating radii when using the basic boom length specified in 3.20.1.1, and for each 10-foot increment in boom lengths up to and including 60 feet. This plate shall also include capacity information for a 15-foot jib.

b. Indication that the rated loads are gross hook loads at a radius measured from the center of rotation of the revolving superstructure, and that the weights of any other external auxiliary lifting means are to be included as part of the hook load.

c. Statement that the rated loads are based on the crane standing on a supporting surface that is firm, level, and uniform.

d. Statement of deduction to be made from the rated loads for components of crane attachments, such as jibs.

3.20.2 Operating characteristics. All operational functions for a given front-end attachment shall operate independently and simultaneously for maximum performance. These functions need not include travel.

3.20.2.1 Maneuverability. The crane-shovel shall be capable of traveling in a forward or backward direction, with the revolving superstructure in any position, and capable of being steered either to the right or left with controls from the operator's position. The completely equipped crane-shovel with applicable front-end attachment located either to the front or back of the crane- shovel shall be in accordance with the following mobility requirements:

a. Capability of ascending, descending, and stopping on a 30-percent grade not less than 50 feet in length on smooth, firm, and dry ground.

b. Capability of negotiating a 15-percent side slope on smooth, firm, and dry ground.

c. Capability of travel on smooth, firm, and dry ground at a speed not less than 0.75 miles per hour.

d. Capability of fording and operation on hard-bottomed bodies of water up tc 12 inches without evidence of damage or water leakage, when tested as specified in 4.5.2.37.6.

3.20.2.2 Ground-bearing pressure. The ground-bearing pressure shall be not greater than 11 pounds per square inch (psi).

3.20.2.3 Conditions. All requirements specified herein shall apply to crane-shovels with all lubricating and cooling systems full and all fuel tanks not less than half full, unless otherwise specified.

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3.20.2.4 Backward stability. The center of gravity from the axis of rotation shall be not greater than 70 percent of the radial distance from the axis of rotation to the tipping fulcrum in the least stable direction when the crane- shovel is resting on a firm, level, and uniformly even supporting surface; in working order without load; and with the basic boom at the minimum working radius.

3.21 Human factors engineering. The characteristics of the crane-shovel shall provide for operation by personnel ranging from the small man, clothed, through the large man, arctic clothed, in accordance with SAE J833. Operator's controls shall comply with SAE J898 and SAE J297. The operator enclosure shall be in accordance with SAE J154 and instrument design and location shall be in accordance with SAE J209.

3.22 Noise limits. The noise produced by the crane-shovel shall be in accordance with MIL-STD-1474, category D requirements, with the exception of MIL-STD-1474, 5.2, 5.3, and 5.4, when tested in accordance with 4.5.2.40. The provision of MIL-STD-1474, 4.3 and 4.4 shall apply if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the contracting officer. Hazard signs shall be in accordance with MIL-STD-1474, 4.3, except the sign will state "HEARING PROTECTION REQUIRED WITHIN (specify) FEET." The sign shall be readable at the distance indicated on the sign.

3.23 Safety. All exposed parts which are subject to high operating temperatures shall be insulated, fully enclosed, or guarded. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. All moving parts that are a hazard to operating or maintenance personnel, subject to breakage or entanglement of wire ropes, shall be fully enclosed or guarded. Provisions shall be made to minimize dripping of grease or oil onto the superstructure floor. Protective devices shall not impair the operating functions of the crane-shovel. All guards and protective devices for nonelectrical components shall be made from metal. Parts containing asbestos will not be used in the manufacturing of this vehicle.

3.24 Environmental and use requirements for nonwinterized crane-shovel.

3.24.1 Operating temperature requirements. The crane-shovel shall perform as specified herein in any ambient temperature from -25 to +125 ^{o}F (-32 to +52 ^{o}C).

3.24.2 Storage temperature requirements. The crane-shovel shall not be damaged by storage under the following conditions:

Air temperature
High - 160 ^{o}F (71 ^{o}C) - time period, 1 hour daily.
Low - -50 ^{o}F (-46 ^{o}C) - time period, 6 hours; no solar radiation.

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3.24.3 Elevation conditions. The crane-shovel shall perform as specified herein at any elevation from sea level and a maximum ambient temperature of 125 °F to 5,000 feet above sea level and a temperature of 107 °F (42 °C).

3.24.4 Use. The crane-shovel shall withstand the usage encountered in military operations, such as operation over unimproved terrain, fording, and operation under all weather conditions specified herein without breakage or permanent deformation.

3.24.5 Reliability. The reliability shall be 68 hours when tested as specified in 4.5.2.42.

3.25 Electromagnetic interference. The electromagnetic interference emission of the crane-shovel shall be in accordance with MIL-STD-461, class C1.

3.26 Weights and dimensions. The physical characteristics of the crane-shovel without boom shall be as specified in table VIII.

TABLE VIII. Crane-shovel dimensions.

Overall height, maximum	130 inches
Overall width (with catwalk removed), maximum	124 inches
Overall length, maximum	192 inches
Ground clearance, minimum	9.50 inches

The gross weight of the crane-shovel in operable condition and fully serviced shall be not less than 44,000 pounds and shall not exceed 54,000 pounds.

3.27 Treatment and painting. The portions of the crane-shovel normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage green 383 in accordance with MIL-C-46168. When specificed (see 6.2), the crane shovel shall then be overcoated in accordance with the Government furnished camouflaged patterns, with MIL-C-46168 top coat of the colors specified in the camouflage patterns, except the first article may be camouflage green 383.

3.27.1 Camouflage pattern data. When specified (see 6.2), the contractor shall provide camouflage pattern data.

3.28 Identification marking. The crane-shovel shall be identified in accordance with MIL-STD-130. The marking shall be applied on plates in accordance with MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

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3.28.1 Identification plates. Each crane-shovel and attachment shall be equipped with instruction plates, labels, or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating or servicing the crane-shovel. All plates shall be furnished by the contractor and permanently affixed by screws, bolts, or rivets in a conspicuous, protected location. Plates for the crane-shovel shall be in accordance with MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates for the attachments shall be in accordance with MIL-P-514, type I, style 4, composition C, of type I, grade A, class 1 material, and shall include the following information:

(Attachment nomenclature)
FOR USE WITH
(Name of crane-shovel manufacturer)
(Model number or numbers of crane-shovel)
*(Range of applicable crane-shovel serial numbers)
Weight
(Center of gravity location)

*The range of crane-shovel serial numbers will encompass lowest machine serial number and the highest known machine serial number designed to use the attachment without modification. When an attachment is common to more than one crane-shovel model number, all applicable model numbers shall be listed.

3.28.2 Information plates. Plates for transportation data shall show the silhouette of the crane-shovel in transport position indicating locations and capacity of lifting and tiedown attachments, dimensions, location of center of gravity, length and size of slings required, and the recommended tiedown arrangement for the crane-shovel. The lettering shall be not less than 0.25 inch in height. The contractor shall install the electrical power warning decals furnished by the contracting officer (see 6.13). Plates indicating contents and capacity of fuel and oil reservoirs shall have letters not less than 0.50 inch in height and shall be attached in a visible location adjacent to the filler openings. Subassembly plates shall include the name of the part with pertinent description of lifting and tiedown provisions, weight and center of gravity of the front-end attachments and counterweight.

a. Crane-shovel boom.

b. Cab.

c. Counterweight.

d. Crawler base.

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e. Revolving superstructure.

f. Swing circle or roller path.

3.29 Stenciling and vehicle weight classification.

3.29.1 Stenciling. Lifting and tiedown attachments shall be identified with 1-inch letters indicating "sling point" or "tiedown". Unless otherwise specified herein, stenciling shall be in accordance with MIL-T-704. Registration numbers shall be marked on the crane-shovel in accordance with MIL-STD-642. The color shall be white.

3.29.2 Crane-shovel weight classification. Unless otherwise specified (see 6.2), the contractor shall complete the Government-furnished vehicle weight classification forms and apply the vehicle weight classification number to the crane-shovel as specified (see 6.10).

3.30 Operating accessories. Two 20-foot-long chains in accordance with RR-C-271, type I, class 2, of 0.75-inch stock shall be furnished for each crane-shovel. Each chain shall be provided with a 6-inch-inside-diameter ring on one end and a grab hook on the other end. Provision shall be made for carrying the chain on the crane-shovel.

3.31 Items to be furnished with crane-shovel. No shipment of crane-shovels shall be made unless repair parts, operating tools, special maintenance tools, and accessories are included with the shipment or unless approval for shipment of the crane-shovels without such items has been received from the contracting officer.

3.32 Government-furnished property. When specified (see 6.2), the following property will be furnished by the Government (see 6.7):

Item No.	Description	Identification	Quantity
1.	Pattern drawings	Camouflage patterns	As required

3.33 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.8):

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Item no.	Description	Identification (FSN)	Quantity
1.	Lead section, top, pile driver: 15-ft. long.	3815-00-983-8029	1
2.	Lead section, lower, pile driver: 10-ft. long.	3815-00-221-2215	4
3.	Boom, crane, 30-ft: complete boom, both sections.	3815-00-541-9839	1
4.	Boom extension, middle, crane: 10-ft. extension.	3815-00-540-4756	3
5.	Boom jib, crane: 15-ft. jib.	3815-00-542-4839	1
6.	Adapters, piledriver lead.	3815-00-554-0083	1 set
7.	Catwalk, piledriver: telescoping 3 sect; 8 to 23 ft.	3815-00-190-3308	1
8.	Fairlead, roller and sheave: for 12-1/2-ton crane-shovel, crawler-mounted and 20-ton crane-shovel, truck-mounted.	3815-00-554-0126	1
9.	Bucket, dragline: 3/4 cu. yd.	3815-00-186-3571	1
10.	Bucket, clamshell: 3/4 cu. yd.	3815-00-186-3556	1
11.	Tagline, crane and crane-shovel: 3/4 to 1-cu. yd. bucket	3815-00-249-8387	1

3.34 Servicing and adjustment. The contractor shall service and adjust all crane-shovels for operational use including at least the following: Adjustment of lights; adjustment of the clutch and brake systems and mechanical components of the crane-shovel engine, transmission and drive train; servicing of the cooling system with a 50-50 solution, by volume, of ethylene-glycol-type anti-freeze in accordance with MIL-A-46153, and water; and cleaning, bleeding, and filling of the hydraulic system.

3.35 Workmanship. All parts, components, and assemblies of the crane-shovel including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be cleaned and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful material. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required.

3.35.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the metal, may be used instead of shearing and sawing. All bends shall be made by controlled means to ensure uniformity of size and shape. Precautions shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces

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shall be free from burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.35.2 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Flat washers or lockwashers shall be provided where necessary. Matching threaded areas securing bolts shall be in accordance with SAE J429, or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All bolts shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

3.35.3 Riveted connections. Rivet holes shall be accurately formed and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full, concentric with the rivet holes, in full contact with the surface of the members, and shall be in accordance with SAE J492.

3.35.4 Welder certification. Prior to welding all welders performing work on this contract must be qualified, tested, and certified to one of the following codes:

a. Structural Welding Code, AWS D1.1 Welding Qualifications, Section 5 and 6.

b. Welding Qualifications of the ASME, Section IX.

c. Specification for welding Earthmoving and Construction Equipment, ANSI/AWS D14.3 Qualification, Section 5.

The Government may require the recertification of welders, at its discretion, if a problem with weld quality or certification data exists. Records of welder certifications or the test results shall be available for Government review and maintained for a period of four years after contract completion.

3.35.5 Welding procedures. All welding shall be performed in accordance with written procedures as defined by one of the following codes:

a. Structural Welding Code, AWS D1.1

b. ASME Boilers and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications.

c. Specification for Welding Earthmoving and Construction Equipment ANSI/AWS D14.3.

All welds shall be inspected and tested in accordance with the criteria for workmanship and welding quality as stated in one of the above codes. Copies of written welding procedures shall be available at the work station for operator reference and inspection.

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3.35.6 Mechanical properties. Mechanical properties of all welds shall be such that welds will transmit stress without failure or permanent deformation when subjected to any test specified in this specification.

3.35.7 Machine work. Tolerance and gages for metal fits shall be in accordance with the limitations specified herein.

3.35.8 Castings and forgings. All parts, components, and assemblies of the crane-shovel which include castings and forgings shall be clean of harmful material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.35.9 Preparation of liquid and air systems. The hydraulic and water systems shall be purged to remove any foreign material in the lines or components after the assembly of the crane-shovel is complete.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of te inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (4.4).

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c. Inspection comparison (see 4.6).

d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination.

4.3.1.1 Pretest examination. The first article crane-shovel shall be subjected to the examination marked "X" under column 4 of table IX. When winterized crane-shovels are furnished, the winterized model shall also be subjected to the examinations marked "X" under column 4, of table X. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Post-test examination. Upon successful completion of all tests specified in 4.3.2, the crane-shovel shall be subjected to the examinations marked "X" in column 5 of table IX and 4.5.2.44. When winterized crane-shovels are furnished, the winterized model shall also be subjected to the examinations marked "X" under column 5 of table X. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. One first article crane-shovel shall be subjected to test marked "X" in column 1 of table XI, with the exception of the reliability test. The other two first article crane-shovels shall undergo the reliability test. When winterized crane-shovels are specified (see 3.10), in addition to nonwinterized crane-shovels, two first article models shall be furnished for examination and test. One of these first article models shall be nonwinterized and one winterized. The winterized first article crane-shovel shall be subjected to the tests marked in column 1 of table XI and to the test in table XII. Failure of any tests shall be cause for rejection. All first article test shall be run using MIL-T-83133, grade JP-8 fuel.

4.3.3 Inspection failure. Failure of a first article crane-shovel to meet any requirement specified herein, as a result of the examination and tests specified in 4.3.1 and 4.3.2, shall be cause for rejection. The Government has the right to refuse acceptance of crane-shovels until evidence has been provided by the contractor that corrective action has been taken. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government. Any deficiencies found as a result of the first article inspection will be considered evidence that all crane-shovels accepted prior to the completion of first article inspection are deficient.

4.4 Quality conformance inspection.

4.4.1 Examination. Each crane-shovel shall be examined for the defects marked "X" in column 6 of tables IX and X. Presence of one or more defects shall be cause for rejection.

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4.4.2 Tests. Each crane-shovel shall be subjected to the tests marked "X" in column 2 of table XI. Failure of any test shall be cause for rejection. The test shall be run on either VV-F-800, grade DF-2 or MIL-T-83133, grade JP-8 fuel.

4.5 Inspection procedure. Prior to examination or tests, the crane-shovel shall be completely lubricated with military oils and military greases as specified in 3.11. Oils shall be those designated for use in the ambient temperatures at which the tests are being performed.

4.5.1 Examination. Examination shall be in accordance with table IX. In addition to the examinations in table IX, the winterized crane-shovel shall be examined in accordance with table X.

TABLE IX. Examination schedule.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
101.	Any component missing, damaged, or not as specified.	3.1	X	X	-
102.	Compatibility not as specified.	3.1.1	X	X	-
103.	Material not as specified.	3.3	X	-	X

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TABLE IX. Examination schedule - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
104.	Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environments as specified.	3.3.1	X	-	X
105.	Dissimilar metals as specified in MIL-STD-889 are not effectively insulated from each other as specified.	3.3.1.1	X	-	X
106.	Contractor does not have documentation available for identification of material, material finishes, or treatments.	3.3.1.2	X	-	X

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TABLE IX. Examination schedule - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
107.	Used, rebuilt or remanufactured components, pieces, or parts incorporated in the crane-shovel.	3.3.2	X	-	X
108.	Interchangeability of parts not as specified.	3.3.3	X	-	X
109.	Threaded parts not as specified.	3.3.9	X	-	-
110.	Maximum unit stress exceeding limits as specified.	3.4	X	-	-
111.	Superstructure not as specified.	3.5	X	-	-
112.	Platform, catwalks, footholds or handholds not as specified.	3.5	X	-	-
113.	Piledriver lead catwalk clips not as specified.	3.5	X	-	-

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TABLE IX. Examination schedule - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
114.	Counterweight not as specified.	3.5	X	-	-
115.	Functional requirements not as specified.	3.5.1	X	-	-
116.	Forward deck machinery protection not as specified.	3.5.2	X	-	-
117.	Cab not as specified.	3.5.3	X	-	-
118.	Cab heating not as specified.	3.5.3.1	X	-	-
119.	Superstructure cab disassembly not as specified.	3.5.4	X	X	-
120.	A-frame not as specified.	3.5.5	X	X	-
121.	Power load-lowering device not as specified.	3.5.6	X	-	-
122.	Crane-boom hoist not as specified.	3.5.7	X	-	-

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TABLE IX. Examination schedule - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
123.	Drums not as specified.	3.5.8	X	-	-
124.	Hoist drum brakes not as specified.	3.5.8.1	X	-	-
125.	Hydraulic components not as specified.	3.5.8.2	X	-	-
126.	Drum clutches not as specified.	3.5.8.3	X	X	-
127.	Rotating frame not as specified.	3.5.9	X	X	-
128.	Swing-gear, lock, or brake mechanism not as specified.	3.5.10	X	X	-
129.	Power transmission clutch not as specified.	3.5.11.1	X	-	-
130.	Torque converter not as specified.	3.5.11.2	X	-	-

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TABLE IX. Examination schedule - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
131.	Power transmission chains not as specified.	3.5.11.3	X	X	-
132.	Electrical system not as specified or does not function.	3.5.12	X	X	X
133.	Fungus- and moisture-resistance not as specified.	3.5.12.1	X	X	X
134.	Wiring not as specified.	3.5.12.2	X	X	-
135.	Floodlights not as specified.	3.5.12.3 thru 3.5.12.3.1.2	X	X	-
136.	Interior cab lights not as specified.	3.5.12.3.2-	X	X	-
137.	Warning horn not as specified.	3.5.12.4	X	X	-
138.	Trouble light not as specified.	3.5.12.5	X	X	-

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TABLE IX. Examination schedule - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
139.	Instrument panel not as specified.	3.5.12.6	X	X	-
140.	Operator controls not as specified.	3.5.13	X	X	-
141.	Crane controls not as specified.	3.5.13.1	X	X	-
142.	Engine hand controls not as specified.	3.5.13.2	X	X	-
143.	Accumulator not as specified.	3.5.13.3	X	X	-
144.	Engine not as specified.	3.6	X	X	-
145.	Power and speed rating not as specified.	3.6.1	X	X	-
146.	Lubrication system not as specified.	3.6.3	X	X	-
147.	Fuel system not as specified.	3.6.4	X	X	-
148.	Cold starting aids not as specified.	3.6.5	X	X	-

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TABLE IX. Examination schedule - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
149.	Coolant system not as specified.	3.6.6	X	X	-
150.	Air induction system not as specified.	3.6.7	X	X	-
151.	Engine exhaust system not as specified.	3.6.8	X	X	-
152.	Governor not as specified.	3.6.9	X	-	-
153.	Starting system not as specified.	3.6.10	X	-	-
154.	Charging system not as specified.	3.6.11	X	-	-
155.	Batteries not as specified.	3.6.12	X	-	-
156.	Battery container not as specified.	3.6.12.1	X	-	-
157.	Emergency starting charging receptacle or cable not as specified.	3.6.13	X	X	-

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TABLE IX. Examination schedule - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
158.	Contamination level not as specified.	3.7	X	X	-
159.	Crawler base not as specified.	3.8, 3.8.1	X	-	-
160.	Track assembly not as specified.	3.8.2	X	-	-
161.	Side frames not as specified.	3.8.3	X	-	-
162.	Crawler drive not as specified.	3.8.4	X	-	-
163.	Axle not as specified.	3.8.5	X	-	-
164.	Front-end attachment not as specified and components missing.	3.9, 3.9.1	X	X	X
165.	Crane boom not as specified.	3.9.1.1	X	X	-
166.	Pendant equalizer assembly not as specified.	3.9.1.2 thru 3.9.1.2.3	X	X	-

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TABLE IX. Examination schedule - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
167.	Boom safety stop not as specified.	3.9.1.3	X	X	-
168.	Boom angle indicator, chart, or overload indicating system not as specified.	3.9.1.4, 3.9.1.4.1	X	-	X
169.	Crane hook block not as specified.	3.9.1.5	X	X	-
170.	Rope not as specified.	3.9.1.6	X	X	-
171.	Rope sockets not as specified.	3.9.1.6.1	X	-	X
172.	Rope length or rope length chart not as specified.	3.9.1.6.2	X	-	X
173.	Secondary front-end attachment not as specified or components missing.	3.9.2	X	-	X
174.	Boom extension not as specified.	3.9.2.1	X	-	-

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TABLE IX. Examination schedule - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
175.	Boom jib not as specified.	3.9.2.2	X	-	-
176.	Clamshell bucket not as specified.	3.9.2.3	X	-	-
177.	Tagline not as specified.	3.9.2.4	X	-	-
178.	Dragline bucket not as specified.	3.9.2.5	X	-	-
179.	Dragline fairlead not as specified.	3.9.2.6	X	-	-
180.	Wrecking ball not as specified.	3.9.2.7	X	-	-
181.	Piledriver leads, boom adapter, and catwalk not as specified.	3.9.2.8	X	X	-
182.	Piledriver hammer not as specified.	3.9.2.9	X	-	-
183.	Shovel front not as specified and components missing.	3.9.2.10 thru 3.9.2.10.5	X	X	X

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TABLE IX. Examination schedule - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
184.	Backhoe not as specified and components missing.	3.9.2.11 thru 3.9.2.11.2	X	X	X
185.	Lubrication, lubricants, components, and fuels not as specified.	3.11 thru 3.11.5.2, 3.12	X	X	X
186.	Lifting and tiedown attachments not as specified.	3.13	X	X	-
187.	Antirotation devices not as specified.	3.14	-	X	X
188.	Toolbox not as specified.	3.15	X	X	-
189.	Maintainability not as specified.	3.16, 3.17	X	X	-
190.	Filling, draining, and checking provisions not as specified.	3.18	X	X	X
191.	Transportability not as specified.	3.19	X	-	-

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TABLE IX. Examination schedule - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
192.	Air transportability not as specified.	3.19.2	X	-	-
193.	Rail transportability not as specified.	3.19.3	X	-	-
194.	Highway transportability not as specified.	3.19.4	X	-	-
195.	Water transportability not as specified.	3.19.5	X	-	-
196.	Crane capacity not as specified.	3.20.1.1	X	-	-
197.	Shovel operating range not as specified.	3.20.1.2	X	-	-
198.	Backhoe working range not as specified.	3.20.1.3	X	-	-

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TABLE IX. Examination schedule - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
199.	Clamshell and dragline not compatible as specified.	3.20.1.4	X	-	-
101A.	Operating characteristics not as specified.	3.20.2	X	X	X
102A.	Maneuverability not as specified.	3.20.2.1	X	-	X
103A.	Ground pressure, conditions, and backward stability not as specified.	3.20.2.2, 3.20.2.3, 3.20.2.4	X	-	-
104A.	Human factors engineering not as specified.	3.21	X	X	-
105A.	Noise limits not as specified.	3.22	X	X	X
106A.	Safety characteristics not as specified.	3.23	X	X	-

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TABLE IX. Examination schedule - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
107A.	Environmental and use capabilities not as as specified.	3.24 thru 3.24.5	X	X	-
108A.	Electromagnetic interference not as specified.	3.25	X	X	X
109A.	Weights and dimensions not as specified.	3.26	X	-	-
110A.	Treatment, painting, paint color, and camouflage pattern not as specified.	3.27, 3.27.1	X	X	X
111A.	Operating accessories not as specified.	3.30	X	X	X
112A.	Items to be furnished with crane-shovel not as specified.	3.31	X	-	X

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TABLE IX. Examination schedule - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
113A.	Servicing and adjustments not as specified.	3.34	X	-	X
114A.	Workmanship not as specified.	3.35 thru 3.35.9	X	X	X
Minor					
201.	Crane-capacity plate not as specified.	3.20.1.5	X	X	X
202.	Identification marking and plates, and information plates not as specified.	3.28, 3.28.1, 3.28.2	X	X	X
203.	Stenciling not as specified.	3.29, 3.29.1	X	X	X
204.	Weight classification forms and numbers not as supplied.	3.29.2	X	X	X

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TABLE X. Examination schedule (winterized model).

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
115A.	Winterized components not in protected locations and not protected.	3.10	X	-	-
116A.	Winterized components not as specified.	3.10	X	-	-
117A.	Defrosters not as specified.	3.10.1, 3.10.4	X	-	-
118A.	Thermostatic controls not furnished and do not operate as specified.	3.10.2, 3.10.3, 3.10.4, 3.10.5.2	X	-	-
119A.	Engine heater not as specified.	3.10.2	X	-	-
120A.	Coolant heater not as specified.	3.10.3	X	-	-
121A.	Positioning of winterized components not as specified.	3.10.4	X	-	-

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TABLE X. Examination schedule (winterized model) - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
122A.	Cab heating system not as specified.	3.10.4	X	-	-
123A.	Heater mounting and enclosure not as specified.	3.10.4	X	-	-
124A.	Battery winterization not as specified.	3.10.5	X	-	-
125A.	Battery boxes not as specified and not of corrosion-resistant steel. Battery boxes not provided with drainage.	3.10.5.1	X	-	-
126A.	Thermal damage (caused by heating media) to batteries.	3.10.5.2	-	X	-
127A.	Battery heater not as specified.	3.10.5.2	X	-	-

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TABLE X. Examination schedule (winterized model) - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
128A.	Boots and grommets not as specified.	3.10.6	X	-	-
129A.	Window wiper not as specified.	3.10.6.1	X	-	-
130A.	Insulation in both cabs and engine housings missing. Thickness, including insulation under the floormats not as specified.	3.10.6.2, 3.10.6.3, 3.10.6.4	X	-	X
131A.	Cab not weatherstripped as specified.	3.10.6.3	X	-	-
132A.	Floormat not as specified.	3.10.6.4	X	-	-
133A.	Winterfront as specified.	3.10.7	X	-	-
134A.	Wiring system and components not as specified.	3.10.8	X	-	X

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TABLE X. Examination schedule (winterized model) - Continued.

Number	Defect	Requirement paragraph	First article		Quality conformance
			Pre-test	Post test	
1	2	3	4	5	6
Major					
135A.	Air cleaner does not function.	3.10.9	X	-	X
136A.	Cold starting aids not as specified.	3.10.10	X	-	X
137A.	Hose and hose assemblies not as specified.	3.10.11	X	-	X
138A.	Fuels, lubricants, hydr fluids and antifreeze not as specified.	3.10.12	X	-	-
139A.	Latches, handles, and access doors not operable with wearing artic mittens.	3.10.13	X	-	-

4.5.2 Tests. Tests shall be in accordance with table XI for non-winterized models. In addition, winterized models shall also be subjected to tests specified in table XII.

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TABLE XI. Test schedule.

First article	Qual. conf.	Test	Test paragraph	Requirement paragraph
1	2	3	4	5
X	X	Test conditions	4.5.2.1	3.20.2.3
X	-	Engine	4.5.2.2	3.6
X	-	Maximum Working Stress	4.5.2.3	3.4
X	-	Counterweight removal	4.5.2.4	3.5
X	-	Rain	4.5.2.5	3.5
-	X	Brake and clutch	4.5.2.6	3.5
X	X	Swing	4.5.2.7, 4.5.2.7.1	3.5
-	X	Hoisting	4.5.2.8	3.5
-	X	Boom hoisting	4.5.2.9	3.5
X	-	Power and speed test	4.5.2.10	3.5.1, 3.24.3
X	-	Boom hoist cable angle	4.5.2.11	3.5.5
X	-	Power lowering	4.5.2.12	3.5.6
X	-	Clutch capacity	4.5.2.13	3.5.6
X	-	Backing down	4.5.2.13.1	3.5.6
X	-	Clutch control	4.5.2.13.2	3.5.8.3, 3.5.13.3
X	-	Swing lock and swing brake	4.5.2.14	3.5.10
X	-	Torque converter stall (when finshed)	4.5.2.15	3.5.11.2
X	X	Lights and horn	4.5.2.16.1	3.5.12.3, 3.5.12.4
X	-	Accumulator	4.5.2.17	3.5.13.3
X	-	Cooling	4.5.2.18	3.6.6
X	-	Emergency starting	4.5.2.18.1	3.6.13
X	-	Boom stop and pile lead	4.5.2.19	3.9.1.3
-	X	Boom stop	4.5.2.20	3.9.1.3
X	-	Overload indicating system	4.5.2.21	3.9.1.4.1
-	X	Overload system	4.5.2.22	3.9.1.4.1
X	-	Rope strength	4.5.2.23	3.9.1.6
X	-	Lubrication	4.5.2.24	3.11, 3.17
X	-	Maintenance package	4.5.2.25	3.11, 3.17

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TABLE XI. Test schedule - Continued.

First article	Qual. conf.	Test	Test paragraph	Requirement paragraph
1	2	3	4	5
X	-	Lifting and tie-down provision strength	4.5.2.26	3.13
X	-	Ease of maintenace	4.5.2.27	3.16
X	-	Maintenance ratio	4.5.2.28	3.17
X	-	Disassembly and re-assembly for trans-port	4.5.2.29.1	3.19.1
X	-	Air loading	4.5.2.29.2	3.19.2
X	-	Rail impact	4.5.2.29.3	3.19.3
X	-	Craning	4.5.2.30	3.20.1
X	-	Lifting capacity	4.5.2.31	3.20.1.1
X	-	Shovel front	4.5.2.32	3.20.1.2
X	-	Backhoe	4.5.2.33	3.20.1.3
X	-	Clamshell and dragline	4.5.2.34	3.20.1.4
X	-	Crane capacity plate determination	4.5.2.35	3.20.1.5
-	X	Off road	4.5.2.36	3.20.2
X	-	Maneuverability and use	4.5.2.37	3.20.2.1, 3.24.4
X	-	Grade	4.5.2.37.1	3.20.2.1a.
X	-	Slope	4.5.2.37.2	3.20.2.1b.
X	-	Brake	4.5.2.37.3	3.20.2.1a.
X	-	Steering	4.5.2.37.4	3.20.2.1
X	-	Propelling test	4.5.2.37.5	3.20.2.1c.
X	-	Fording	4.5.2.37.6	3.20.2.1d.
X	-	Ground-bearing pressure	4.5.2.38	3.20.2.2
X	-	Backward stability	4.5.2.39	3.20.2.4
X	X	Noise level	4.5.2.40	3.22
X	-	High temperature operation (non-winterized unit only)	4.5.2.41 through 4.5.2.41.1.1.1	3.24.1
X	-	Low temperature operation	4.5.2.41.1.2	3.24.1
X	-	High temperature storage	4.5.2.41.2, 4.5.2.41.2.1	3.24.2, 3.24.3

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TABLE XI. Test schedule - Continued.

First article	Qual. conf.	Test	Test paragraph	Requirement paragraph
1	2	3	4	5
X	-	Low temperature storage	4.5.2.41.2.2	3.24.2
X	-	Reliability	4.5.2.42	3.24.5
X	-	Electromagnetic interference	4.5.2.43	3.25
X	-	Post test disassembly	4.5.2.44	

TABLE XII. Test schedule (winterized model).

First article	Test	Test paragraph	Requirement paragraph
1	2	3	4
1	High temperature operation	4.5.2.41.4, 4.5.2.41.4.1	3.10
2	Cold starting and running	4.5.2.41.4.2	3.10

4.5.2.1 Test conditions. Prior to test, the crane-shovel shall be lubricated with oils and greases specified in 3.11.5. Oils shall be those designated for use in the ambient temperature at the place of test. Unless otherwise specified in a test, tests shall be performed without shelter and in existing environmental conditions. The crane-shovel shall operate as specified herein in accordance with the maintenance schedule prepared and submitted by the contractor.

a. The front-end equipment required for the tests specified herein but not included as Government-loaned equipment (see 3.33) shall be furnished by the contractor at no cost to the Government. The equipment furnished by the contractor for the test shall be in accordance with the applicable military specification for the equipment.

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b. The test area shall be examined to determine that:

(1) There are no obstructions which may cause interference to raising the boom to its maximum height in any position or rotating the boom horizontally through 360 degrees.

(2) Grade does not exceed 1 percent in any direction.

(3) That the test is to be performed on a hard surface.

c. The crane-shovel shall be checked prior to and during the tests to determine the following:

(1) Ensure that coolant and lubrication systems are full.

(2) Check that fuel tanks are maintained between full and half-full level throughout these tests. Check fuel gages for accuracy at empty, one-half full, and full levels.

(3) Adjust clutches and brakes in accordance with the manufacturer's instruction manual. Inspection shall be made to ensure there is no dragging of the clutches or brakes with the controls in the disengaged position.

(4) Examine all cable anchors, sheaves, revolving base rollers, drive chains, rotating gears, rotating drums, and ropes to ensure there is no rubbing or interference. This examination shall be made at all operating positions of the front-end attachments and shall include the boom raised to maximum height.

(5) All lights shall be turned on during all operational tests.

d. The component tests 4.5.2.2 and 4.5.2.17 shall be conducted prior to the first article tests of the complete crane-shovel.

4.5.2.2 Engine. Prior to installation of the engine in the first article, the engines shall be tested in accordance with MIL-STD-1400, test method 2000. Nonconformance to 3.6 shall constitute failure of the test. Upon successful completion of this test and prior to installation of the engine in the first article, the fuel injection pumps, governors, and turbocharger, if furnished, shall be sealed by a Government representative at the settings established by this test. Seals for each engine shall also be placed by a Government representative to prevent separation of the crankcase from the engine block and the rocker-arm cover from the engine head. Any changes made to these settings or removal or alteration of the seals by other than the Government inspector or without his expressed consent will invalidate all tests previously performed on these engines.

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4.5.2.3 Maximum working stress. The maximum working stress test shall be conducted as follows:

a. Preparation of crane.

(1) One-half full to full fuel tanks.

(2) Full coolant and lubricant.

(3) Installation of crane boom, shovel front, and backhoe.

(4) All exterior surfaces cleaned of grease, oil, and dirt.

b. Installation of strain gages. Strain gages shall be installed in accordance with the American Society for Non-Destructive Testing Handbook, vol. II, section 54, Resistance Strain Gage Testing. Where directions of principal stress are questionable, two or more gages or strain-gage rosettes may be employed. During test operation or visual examination, any points which show visual evidence of weakness due to design or paint checking should be gage tested in addition to those points specified in 4.5.2.3.d. Particular note shall be made of residual stresses in the structural members due to dead loading. These stresses shall be relieved prior to zeroing the strain-recording instrument (for example, resting the crane boom on a support with cable slacked).

c. Instrumentation operation. Operation of strain-gage recording or indicating apparatus shall be in accordance with the instrument manufacturer's recommended procedure. Standard commercial instruments specifically designed for resistance strain gages shall be considered acceptable provided the manufacturer guarantees an accuracy to within ±10 microinches at a scale reading of 600 microinches.

d. Location of strain gages. Strain gages shall be located approximately as follows:

Superstructure:

S-1	*Front hook roller bracket.
S-2	*Rear hook roller bracket.
S-3	Hook roller path - top side, outer edge, radially.
S-4	*Hook roller path - top side, outer edge, tangential.
S-5	Boom foot brackets.
S-6	A-frame compression members - top midsection.
S-7	A-frame compression members - side midsection.
S-8	A-frame tension members - rear midsection.
S-9	A-frame tension members - side midsection.

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S-10	A-frame spreaders - top midsection.
S-11	A-frame spreaders - front midsection.
S-12	A-frame head shaft - top midsection.
S-13	A-frame head shaft - front midsection.
S-14	Other critical points on A-frame.
S-15	*Drum mountings - at points of critical stress.
S-16	*Rotating base - at points of critical stress.
S-17	*Pendant equalizer - at points of critical stress.

Note any cutaway sections or direction changes in structural members which may be points of stress concentration.

Shovel front:

SF-1	Boom head shaft - top side.
SF-2	Boom head shaft - front side.
SF-3	*Dipper stick - top side to rear of bucket.
SF-4	Other critical points of shovel-front attachment.

Backhoe:

H-1	Sub A-frame - midsection main compression members.
H-2	Sub A-frame - midsection spreaders.
H-3	Backhoe boom - midsection bail sheave shaft.
H-4	Other critical points of backhoe attachments.

*Recommended rosette gages.

e. Procedure. The wire rope required for making the gantry test shall be supplied by the contractor. Strain test for each strain gage shall be repeated for at least three operating test cycles with applicable attachment. Superstructure tests shall be conducted in conjunction with crane boom tests and with other applicable attachments as specified herein only if the boom cable arrangement for these attachments is different than that required for the crane boom. In addition to the strain-gage tests, visual examination shall be made immediately after each test cycle to note any indication of yield or failure (this shall be construed to be any form of permanent deformation to include bending, weld failures, cracks, and scoring of bearings, pins, or bushings). Where rotation of drums, sheaves, or gearing becomes eccentric due to test loading, shafts shall be removed and checked with a straightedge for permanent deformation. Any zero shift in the instrument during a test cycle, where failure of the instrument to return to zero after loading occurs, may be an indication of yield along lines of principal stress other than the longitudinal axis of the gage.

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(1) Pendant equalizer. Strain tests of the pendant equalizer shall be conducted while hoisting boom from ground, while hoisting, swinging, and lowering capacity loads (see 6.9.4) at minimum and maximum radii over the side or rear, whichever is the most stable. The strain tests shall be conducted while booming out and in, with capacity loads at minimum rated radius to the tipping point, and boom out and in a load that will induce tipping at the maximum rated radius. In addition, strain tests shall be conducted for those components not receiving their maximum stress in the most stable direction while hoisting, swinging, and lowering capacity loads at minimum and maximum radii over the side or rear, whichever is the least stable direction. Maximum strains shall be recorded during each testing cycle on the strain-gage locations specified in 4.5.2.3.d. Swinging of capacity load over the side shall be through a cycle of 180 degrees and over the rear through a cycle of approximately 10 degrees on either side of the longitudinal axis. A 60-foot boom shall be lifted from ground level up to 45 degrees. Strain-gage readings shall be taken on all superstructure and pendant equalizer components.

(2) Shovel-front attachment. Strain tests of the shovel-front attachment shall be conducted with the shovel boom at an angle of 45 degrees to the horizontal, the dipper stick at an angle of 45 degrees to the horizontal, and the bucket teeth hooked beneath a stationary object (simulating quarry excavation). Hoist the shovel to engine stall or incipient vehicle tip, whichever occurs first, and record maximum induced strains. This operation shall be conducted operating over the side and over the rear.

(3) Backhoe attachment. Strain tests of the backhoe attachment shall be conducted excavating in heavy undisturbed clay to maximum rated digging depth of the machine. Bucket retrieving shall be limited by engine stall or incipient vehicle tip, whichever occurs first.

Note: Strain gaging is not required on front-end attachments which are provided for testing only, either as Government-loaned property or furnished by the contractor at no cost to the Government. Strain gage testing is not required on crane booms that comply with MIL-B-52203.

f. Records. In order to provide uniform records, data shall be maximum stress recorded during static tests and maximum stress obtained from strip recorder during dynamic tests. Data shall be recorded as follows:

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MAXIMUM WORKING STRESS

Contract Order No. _____ Place _____

Front-End Attachment _____ Date _____

Recording Instruments _____ Recorder _____

Gage Type _____

Gage Factor _____

Gage Resistance _____

Gage No.	Component	Gage	Location	Material 1/	Operation	Recorded Strain Microinches									Maximum 2/ psi 3/ (Max)
						Test 1			Test 2			Test 3			
						(a) Zero Reading	(b) Test Reading	(b-a) Strain	(a) Zero Reading	(b) Test Reading	(b-a) Strain	(a) Zero Reading	(b) Test Reading	(b-a) Strain	

1/ Modulus of elasticity (E).
2/ Where rosette gages are used, see manufacturer's instruction for exact strain.
3/ Stress = Actual Strain (XE).

g. Qualification. The yield strength and modulus of elasticity of the material of the structural member shall be obtained from recognized

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standards or from certified laboratory tests of the material concerned. Maximum unit stress imposed by the required loading exceeding 60 percent of the yield point for any structural member shall constitute failure of this test.

4.5.2.4 Counterweight removal. Remove the counterweight from the crane-shovel and place it on a truck or trailer. Reverse the operation and replace the counterweight to its original position. This test shall be accomplished by not more than two men using only those lifting provisions of the crane-shovel. Use of equipment or machinery other than as specified above or inability to perform the removal or replacement of the counterweight in not more than 30 minutes shall constitute failure of this test.

4.5.2.5 Rain. Subject the crane-shovel to a simulated rainfall of 4-inches ±1 inch per hour as measured by a U.S. Weather Bureau-type gage. The simulated rainfall shall fall on the crane-shovel at an angle of 45 degrees, ±5 degrees and impinging with a force equivalent to a 20 mile-per-hour wind. The simulated rainfall shall disperse uniformly over the surfaces of the crane-shovel simultaneously for not less than 30 minutes on each surface. The engine shall be started (without removing accumulated water) 26 minutes after beginning the rain test, and idle for the remainder of the test. At the completion of the test (within 30 minutes) the door shall be opened, the inside of the cab shall be examined, and all craning functions shall be performed (including lifting and hoisting a capacity load). Evidence of water penetration such as free water in the operator's area and machinery area, inability of the engine to start and idle, or inability to perform craning functions as specified herein shall constitute failure of this test.

4.5.2.6 Brake and clutch test. Perform all hoisting and swinging functions for a period of 30 minutes with a 6,000-pound load and with the hoist and swing brakes partially set.

4.5.2.7 Swing test. Without test load revolve the superstructure clockwise for 15 minutes; repeat the test counterclockwise.

4.5.2.7.1 Swing. Preliminary to this test, check full-load engine speed to see that it does not exceed governed speed as specified in 3.6.1.

a. With a load equal to 50 percent of maximum line pull at 160 feet per minute and the boom at a 40-degree angle, time with a stopwatch 10 revolutions of the loaded superstructure at full throttle. Repeat twice. Swing speed less than 3.5 rpm shall constitute failure of this test.

b. With the crane superstructure loaded as specified in a above, rotate the superstructure at full throttle for 10 minutes in each direction to determine heat in rollers. Thermocouples or contact pyrometer

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may be used to determine temperature. A temperature of more than 175 °F (79 °C) shall constitute failure of this test.

c. With the attachments, loading, and radius as specified in 3.5.8.3, the crane-shovel shall make not less than three and one-half 90 degree passes per minute with the superstructure for 1.50 hours. Thermocouples or contact pyrometer may be used to measure the temperature in the swing clutches. A maximum temperature of more than 500 °F (260 °C) or fading shall constitute failure of this test.

4.5.2.8 Hoisting test. With a 6,000-pound load at a 20-foot radius on a single line attached to the primary hoist drum: (a) Hoist and lower on the brake for 50 cycles; (b) hoist and power lower for 50 cycles. Repeat these tests with the line attached to the secondary drum.

4.5.2.9 Boom hoisting. Raise a 60-foot boom from ground to an angle of 60 degrees from the horizontal and lower to ground level for 10 cycles. The contractor, at his option, may perform this test with a 30-foot boom and a hook load with a combined total weight not less than the 60-foot boom.

4.5.2.10 Power and speed test. This test shall be conducted at an actual or simulated altitude of 5,000 feet. If the option is taken to simulate altitude conditions for test purposes, such simulation shall be accomplished in the following manner. In the vicinity of the engine, mount a container of sufficient capacity and inlet orifice size to assure a steady air pressure to the engine equal to 24.89 inches of mercury. Connect the outlet of the container to the air-cleaner so air is inducted into the engine in the normal manner through the air-cleaner filter. Determine the intake air pressure with a U-tube manometer tapped into the container at a point where static air pressure is maintained in the container. The intake and exhaust systems shall be complete and unaltered. Exhaust back pressure shall not be reduced for this test.

a. Assemble a test load of not less than 15,625 pounds and boom at minimum radius.

b. By trial lifts, adjust the load to require two parts of hoist line and provide steady hoisting rate near governed engine speed. Single sheave block may be used. Establish a mark on the hoisting rope at a point near the primary hoist drum where the load is 3 feet clear of the ground. Establish a second mark on the rope when the load has been lifted to a point where the blocks are approximately 3 feet apart. Lower the load and measure the lineal feet of rope between the two marks. Test shall be started with not less than three wraps on the hoist line drum.

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c. Using a stopwatch accurate to ±0.2 second in 1 minute, time the interval required for the two marks on the hoisting rope to pass a selected point while the load is being lifted with the engine operating at governed speed. At least three checks shall be made and averaged.

d. Compute the hoist line speed in feet per minute. Multiply this computed line speed by the load weight in pounds divided by the number of parts of hoisting line.

e. Assemble a test load equal to the line pull determined by conditions specified in 3.5.1 (line power in foot-pound per minute divided by the corresponding minimum line speed in feet per minute). Lift this load with a single hoist line and simultaneously rotate the superstructure.

f. With basic length of boom, rest boom tip on ground to insure that three wraps of cable are left on boom hoist drum (see 3.9.1.6.2).

g. Evidence of any of the following shall constitute failure of this test:

(1) Hoisting drum line speed and hoisting drum line power not within limits specified in 3.5.1.

(2) Failure to lift the specified load and simultaneously rotate the superstructure.

(3) Less than three wraps of cable on boom hoist drum with boom tip on ground.

4.5.2.11 Boom hoist cable angle. With the boom at an angle of 82 degrees from the horizontal, measure the fleet angle made by boom hoist cable with all sheaves on the A-frame and pendant equalizer where a divergence is made by the cable across the cable groove. The measurement shall be made with a metal template not less than 5 feet in length. Any fleet angle greater than 6 degrees shall constitute failure of this test.

4.5.2.12 Power lowering. Inspection shall be made of the power-down system for lubrication, ease of operation of controls, safety precautions, and for workmanship.

a. The crane shall be equipped with a boom of appropriate length for the power load lowering tests.

b. Position the crane superstructure in the most stable craning position, and the boom reeved with a load line. The test load assembled and used for this test shall weigh, not including the block and

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cable, the same ratio of the maximum capacity load (see 6.9.4) as the ratio of the parts of load line to the number of lines used in tipping tests (see 4.5.2.31) (if a 4-part load line is standard reeving with a capacity load, a 2-part load line will be used with a .50 capacity load).

c. Position the boom for a load radius of 12-feet ±2 inches.

d. With the hoist drum brake unlocked, hoist the load 20 feet ±3 inches above the ground.

e. Adjust the hoist drum brake to hold a maximum capacity load and apply the hoist drum brake until the brake will hold the load suspended.

f. Prepare the crane for power-down operation on the load line.

g. With brake set and locked for holding the load and the engine set at an idle speed, bring the power-down clutch into full engagement. If this stalls the engine, increase the idle speed until the engine will override the hoist brake.

h. As soon as the load reaches the ground, immediately repeat operations specified in d, f, and g above.

This cycle shall be repeated 15 times if a 4-part line and the maximum capacity load is used, or 30 times if a 2-part line and one-half the maximum capacity load is used.

i. Check the temperatures of all brake linings employed in this operation with a contact pyrometer, or at least four thermocouples imbedded within each lining near the working face and uniformly spaced around the perimeter.

j. Assemble a load equal to one-half maximum capacity load, and repeat tests specified in c through i above.

k. By the use of a tachometer on the engine and a tachometer on the main hoist drum shaft, check the ratio of the engine speed to the main hoist drum shaft.

The following shall constitute failure of this test: failure to override the main hoist drum brake; temperatures of the brake linings exceeding 600 °F (316 °C) or the maximum recommended by the manufacturer of the lining for the application, whichever is less; main hoist drum speed greater than 50 percent or less than 30 percent of the engine speed; inability to accomplish power lowering without pumping the control lever; and evidence of leakage, corrosion, or any malfunction, permanent deformation, or

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breakage. After completion of this test, remove the brake bands and examine the brake linings for evidence of failure due to (1) cracking, (2) internal rupture (indicated by distortion of rivet holes), or (3) wear (any amount beyond crazing of surface not to exceed 0.0625-inch). Evidence of damage or weakness described above shall constitute failure of this test.

4.5.2.13 Clutch capacity. With a 30-foot boom and at a short radius, hoist to a height of 5 feet above the ground a load that will develop a line pull in pounds equal to the maximum line pull developed by the hoist drum. Apply the hoist-drum brake to hold the load. Stop the engine and jam lock a meshing set of gears between the engine and the power-down clutch. Engage the power-down clutch for load lowering, release the hoist-drum brake, and let the load be supported on the power-down clutch which in turn shall be fixed by virtue of the locked gears. As the brake is released, examine the power-down clutch for slippage. Any slipping of the clutch, malfunction, permanent deformation, or breakage shall constitute failure of this test.

4.5.2.13.1 Backing down. Suspend on a single line a load that will develop the single line pull equal to 133 percent of rated load at recommended reeving. Stop the engine, and jam lock a meashing pair of gears, knuckle-in the primary hoist drum clutch, and release the brake holding the load. After the brake is released, inspect the clutch for slippage. Any clutch slippage shall constitue failure of this test.

a. Repeat the backing-down procedure using the secondary hoist drum clutch.

b. Suspend the same load as used above on a single line. Stop the engine and knuckle-in both the secondary and primary drum clutches, disengage the master clutch, and lower the load on the brake backing down the hoisting machinery. An inspection shall be made for ease of control operation and smoothness of load lowering and machinery operation. Any binding, chattering of machinery, slipping of the clutches, or unusual effort required for the operation shall constitute failure of this test.

4.5.2.13.2 Clutch control. When a power hydraulic control system is furnished for the superstructure (see 3.5.13.3) the following test shall be performed. Without engine power, a clutch shall be engaged and released for not less than 15 cycles. Inability of the clutch to engage and release at least 15 times using only the accumulator pressure, without engine power, shall constitute failure of this test.

4.5.2.14 Antirotation test.

a. Install the antirotation rods and tie the crane-shovel down to prevent tipping. Release the swing brake and lock of the crane-shovel. Reeve a line to a point on one end of the revolving super-

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structure to exert a moment in a horizontal plane both clockwise and counterclockwise. The moment shall be not less than twice the gross weight of the superstructure (without boom) times the distance from the center of gravity to the center of rotation.

b. Repeat the procedure with the swing lock engaged and the antirotation rods and swing brake released with a moment equal to one-half the moment in a above.

c. Repeat the procedure with a moment of not less than 0.30 of the torque used above with the swing brake set and the antirotation rods and swing lock released.

Breakage or distortion of the rods, brackets, attaching members, welds or inability of either the lock or brake independently to hold the superstructure in place shall constitute failure of this test.

4.5.2.15 Torque-converter stall. The torque-converter shall be stalled at full throttle of the crane-shovel engine for not less than 30 seconds. The crane-shovel shall then be operated for a period of 15 minutes. Oil temperature exceeding 250 °F (121 °C) but not less than 180 °F (82 °C) and in an ambient temperature of not less than 30 °F (-1 °C), and the inability of the torque-converter oil to return to a stabilized operating temperature within 15 minutes shall constitute failure of this test. This test shall be conducted during the cooling test specified in 4.5.2.18.

4.5.2.16 Electrical.

4.5.2.16.1 Lights and horn. Operate each of the switches controlling all lights and the horn button not less than five times. Measure the light intensity of the instrument panel light. Failure of the switches to operate, lights to burn clearly, horn to function properly, and light intensity of instrument panel light less than that specified shall constitute failure of this test.

4.5.2.17 Accumulator.

a. When a power hydraulic control system is furnished for the crane-shovel (see 3.5.13.3), the accumulator charge pressure shall be measured prior to performing any tests.

b. After completion of all tests, the accumulator charge pressure shall be measured. An accumulator charge pressure less than that measured in a above shall constitute failure of the test.

4.5.2.18 Cooling test. The crane-shovel shall be tested in accordance with test method no. 2 of appendix A. Extrapolated water temperature

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exceeding 210 °F (99 °C) leaving engine cooling jacket or failure of the controls and instruments to operate shall constitute failure of this test.

4.5.2.18.1 Emergency starting. Disconnect the crane-shovel batteries. Start the engine using an emergency charging receptable with a 20-foot cable as specified in 3.6.13, connected to a separate power source. Failure of the engine to start within 1 minute shall constitute failure of this test.

4.5.2.19 Boom stop and pile lead.

a. Calculate the load imposed on the boom stop and attaching points when bending a boom over the stops. Subject the boom stop to this load. The basic boom shall be lowered until the boom point touches the ground; then raise the point until the boom makes an angle of not less than 25 degrees with the horizontal. Repeat this cycle for a total of not less than 12 times, all at maximum boom hoist speed.

b. The pile-driving leads, boom adapter, and catwalk specified in 3.9.2.8 or 3.33 shall be completely attached to the boom and revolving superstructure and be ready for piledriving operation. The mounting shall be checked.

c. Hoist the boom to engage the boom stop. Any noticeable binding or chattering of the boom stop when raising boom or incorrect fitting or mounting of the piledriving leads shall constitute failure of this test.

4.5.2.20 Boom stop test. Failure of the boom stop to withstand force required for completely bending the boom and to stop the boom at 82 degrees shall constitute failure of this test.

4.5.2.21 Overload indicating system. Utilizing a 30-foot boom set at first for the minimum and then the maximum operating radius, perform the following tests:

a. Assemble a load of a size which will not activate the visual signal.

b. Lift the load 2 feet off the ground.

c. Increase the load by small increments and determine the load required to activate the visual signal.

d. Continue to increase the load and determine the load required to activate the audible signal.

e. Reduce the load and determine the load at which the audible signal ceases.

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f. Continue to reduce the load to determine the load at which the visual signal stops.

g. Lower the load to the ground and repeat the above procedure for the 40-, 50-, and 60-foot booms.

h. Failure of the overload indicating system and its components to meet the requirements of 3.9.1.4.1, inability to complete the test, or malfunction shall constitute failure of this test.

4.5.2.22 Overload system. Perform the test specified in 4.5.2.21 using the minimum operating radius and a 30-foot boom. Failure of the overload system to operate as specified in 3.9.1.4.1 shall constitute failure of this test.

4.5.2.23 Rope strength. Measure the maximum load on all ropes at 30-feet operating radius with rated load. Measure the maximum load on all ropes while raising a 60-foot boom from the ground. A safety factor less than specified in 3.9.1.6 shall constitute failure of this test.

4.5.2.24 Lubrication test. Lubricate the crane-shovel as recommended by the contractor. Inability to conform to the following shall constitute failure of this test:

a. That the seals retain lubricant.

b. That all fittings are accessible without removal of parts or components of the crane-shovel.

c. That fittings comply with 3.11.2 and 3.11.3.

d. That fittings are protected from damage due to operation or movement of the crane-shovel.

e. That provision has been made for lubrication of gears, chains, and sprockets.

4.5.2.25 Maintenance package. During initial production testing, maintenance support elements such as manuals (when furnished), lubrication charts, repair parts, and special tools (see 3.11 and 3.16) shall be evaluated. Failure of the crane-shovel to be properly maintained using the support elements shall constitute failure of this test and shall require revision of the document. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the reliability failure (see 4.5.2.28 and 4.5.2.42).

4.5.2.26 Lifting and tiedown provision test. To determine conformance with 3.13, lifting and tiedown provisions, markings, and the connecting

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structural members of all components of the disassembled crane-shovel shall be inspected for conformance with MIL-STD-209. All lifting and tiedown provisions shall be statically tested to ensure that the provisions and connecting structural members meet the criteria of MIL-STD-209. Test results that show the provisions do not meet the criteria of MIL-STD-209 shall constitute failure of this test.

4.5.2.27 Ease of maintenance. Maintenance operations specified in 3.16 shall be accomplished on the crane-shovel. Failure to accomplish the operation in the time specified with the tools specified shall constitute failure of this test. The use of a hoist with its operator will be permitted during engine removal and replacement.

4.5.2.28 Maintenance ratio. Maintenance operations required during first article testing shall be accomplished in accordance with 3.16. The ratio of all the manhours of maintenance required to the hours of operation performed shall be computed. Nonconformance to 3.17 shall constitute failure of this test.

4.5.2.29 Transportability inspection. To determine conformance to 3.19, the crane-shovel shall be inspected for agreement with the transportability criteria defined in MIL-STD-1366 and MIL-HDBK-157. Not conforming to the criteria shall constitute failure of this inspection.

4.5.2.29.1 Disassembly and reassembly verification. To determine conformance with 3.19.1, all components of the crane-shovel specified in 3.19.1 shall be disassembled and reassembled, using a crane, to verify capability to meet the time requirement. Inability to disassemble or reassemble in two hours using 2 people shall constitute failure of this verification.

4.5.2.29.2 Air transportability test. All components of the disassembled crane-shovel shall first be inspected and calculations provided for meeting the dimensional and weight criteria of MIL-STD-1791. If the inspection or calculations show possible interference, the crane-shovel shall be loaded, tied down, and unloaded in a C-5 aircraft to determine conformance with 3.19.2. Non-conformance to the criteria in MIL-STD-1791 or inability to be loaded and tied down in the C-5 and unloaded, shall constitute failure of this test.

4.5.2.29.3 Rail impact test. All major components of the disassembled crane-shovel shall be tested in accordance with 4.5.2.26 prior to the rail impact test. To determine conformance to 3.19.3, all components of the disassembled crane-shovel shall be inspected for agreement with the dimensions of the GIC diagram, and shall be subjected to the rail impact test proposed for the modified MIL-STD-810 (see appendix B). Each component of the disassembled crane-shovel shall be inspected after each impact for spillage of lubricants, fuel or water; structural damage; electrical shorts; and normal operations and performance. Performance degradation or permanent

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deformation of any part is considered a deficiency and a failure. Spillage of lubricants, fuel or water; structural damage; or electrical shorts shall also constitute a failure of this test.

4.5.2.29.4 Transport by highway verification. To determine conformance to 3.19.4, the disassembled crane-shovel and its prime mover combinations shall be inspected for agreement with dimensional and weight highway permit limits of the individual States, as described in Appendix C. Inability of the disassembled crane-shovel and its prime mover combinations to meet highway permit limits shall constitute failure of this verification.

4.5.2.29.5 Shipment by water. The crane-shovel shall be inspected as defined in 3.19.5. Not conforming to the criteria shall constitute failure of this inspection.

4.5.2.30 Craning. Using various loads, including rated loads determined in 4.5.2.31, perform craning operations for 50 hours. Inability to perform craning, inability to complete the test, malfunction, evidence of any damage, or failure of any crane component shall constitute failure of this test.

4.5.2.31 Lifting capacity tests. Lifting capacities of the crane-shovel shall be determined by use of actual weights or load cells using a standard length boom in accordance with 3.20.1.1. With the boom and crane-shovel in the least stable direction, determine the tipping load with the boom at the minimum working radius, the boom radius equal to boom length, and at three intermediate points (see 6.9.2 and 6.9.5). Overload indicating system may be disconnected during this test.

a. Rated lifting load computation. A curve shall be plotted using the tipping loads and radii measured above as points on the curve. A curve representing 75 percent of the tipping loads obtained shall be plotted.

b. Rated loads less than those shown in table V shall constitute failure of this test.

4.5.2.32 Shovel-front attachment. Perform shovel-front excavating operations at maximum capacity and include equal portions of time operating at each of the ranges specified in table VI. The total period of operation shall be not less than 40 hours. The operations shall be conducted in a gravel pit or rock quarry. The crane-shovel shall average not less than 60 cubic yard (cu. yd.) per hour. Failure to perform as specified, any evidence of malfunction, damage, or permanent deformation shall constitute failure of this test.

4.5.2.33 Backhoe. Perform maximum-capacity, heavy-duty trenching operations in undisturbed heavy clay soil at minimum depths specified in table VII for not less than 40 hours. The crane-shovels shall average not less

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than 60 cu. yd. per hour. Failure to perform as specified, any evidence of malfunction, damage, or permanent deformation shall constitute failure of this test.

4.5.2.34 Clamshell and dragline. Perform maximum capacity, heavy-duty clamshell and dragline operations in undistrubed heavy clay soil. The operation shall be performed with the clamshell and dragline buckets for not less than 16 hours (8 hours for each bucket). The crane-shovel shall average not less than 55 cu. yd. per hour. Inability to operate with clamshell and dragline buckets, inability to complete test, malfunction, evidence of any damage or failure of any component shall constitute failure of this test.

4.5.2.35 Crane capacity plate determination. Repeat the procedure specified in 4.5.2.31 for each additional 10-feet of boom length up to 60 feet. Plot 75 percent of the load values determined for 10-foot increments for each 10-foot section of boom length. A capacity chart for the jib shall be included. The overload indicating system may be disconnected during this test.

4.5.2.36 Off-road test. Each completely assembled crane-shovel shall be off-road tested for a distance of not less than 1 mile. This 1 mile shall include the following:

a. One grade not less than 30 percent from the horizontal and not less than 25 feet long. The crane-shovel shall be driven uphill and downhill over this grade.

b. The crane-shovel shall be driven backward not less than 200 feet.

c. The brake shall be tested by parking on the 30-percent grade.

Failure of the crane-shovel to accomplish any part of this test shall constitute failure of this test.

4.5.2.37 Maneuverability test. With fuel tank not less than 0.75 full perform the following:

4.5.2.37.1 Grade. The crane-shovel shall ascend and descend a grade of not less than 30 percent and 50 feet in length. This test shall be conducted with each attachment listed in column 1 mounted on the crane-shovel, the counterweight located as listed in column 2, and the crawler track travel as shown in column 3 in table XIII.

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TABLE XIII. Grade test.

Attachments	Counterweight location	Crawler track travel
Col. 1	Col. 2	Col. 3
Boom, 30 feet with block Backhoe Shovel	Forward Back Back	Forward Forward Forward

Inability of the crane-shovel to negotiate the grade, permanent deformation, fuel leakage, or inability to complete the test shall constitute failure of this test.

4.5.2.37.2 Slope. With the operator on the high side, propel the crane-shovel on a side slope of not less than 15-percent grade and 100 feet in length. Repeat the above with operator on the low side. Inability of the crane-shovel to negotiate the side slope, permanent deformation, fuel leakage or inability to complete the test shall constitute failure of this test.

4.5.2.37.3 Brake. With the crane-shovel equipped with the shovel front attachment, the crawler track in forward travel, ascend and descend the grade specified in 4.5.2.37.1. Stop three times in each direction and engage and lock the track-locking mechanism. Inability of the track-locking mechanism to hold the tracks stationary or permanent deformation shall constitute failure of this test.

4.5.2.37.4 Steering. Steer the crane-shovel to the right and left to prove the steering capability. Inability to steer shall constitute failure of this test.

4.5.2.37.5 Propelling test. Self-propel the crane-shovel with 30-foot boom and hook block for 40 miles at an average speed of not less than 0.75 miles per hour. Terrain to be covered shall consist of 20 miles of unimproved road and 20 miles of off road. Off-road terrain shall include not less than a total of 1 mile of 10-percent grade. Inability to complete the test, permanent deformation, breakage, or malfunction shall constitute failure of this test.

4.5.2.37.6 Fording. Water fording capabilities shall be as follows for the first article model: (a) Bring crane-shovel to operating temperature. (b) Operate a minimum of 2 hours in not less than 12-inch deep water. (c) Sample oil from engine and all drive components. (d) Repeat the above

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operations four times. The first article test items shall be subjected only to (a) and (b) above. Inability to complete or nonconformance to 3.20.2.1.d shall constitute failure of this test.

4.5.2.38 Ground bearing pressure test. The ground bearing pressure is derived by dividing the total working weight by the bearing area in square inches. The weight shall be determined by platform scales, by calibrated hydraulic or electronic devices, and the bearing area used shall be the center-to-center distance between crawler end sprockets multipled by twice the width of the track shoe. Ground bearing pressure over 11 psi shall constitute failure of this test.

4.5.2.39 Backward stability test. With the boom of the length specified in 3.9.1.1, at its minimum working radius, in the least stable direction, the location of the center of gravity shall be determined as follows for conformance to 3.20.2.4.

$$X = \frac{Rd}{W}$$

Where: X = Distance between centerline of crawler track roller opposite lifting side of crane-shovel to center of gravity.
R = Weight measured when crawler track opposite lifting side of crane-shovel is on firm ground and the crawler track on lifting side of crane-shovel is on a platform scale.
d = Distance between centerlines or on center of outer flange of the crawler track rollers.
W = Weight measured when entire crane-shovel is on platform scales.

Failure to comply with the requirement in 3.20.2.4 shall constitute failure of this test.

4.5.2.40 Noise Level Test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating at 2/3 and full load. The noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.41 Climatic.

4.5.2.41.1 Operating temperature.

4.5.2.41.1.1 High temperature operation. Subject the nonwinterized crane-shovel to an ambient temperature of 125 °F (52 °C) until the temperature of the crane-shovel has stabilized. Start the engine and operate it at maximum

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governed speed for a total of 1 hour. During this hour, individually engage and disengage clutches instantly under maximum governed engine speed. Actuate the brake controls. Move the superstructure clockwise and counterclockwise and the crane forward and backward within the limitations of available space. Stop the engine at the end of 1 hour; restart the engine after 5 minutes and operate for 10 minutes without load. Failure of the engine to start and operate within the limits of applicable engine specifications or failure of any component to function shall constitute failure of this test.

4.5.2.41.1.1.1 Thermocouple location. The crane-shovel shall have not less than the following thermocouples installed prior to the high temperature test.

a. Engine coolant temperature: One in the engine-cooling-jacket discharge pipe and another in the coolant inlet to the engine.

b. Fuel temperature: In the tank.

c. Battery temperature: In the two center cells of each battery between the plates and at mid-depth of the electrolyte.

d. Ambient temperature measured at an angle of 45 degrees and not less than 3 feet from each corner of the crane-shovel at engine-crankshaft level.

e. Engine oil gallery.

f. Torque converter sump. (When torque converter is furnished.)

Record at 10-minute intervals temperature readings of the thermocouples listed above during the operations specified in 4.5.2.41.1.1.

4.5.2.41.1.2 Low temperature. The following test procedure shall apply and all data shall be recorded for the low-temperature operation -25 ^{o}F (-32 ^{o}C). The crane-shovel shall have installed, prior to the test, thermocouples as specified in 4.5.2.41.1.1.1a. through f. The temperatures shall be recorded prior to engine starting, immediately after engine starting, and at 10-minute intervals during engine warmup and cycling:

a. Check the specific gravity of each battery to ensure full charge.

b. Subject the crane-shovel to an ambient temperature of not less than -25 ^{o}F (-32 ^{o}C) until all parts, lube oil, coolant, fuel, and batteries have stabilized.

c. Maintain an ambient temperature of -25 ^{o}F (-32 ^{o}C) or lower throughout the entire test.

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d. Start the crane-shovel engine. Engine should run on diesel fuel in not more than 5 minutes after beginning of first cranking.

e. One operator wearing arctic clothing including mittens shall perform all priming and starting operations.

f. Crank the engine using batteries that were cold soaked and stablized at not less than -25 ^{o}F (-32 ^{o}C).

g. Supplementary batteries or other electrical cranking energy shall not be used.

h. Glow plugs or other accessories working off the cranking batteries and auxiliary fluid priming systems which are permanently mounted on the crane-shovel used in starting the engine may be used.

i. Smooth engine running should be obtained within 15 minutes after starting without unnatural or continued control manipulation and without use of primers and priming fluids.

j. Start and run engine for 15 minutes. After 15 minutes operate the engine as follows:

(1) 10 minutes idling.

(2) 10 minutes half throttle.

(3) 5 minutes full throttle.

Perform the 25-minute cycle three times.

k. During engine cycling and while wearing arctic mittens and boots, the operator shall perform each of the following:

(1) Engage and disengage power train to the crawler, steering, and brake mechanism.

(2) Operate through two complete cycles all brakes, shafts, and cylinders.

(3) Mount and dismount from the operator's position on the crane-shovel.

l. Determine the functioning of the crane-shovel as follows:

(1) Individually engage and disengage superstructure clutches under maximum governed engine speed.

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(2) Actuate brake controls.

(3) Move superstructure clockwise and counterclockwise within limitations of available space.

(4) Drive the crane-shovel forward and backward within the space available.

Failure of the engine to start and operate within the limits of applicable engine specification or failure of any component to function shall constitute failure of this test.

Upon completion of the cycling period, shut down the engine and with the crane-shovel still in the chamber at a temperature of not less than -25 °F (-32 °C), restart the engine by using outside power. The operator shall be wearing arctic mittens during this operation. Failure to start the engine with outside power shall constitute failure of this test.

4.5.2.41.2 Storage.

4.5.2.41.2.1 High temperature storage. Subject the crane-shovel to a high ambient temperature until all parts have stablized at 160 °F (71 °C). The temperature of all parts shall then be maintained continuously at 160 °F (71 °C) for 1 hour. Examine the crane-shovel visually and without disassembling. Any evidence of damage, leaks, cracks, permanent deformation, or discoloration shall constitute failure of this test.

4.5.2.41.2.2 Low temperature storage. Subject the crane-shovel to a low ambient temperature of -50 °F (-46 °C) until all parts have stablized at -50 °F (-46 °C). The temperature of all parts shall then be maintained continuously at -50 °F (-46 °C) for at least 6 hours. Examine the crane-shovel visually and without disassembling. Any evidence of damage, leaks, cracks, permanent deformation, or discoloration shall constitute failure of this test.

4.5.2.41.3 Winterization. In addition to the tests specified in table XI (excluding the 125 °F (52 °C) test), the winterized crane-shovel shall be subjected to the following tests as indicated in table XII.

4.5.2.41.3.1 High temperature operation. Subject the crane-shovel to an ambient temperature of 105 °F (41 °C) until the temperature of the crane-shovel has stabilized. The crane-shovel shall be tested with the winterfronts in open position and the engine side panels removed. Start the crane-shovel engine and operate it at maximum governed speed for 1 hour. During this hour, individually engage and disengage clutches instantly under maximum governed speed. Actuate the brake controls. Move the superstructure clockwise and counterclockwise within the limitations of available space. Stop the engine at the end of 1 hour; restart the engine after 5 minutes

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and operate for 10 minutes without load. Failure of the engine to start and operate within the limits of applicable engine specifications or failure of any component to function shall constitute failure of this test.

4.5.2.41.3.2 Cold starting and running. The crane-shovel, when winterized, shall be tested in accordance with test method no. 1 of the appendix A. One or more of the following shall constitute failure of this test:

a. Engine does not start within 5 minutes after 4 hours of standby heat.

b. Smooth engine running should be obtained within 15 minutes after starting without unnatural or continued control manipulation, and without the use of primers and priming fluids.

c. Engine oil sump temperature lower than 50 °F (10 °C) prior to engine starting.

d. Battery temperature lower than 50 °F (10 °C) prior to engine starting.

e. Engine coolant temperature at heater inlet lower than 50 °F (10 °C) prior to engine starting.

f. Battery compartment temperature less than 50 °F (10 °C) prior to engine start.

g. Cab temperature less than 35 °F (2 °C) at operator's foot pedals within 1 hour after starting engine.

h. 75 percent of each window and windshield not frost free within 1 hour after starting engine.

i. Lubricating oil temperature not stabilized in the engine lubricating oil gallery within the range of 130 to 230 °F (54 to 110 °C) within 1 hour after starting and not remaining within this range throughout the 1-hour test period.

j. Coolant temperature not stabilized in the engine-cooling-jacket discharge pipe within the range of 140 to 210 °F (60 to 99 °C) within 1 hour after starting and not remaining within this range throughout the 1-hour test period.

k. Cab temperature at operator's foot pedal after cab heaters have operated 1 hour could not be maintained at 35 °F (2 °C).

l. Instruments and gages do not function properly.

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m. Thermal damage to batteries.

n. Battery charging system does not show positive charge.

o. While the operator is wearing arctic mittens and boots:

(1) Power train cannot be engaged to the crawler tracks.

(2) All clutches, brakes, shafts, drums, and hydraulic cylinders cannot be operated through two cycles.

(3) Windshield wipers cannot be manually or power operated.

(4) All cab doors, engine housing doors, and winterfronts cannot be operated, latched, or unlocked.

(5) Operator cannot safely mount or dismount from the operator's position in the superstructure.

(6) Maintenance checks cannot be performed.

4.5.2.42 Reliability. For purposes of the reliability test, the test values shall be as follows:

Upper: θ_0 = 68 hours
Lower: θ_1 = 45 hours

Each of the two first article crane-shovels shall be subjected to two cycles of testing in accordance with 4.5.2.37.5, 4.5.2.37.6 (b), 4.5.2.30, 4.5.2.32, 4.5.2.33 and 4.5.2.34. The road mileage shall be divided (computed based on speed of 0.75 mph) and a part conducted between each of the operations. Experiencing 15 or more failures shall result in a reject decision.

4.5.2.42.1 Failures. Failure for the purpose of this test shall be defined as any malfunction not deemed nonrelevant or nonchargeable as defined below which causes degradation below the operating condition required to continue tests specified above, and which cannot be remedied in 1 hour (all waiting periods excluded) utilizing resources available to the using unit.

4.5.2.42.1.1 Nonrelevant failures. Nonrelevant failures shall be as specified in a through g:

a. Installation damge.

b. Accident or mishandling.

c. Failures of the test facility or test-peculiar instrumentation.

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d. Equipment failures caused by an externally applied overstress condition, in excess of the approved test requirements.

e. Normal operating adjustments (nonfailures) specified in the approved equipment operating instructions.

f. Secondary failures within the equipment, which are directly caused by nonrelevant or relevant primary failures. The secondary failures must be proved to be dependent on the primary failure.

g. Failures caused by human errors.

4.5.2.42.1.2 Nonchargeable failures. Nonchargeable failures shall be as specified in a through c:

a. Nonrelevant failures.

b. Failures induced by GFE operating, maintenance, or repair procedures.

c. Failures of items having a specified life expectancy and operated beyond the specified replacement time of the item.

4.5.2.43 Electromagnetic interference. The first article model shall be tested in accordance with 3.25. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.5.2.44 Post-test disassembly and examination. Upon completion of all tests, the crane-shovel shall be dismantled and the following components shall be disassembled in detail for a visual examination:

a. Engine. The engine shall be removed from its mounting and disassembled in detail as follows: Remove the cylinder heads, all valves, the two pistons and piston pins from the two cylinders having the lowest pressure reading at the end of the tests. Remove two main bearings and two connecting rod bearings and turbocharger if one is provided with the engine.

b. Torque converter (if furnished).

c. Brakes and clutches.

d. Ring gear and pinion assembly.

e. Track and turntable rollers.

f. Hydraulic system.

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(1) Accumulators.

(2) Cylinders.

(3) Hose and fittings where leakage has occurred.

g. Drums including appropriate clutch and brake operating parts.

h. Winterization system (when furnished).

Additional disassembly and examination shall be made at the option of the contracting officer. A complete description of the condition of all components and parts of the crane-shovel shall be recorded and photographs shall be taken of any deficiency. Any evidence of fretting, overheating, permanent deformation, or any wear of parts, or parts of components, or wear exceeding manufacturer's published tolerances shall constitute failure of this test.

4.6 Quality conformance. The Government may select crane-shovels at any time during the contract production period and subject these crane-shovels to the examination identified by X in column 6 of table IX, and the test identified by X in column 2 of table XI in accordance with the requirements of this specification. The inspection shall be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and shall not include the previously inspected first article model and initial production crane-shovels. In addition to any test specified as part of the quality conformance, the Government reserves the right to conduct any and all other tests contained in this specification. The quality conformance and failure of such additional tests shall have the same effect as failure of those tests previously specified.

4.6.1 Inspection failure. Failure of quality conformance crane-shovel to meet any requirement specified herein as a result of the examination and tests specified in 4.6 shall be cause for rejection. The Government has the right to refuse acceptance of crane-shovels until evidence has been provided by the contractor that corrective action has been taken. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government. Any deficiencies found as result of the initial production inspection will be considered evidence that all crane-shovels accepted prior to the completion of inspection comparison are deficient. Evidence to the contrary shall be furnished by the contractor provided such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging. The preservation, packing and marking shall be inspected in accordance with the requirements of section 5.

5. PACKAGING

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5.1 Preservation. Preservation shall be levels A, B or C as specified (see 6.2).

5.1.1 Level A and B. Preservation shall be in accordance with the applicable level A or B requirements of MIL-C-3580.

5.1.2 Level C.

5.1.2.1 Disassembly and matchmarking. Disassembly and matchmarking shall be in accordance with the level A requirements of MIL-C-3580.

5.1.2.2 Engine. The engine cooling system shall be preserved using the antifreeze and water procedure of MIL-E-10062.

5.1.2.3 Technical publication. Technical publications shall be preserved in accordance with method IC-1 or IC-3 of MIL-P-116 and placed in the operators cab of the crane-shovel.

5.1.2.4 Any tools and repair parts. Any tools and repair parts shall be preserved in accordance with the level B requirements of MIL-C-3580.

5.2 Packing. Packing shall be levels A, B or C as specified (see 6.2).

5.2.1 Levels A and B. Packing shall be in accordance with the applicable level A or B requirements of MIL-C-3580.

5.2.2 Level C. Boxed components and consolidated packed items shall be packed as for level B except that the containers may be commercial equivalents of the containers specified for level B. Boxes of accessories or removed items shall be secured to the crane-shovel in a manner which will not increase the shipping cube. The cranes shall remain operable for shipment.

5.3 Marking. Marking for levels A, B and C shall be in accordance with MIL-STD-129 as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The crane-shovels, together with the various items of front-end operating equipment, are intended for use in handling materials, lifting and transporting loads short distances, soil stripping, earthmoving, ditch digging, and piledriving.

6.2 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. When a first article (see 4.3) will be required for inspection and approval, number of units required, timeframe for submission, and where tests should be performed (see 3.2).

d. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests will be conducted by the contractor (see 3.2 and 4.3).

e. When first article testing will be required, number of first article crane-shovels to be furnished, and where tests will be performed (see 4.3).

f. When engine connection to transmission will be: friction clutch; friction discharging clutch and fluid coupling, or friction disengaging clutch and hydraulic torque convertor (see 3.5.11).

g. Composition of varnish required (see 3.5.12.1.1).

h. When engine will not be provided with cold starting aids (see 3.6.5).

i. When primary front-end attachments will not be as specified (see 3.9.1).

j. When an overload indicating system will be required (see 3.9.1.4.1).

k. When boom extensions are required (see 3.9.2.1).

l. When boom jib is required (see 3.9.2.2).

m. When clamshell bucket is required (see 3.9.2.3).

n. When clamshell bucket is to be furnished with side and corner teeth (see 3.9.2.3).

o. When tagline is required (see 3.9.2.4).

p. When dragline bucket is required (see 3.9.2.5).

q. When dragline fair-lead is required (see 3.9.2.6).

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r. When wrecking ball is required and size of wrecking ball to be furnished (see 3.9.2.7).

s. When piledriver leads, boom adapter, and catwalk will be required (see 3.9.2.8).

t. When piledriver hammers will be required; drop or diesel, and diesel size required (see 3.9.2.9).

u. When shovel front is required (see 3.9.2.10).

v. When backhoe is required (see 3.9.2.11).

w. When winterization system will be required (see 3.10).

x. Grades of fuels and lubricants required at time of delivery if other than specified (see 3.12).

y. Color required when other than as specified (see 3.27).

z. When camouflage pattern data will be required (see 3.27.1).

aa. When vehicle weight classification forms and application of numbers to crane-shovel not required (see 3.29.2).

bb. When Government-furnished property will be furnished (see 3.32).

cc. What Government-loaned property will be furnished (see 3.33).

dd. Level of preservation and packing required (see 5.1 and 5.2).

ee. Level of marking (see 5.3).

6.2.2 Data requirements. When this specification is used for an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below should be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

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Paragraph No.	Data requirements	Applicable DID No.
a. 3.28.1	Color photograph prints	DI-MISC-80192
b. 3.28.1	Camouflage line art data	DI-MISC-80176

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item(s) should be an initial production model. The first article should consist of three or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waiver the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Welding of high-strength constructional steels. When high-strength constructional steels are used in the fabrication of the crane-shovel, the contracting officer should require the contractor to include in the technical publications specified (see 6.11) the identification and location of these steels, and the necessary precautions to be taken by the using agency in the repair and replacement of these steels. These precautions will include but not be limited to:

a. How to avoid excessive restraint in a weld joint that could produce weld-metal or heat-affected-zone cracking.

b. Type of weld filler metal to be used for repair of each different type of steel by the shielded electric-arc (coated electrode) process and by the metal inert gas (MIG) process including:

(1) Type and class of electrode and MIG wire.

(2) Size (diameter) of electrode and MIG wire.

(3) Shielding medium for MIG process.

c. Material handling safeguards for coated electrodes and MIG wire to prevent contamination and moisture absorption, including the treatment of coated electrodes and MIG wire after exposure to contamination and moisture absorption.

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d. Preparation of base metal for welding. Method to be used (grinding or other). Preparation of root for welding second side when thickness of material necessitates that joint be welded from both sides.

e. Base metal temperatures.

(1) Minimum temperature of base metal at start of welding.

(2) Minimum and maximum preheat of base metal at start of welding.

(3) Maximum interpass temperature during welding.

f. Rate of flow of shielding medium for MIG process.

g. Maximum heat input during welding.

h. The minimum ambient temperature at which welding should be done.

i. Post-weld heat-treatment (stress-relieving) requirements (necessary when weldment must retain dimensional stability during machining or stress-corrosion are a factor).

6.5 Information figures. Figures 2 and 3 show a type of labyrinth seal which has been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other labyrinth seal shapes which are in accordance with this specification.

6.6 Lubricants. MIL-STD-838, Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of nonstandard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113, Guide for the Selection of Lubricants, Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development and Engineering Center (ATTN: STRBE-VF), lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under chapter 2 of MIL-HDBK-113.

6.7 Government-furnished property. The contracting officer should arrange to furnish the contractor the property listed in 3.32.

6.8 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.33 unless the item is being purchased under the same contract.

6.9 Definitions. The following definitions apply to the text of this specification.

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6.9.1 Boom radius. The boom radius is the horizontal distance between the centerline of rotation of the crane-shovel superstructure and the centerline of the hook block.

6.9.2 Tipping. The tipping point is that condition where the crawler rollers on the side opposite of the load are lifted from contact with the treads when lifting over the side, or where one-third of the bearing area of properly adjusted treads are lifted from the ground when lifting over front or rear.

6.9.3 Rated load. Rated load shall be a load that is not more than 75 percent of the tipping load for a given radius with the crane-shovel resting on firm, level ground, fully serviced, and with the fuel tanks not less than one-half full.

6.9.4 Capacity loads. Capacity loads are loads which are not less than 133 percent of the loads shown in table V.

6.9.5 Tipping load. Tipping load is the weight required to cause tipping at any specified radius. It should consist of the hook block, slings and other load weight components attached to the hook of the hook block and should not include weight of the crane boom wire rope, hook block, or other components necessary for crane operations.

6.9.6 Gross weight. Gross weight includes the complete crane-shovel, basic crane boom, crane hook block, complete accessories, lubricants, and coolant necessary to completely service the crane-shovel for operation, and fuel tanks at least half-full.

6.10 Vehicle weight classification. The contracting officer should arrange to furnish vehicle classification forms to the contractor and request the return of the completed forms to the Government. The information thus obtained should be used to determine the weight classification number for the crane-shovel. The contractor should then be requested by the contracting officer to apply the weight classification number to the crane-shovel.

6.11 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.12 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including

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any special tools), and instructions on shipment of crane-shovels. A suggested paragraph is as follows:

"Shipment of crane-shovels should include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.13 Warning decals. The contracting officer should arrange to furnish the warning decals, and provide mounting instructions as required in 3.28.2.

6.14 Subject term (key word) listing.

Crane
Crane, ditch digging
Crane, earth moving
Crane, front-end operating
Crane, lifting
Crane, material handling
Crane, piledriving
Crane, shovel
Crane, stripping
Crane, transport

6.15 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.16. AQL certification. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQL's/LTPD's (Acceptable Quality Levels/Lot Tolerances Percent Defectives) from military specifications.

(Certified by/date: 8/3/89).

Custodian:

Army - ME

Review activities:

Army - AT

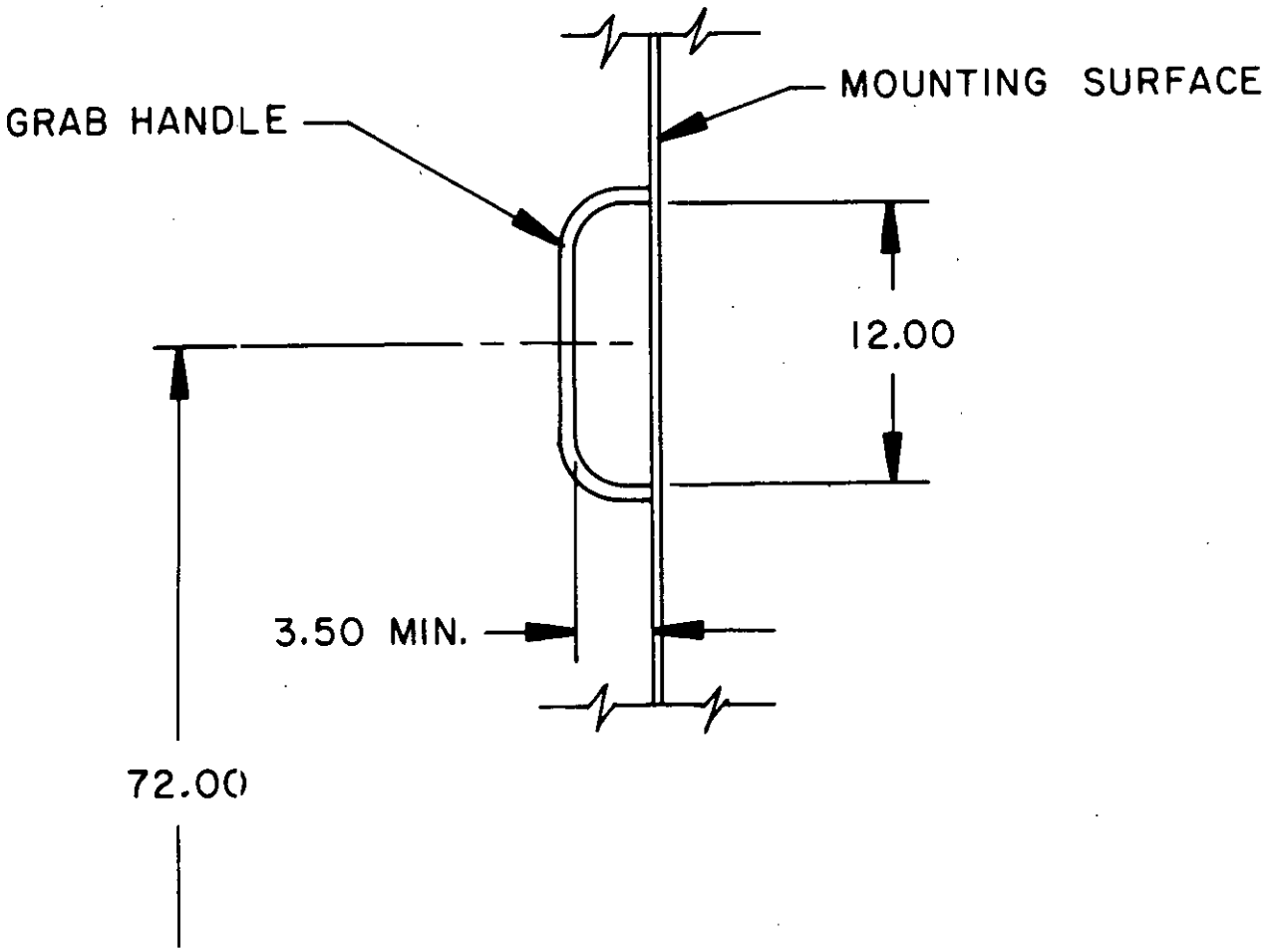
DLA - CS

Preparing activity:

Army - ME

Project No. 3810-A097

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NOTE: ALL DIMENSIONS ARE IN INCHES.

FIGURE 1. Grab handles.

X-813 A

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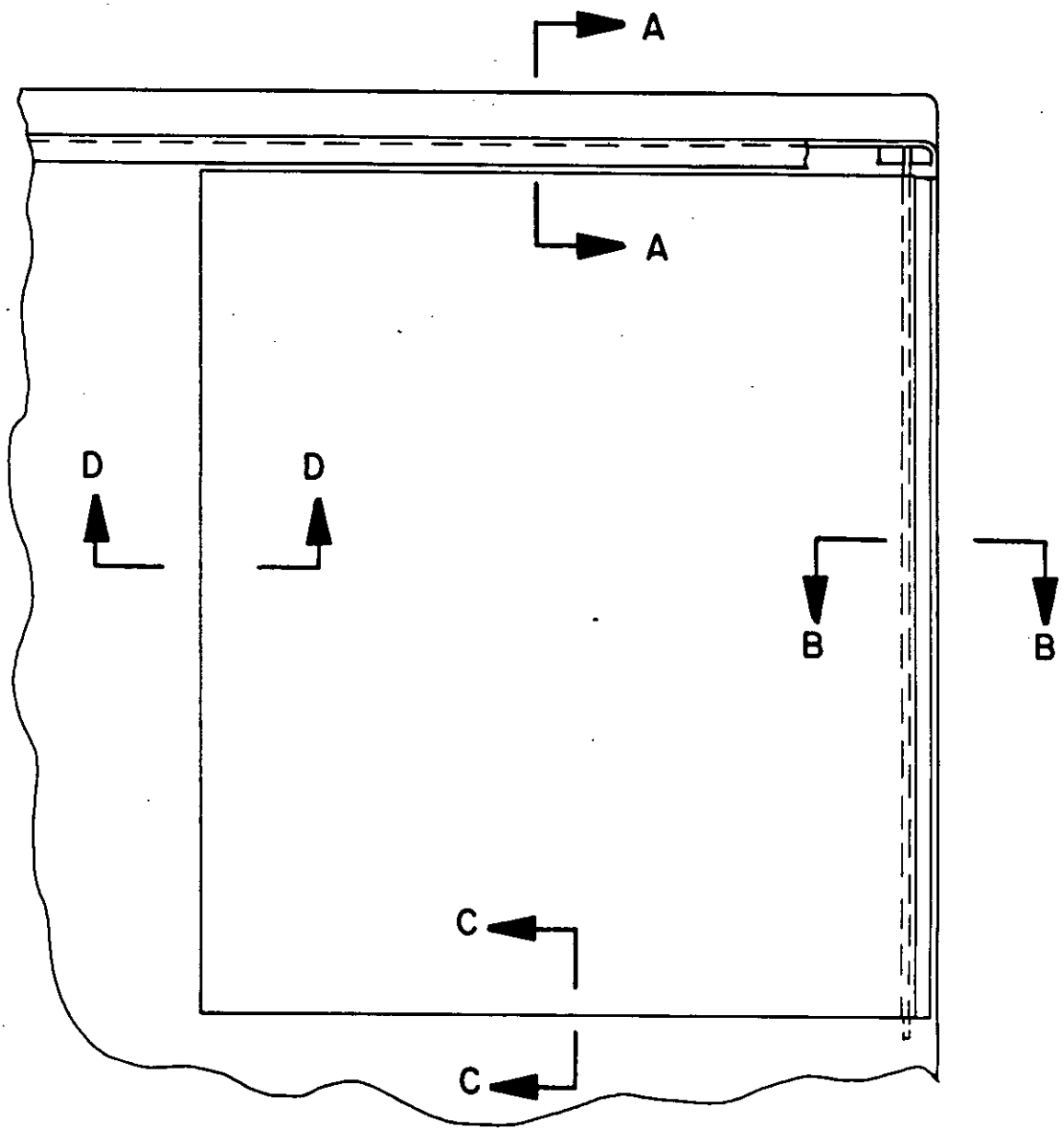


FIGURE 2. Labyrinth seal typical view.

X-2477 A

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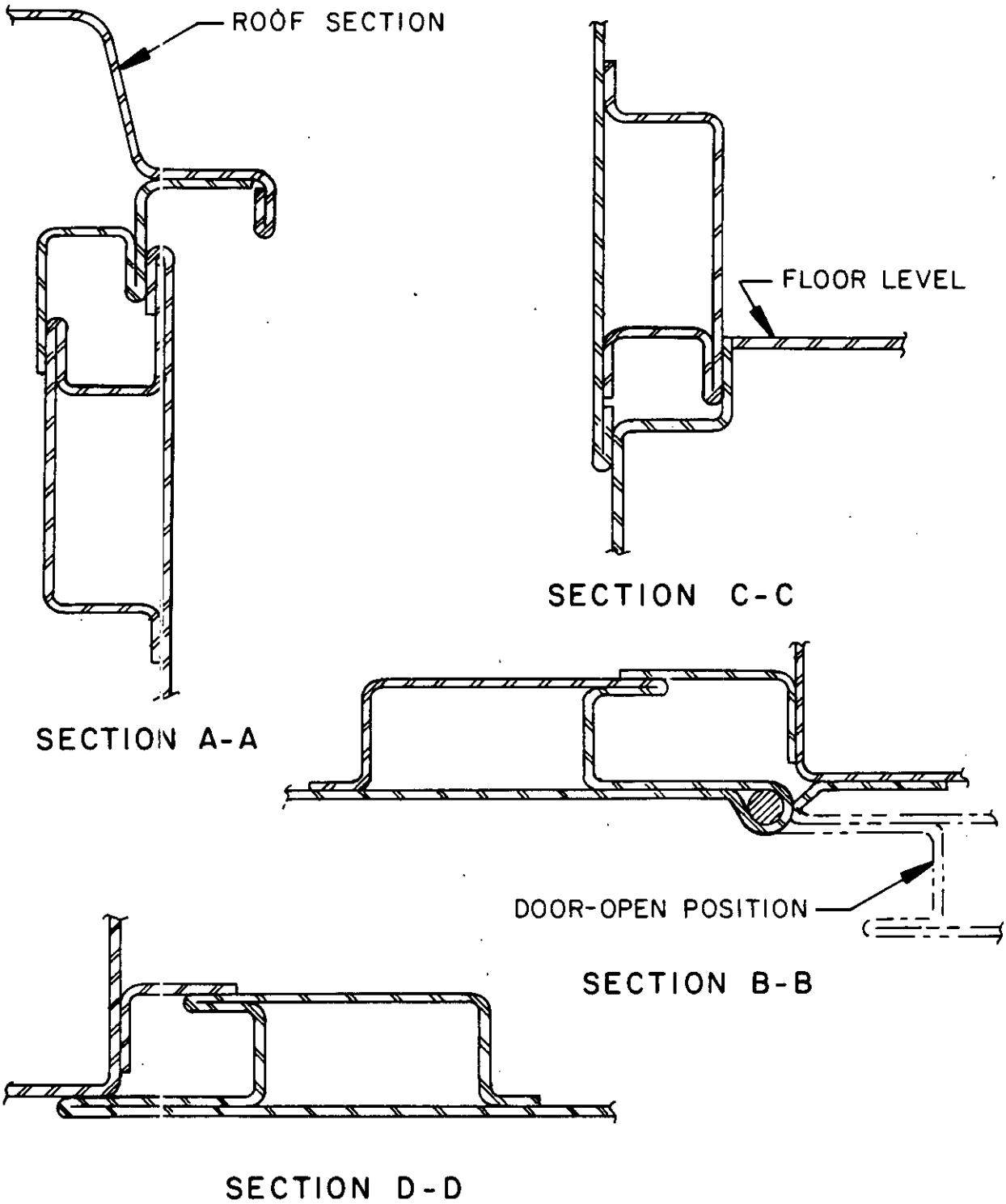


FIGURE 3. Labyrinth seal typical view, sectional views.

X-2478 A

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APPENDIX A

WINTERIZED CRANE-SHOVEL TESTING

10. SCOPE

10.1 Scope. This appendix details the test methods to determine whether crane-shovels procured under this specification are in accordance with the requirements set forth. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. TEST METHOD NO. 1. Winterization test (-50 °F) (-46 °C).

30.1 Test apparatus.

a. Cold chamber.

b. Arctic clothing.

c. Thermocouples.

d. Battery hydrometer.

e. Wattmeter.

f. Voltmeter.

g. Recording instruments (temperature readings), electrical.

30.2 Test procedure.

a. Service the crane-shovel with arctic lubricants in accordance with MIL-L-46167 and antifreeze in accordance with MIL-C-11755.

b. Check the specific gravity of each battery. Specific gravity shall be not lower than 1.275 before beginning of cool-down.

c. During the 4-hour standby heating and starting period, the operator wearing arctic mittens shall perform the following:

(1) Close and latch the housing panels.

(2) Prime and start the engine.

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d. Subject the crane-shovel to an ambient temperature of -50 °F (-46 °C) until all parts are stabilized. Perform the test after stabilization is assured. The thermocouples are located as follows:

(1) In the two center cells of each battery between the plates and at mid-depth of the electrolyte.

(2) In the volumetric center of any hydraulic oil tank of more than 25-gallon capacity.

(3) In the engine lubricating oil gallery (at downstream end).

(4) In the engine coolant in the engine-cooling jacket discharge pipe.

(5) In the chamber at a horizontal angle of 45 degrees from the sides and ends of the crane-shovel and not less than 3 feet from each corner of the crane-shovel at engine-crankshaft level.

(6) In the operator's compartment 6 inches above the floorboards and between the operator's foot pedals.

e. Maintain -50 °F (-46 °C) ambient temperature throughout the test.

f. Preheat the crane-shovel for 4 hours by interconnecting the crane-shovel to outside 115-volt power source using the 25-foot cable supplied with the crane-shovel. Start coolant-type cab heaters not more than 15 minutes after the engine has been started.

g. Supplementary batteries or other electrical cranking energy shall not be used.

h. Glow plugs or other accessories working off of the cranking batteries may be used in the -50 °F (-46 °C) start. Use only the cold starting aids specified in 3.6.5.

i. Within 5 minutes after the initial 4 hours of standby heat start, crane-shovel engine.

j. Operate the engine until:

(1) The lubricating oil temperature is stabilized within the range of 130 to 230 °F (54 to 110 °C).

(2) The coolant temperature is stabilized in the engine-cooling jacket discharge pipe within the range of 140 to 210 °F (60 to 99 °C).

k. Run the engine for 1 hour after these conditions have been met.

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l. Cycle the engine periodically between idling and full-load governed speed during the 1-hour period of engine operation specified in (1) above.

m. Record the temperature at 10-minute intervals during the 4-hour standby heating period at the following locations:

(1) Batteries.

(2) Hydraulic tank.

(3) Engine oil.

(4) Engine coolant.

(5) Ambient temperature.

(6) Operator's compartment 6 inches above the floor at the foot pedal.

(7) Coverter oil at outlet (if furnished).

n. During 1-hour operation, and while wearing arctic clothing, an operator shall perform each of the following:

(1) Operate through two complete cycles all clutches, brakes, shafts, drums, cylinders, and hydraulic components.

(2) Operate windshield wipers.

(3) Operate cab doors, engine housing doors, sidehoods, and winterfronts.

(4) Mount and dismount from the operator's position.

o. After completion of the above 1-hour operation, the operator shall perform the following:

(1) Shutoff the engine.

(2) Restart the engine using outside power.

(3) Disconnect cable to outside power and shutoff engine.

(4) Check oil level on all dipsticks.

40. TEST METHOD NO. 2. Cooling test.

40.1 Test conditions. The crane-shovel, with or without boom, shall be tested in area where the change in the ambient air temperature shall not be

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greater than +5 °F (-15 °C) during a 4-hour period. A dynamometer or other loading device shall be directly connected to the drum shaft, or chain connected to a sprocket on the drum. The test shall be conducted with doors, windows, and access doors to the engine compartment opened. The superstructure shall be serviced with military oils and greases designated for use in the ambient tempertures at which the test is to be run. The coolant system shall be serviced with water.

40.2 Procedure.

a. Instrumentation. Install thermocouples to measure the following temperatures:

(1) Air, ambient.

(2) Air, entering inlet grille.

(3) Air, leaving outlet grille.

(4) Air, within engine compartment.

(5) Air, entering air cleaner.

(6) Oil temperature, engine sump.

(7) Oil temperature, engine gallery.

(8) Water entering radiator (top tank).

(9) Water leaving radiator (bottom tank).

(10) Oil torque converter sump (if applicable).

b. The ambient air temperature shall be measured at the engine crankshaft level, 4 feet from the external face of the radiator on suction type fan blade and 4 feet from the largest accessible opening to engine compartment on blower-type fan blade.

c. With the engine running at low idle speed, record temperature readings of the thermocouples for 10 minutes.

d. With the engine at wide-open throttle and the recorder reading temperatures at 3-minute intervals, load the loading device to capacity as specified in 6.9.4. Maintain this condition and record temperatures until oil and coolant temperatures have stabilized. Stabilization shall be considered achieved when the temperature differential between the top tank and ambient air does not vary not more than 2 °F (-17 °C) for any 30-minute period.

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APPENDIX B

PROCEDURE FOR RAIL IMPACT TEST

10. SCOPE

10.1 Scope. This appendix details the test procedure for the rail impact test of crane-shovels that will be transported by rail. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. PROCEDURE.

30.1 I-3.10 Procedure VIII - Rail impact.

a. Purpose of Test. Procedure VIII is intended to test equipment that will be transported by rail; to determine the effect of normal railroad car impacts that occur during rail shipment, and to verify the structural integrity of the test item and the adequacy of the tiedown system and the tiedown procedures. All test items shall be tested at their maximum gross weight rating.

b. Restrictions. Procedure VIII is not intended for the separate testing of small, individually packaged pieces of equipment that would normally be shipped (and tested) when mounted on a pallet.

c. Test Conditions. This test is conducted by mounting the test item on a rail car in its rail shipment configuration and then performing a series of at least four impacts. The first three impacts shall be at 6.4, 9.7 and 13 km/h (4, 6 and 8 mph), respectively, in the same direction. The fourth shall by conducted at 13 km/h in the reverse direction. All four impacts shall have a tolerance of +0.8, -0.0 km/h. If the test commodity can be shipped in two orientations (such as lengthwise and crosswise on the rail car), the four impacts shall be repeated for each orientation.

d. Rationale. Data for the rail impact test were derived from statistical data on the frequency of impacts with relationship to speed and frequency of occurrence. Brakes are set on the buffer car to provide a more conservative test.

e. Failure Analysis. A test item shall be classified as not having survived the rail impact test and will be deemed a test failure if any item that is attached to or included as an integral part of the test item breaks free, loosens or shows any signs of permanent deformation beyond specification tolerances. A test item that passes this procedure should be capable of rail transport without damage to the item or tiedowns.

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30.2 II-3.8 Procedure VIII - Rail impact.

Step 1. Position from one to five rail cars on a level section of track. Satisfy the following conditions:

a. Adjust the total weight of the car assemblage (see above) to at least 114,000 kilograms (250,000 pounds).

b. Compress the couplers between cars to take up any slack.

c. Set all of the air and hand brakes on the car(s) to be used.

d. Secure any load in or on the car(s) to prevent sliding or shifting; any movement greater than 5 cm shall be justification for retest.

e. The end of the buffer car to be struck must have a standard draft gear.

Step 2. Mount the test item on the test car. The test car shall be equipped with standard draft gears and conventional underframes. The materiel developer is responsible for the development of transportation instructions, and shall coordinate these with, and obtain approval from the Military Traffic Management Command Transportation Engineering Agency (MTMCTEA), well in advance of rail impact testing. Mounting of the test item shall incorporate the standard loading and bracing method as shown in section 6 of the Association of American Railroads (AAR) "Rules Governing the Loading of Department of Defense Materiel on Open Top Cars. No exotic or unusual tiedown methods shall be used; any non-stndard loading and bracing must be approved by the Military Traffic Management Command Transportation Engineering Agency (MTMCTEA), ATTN: MTT-TR, P.O. Box 6276, Ft. Eustis, VA 23606-0276 (or its European equivalent) prior to testing. Substitute (dummy) test items must be of equal weight and weight distribution as the actual test item and similar in shape. The arrangement of the test item and its blocking and tiedown to be tested shall be identical to that proposed and approved by MTMCTEA (if non-standard).

This requirement is mandatory for all equipment developed for use in the U.S. Equivalent European standards may apply for non-U.S. commodities.

Step 3. Situate the test car between the buffer cars and the locomotive, and pull the test car at least 65 meters (200 feet) from the buffer car(s) along a level section of track (a minimum distance to achieve the required locomotive speeds).

Step 4. Position the knuckles of the buffer and test cars for coupling.

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Step 5. Install a timing device to measure the test car speed (± .1 km/h) just prior to impact with the buffer car(s). Suggested methods include electronic timing (microswitches) and radar. The use of torpedoes and a stopwatch is permissible but not recommended because of the inaccuracies involved.

Step 6. Push the test car toward the buffer car(s) and, by using the locomotive's speedometer or other means, release the test car when the desired test speed is reached, thus allowing the test car to freely impact the buffer car(s).

NOTE: Any impacts below the required test speeds shall be repeated. Impacts above the required test speeds shall be accepted providing the requirements of paragraph I-3.10e are satisfied. Retesting shall be accomplished with new tiedown material to eliminate additive effects and, if possible, a new test item.

Step 7. Repeat Step 6 until the test car(s) has impacted the buffer car three times at the same end, once each at speeds of 6.4, 9.7 and 13.0 km/h (4, 6, and 8 mph) +0.8, -0.0 km/h. Reverse the test car and repeat the 13.0 km/hr impact, for a total of 4 impacts.

NOTE: Adjustment of the lading or securing mechanisms, or reconditioning of the bracing or items of securement is not allowed without a complete retest.

Step 8. Repeat steps 1-7 for any other shipping orientation.

Step 9. Record the pertinent information for each impact, to include the following:

a. Complete test item identification.

b. The number and speed of impact.

c. Observations of tiedowns, blocking, fittings, etc.

d. Observations of the test item's physical condition.

e. Results of any operational checks.

NOTE: Cargo requiring extraordinary attention, i.e., nuclear, one-of-a-kind, high value, or key military equipment, may justify changes to the test procedure and criteria; these shall be identified by the developer or Program Manager, and approved by the Commander, Military Traffic Management Command Transportation Engineering Agency (MTMCTEA), ATTN: MTT-TR, P.O. Box 6276, Ft. Eustis, VA 23606-0276 (or its European equivalent).

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APPENDIX C

MOVEMENT PERMITS WITHOUT CERTIFICATION

10. SCOPE

10.1 Scope. Conditions for which states will issue movement permits without certification as essential to national defense.

	LENGTH	WIDTH	HEIGHT	GROSS WEIGHT
ALABAMA	75'	14'	15'5"	150,000(22)
ALASKA	85'	16'	18'	150,000
ARIZONA	(15)	(15)	(15)	(1)
ARKANSAS	75'	12'	13'6"	(1)
CALIFORNIA	135'	14'	AS NEEDED	200,000(1)
COLORADO	100'	14'	14'6"	200,000
CONNECTICUT	100'	14'	14'	120,000
DELAWARE	100'	12'	13'6"	120,000
D.C.	AS NEEDED	AS NEEDED	AS NEEDED	N/A
FLORIDA	(15)	14'	16'	150,000
Georgia	AS NEEDED	14'	AS NEEDED	150,000
HAWAII	70'	12'	13'6"	(3)
IDAHO	100'	14'	14'	110,000(1)
ILLINOIS	145'	14'6"	15'	120,000(1)
INDIANA	95'	14'4"	15'	108,000
IOWA	120'	14'(10)	14'5"	156,000
KANSAS	126'(3)	14'(3)	18'(3)	110,000(3)
KENTUCKY	80'	14'	15'	120,000
LOUISIANA	90'	12'	14'	108,000(6)
MAINE	90'	14'	13'6"	159,000(11)
MARYLAND	85'	14'	15'6"	110,000
MASSACHUSETTIS	70'	12'	14'	100,000
MICHIGAN	150'	14'	15'	NONE
MINNESOTA	85'	14'6"	14'(10)	132,000(11)
MISSISSIPPI	60'	12'	13'6"	80,000
MISSOURI	120'	14'	VARIES	120,000(1)
MONTANA	95'	15'	14'6"	110,000
NEBRASKA	(16)	14'	14'6"(10)	110,000
NEVADA	105'	14'	14'	250,000(1)
NEW HAMPSHIRE	90'	14'	13'6"	120,000
NEW JERSEY	(17)	--	(9)	(9)
NEW MEXICO	78'	12'	14'	130,000
NEW YORK	78'	12'	13'11"	110,000
NORTH CAROLINA	85'	14'	13'6"(10)	122,000(1)

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	LENGTH	WIDTH	HEIGHT	GROSS WEIGHT
NORTH DAKOTA	120'	14'6"	15'6"	96,000(12)
OHIO	135'(10)	14'	(10)	120,000(1)
OKLAHOMA	(10)	12'	(18)	(1)
OREGON	60'	12'	14'	96,000
PENNSYLVANIA	90'	13'	14'6"	136,000
PUERTO RICO	50'	8'	13'6"	--
RHODE ISLAND	80'	11'11"	13'6"	(1)
SOUTH CAROLINA	85'	15'	13'6"	120,000
SOUTH DAKOTA	85'	16'	14'	(1)
TENNESSEE	90'	14'	14'	(1)
TEXAS	95'	12'	14'	(1)
UTAH	120'	15'	16'6"	125,000(1)
VERMONT	84'	10'6"	14'6"	120,000
VIRGINIA	100'	14'	14'	115,000(1)
WASHINGTON	100'	14'	14'	200,000(1)
WEST VIRGINIA	78'(21)	14'	14'	110,000
WISCONSIN	99'	16'	13'6"	150,000
WYOMING	105'	18'	17'	135,000(10)
GUAM	65'	8'6"	13'6"	76,800(1)

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	SINGLE AXLE	TANDEM AXLE	REMARKS
ALABAMA	22,000	44,000	Each case reviewed on an individual basis.
ALASKA	30,000	70,000(8)(6)	Approval of State Maintenance Engineer.
ARIZONA	20,000	34,000	Overdimensional, overweight or nonreducible loads required to obtain permit.
ARKANSAS	20,000	34,000	Required to obtain permit from Permit Officer, Little Rock. See also (3).
CALIFORNIA	20,000	46,000(1)(20)	
COLORADO	22,000	42,000	
CONNECTICUT	22,000	36,000(2)	Dual tandem - 54,000.
DELAWARE	20,000	36,000	With escort weekdays, daylight hrs. Certification night, weekends, and holidays.
D.C.	20,000	38,000	Cases reviewed on an individual basis.
FLORIDA	25,000	50,000	Escort required with stipulated routing. Single trip with nondivisible load. Specified routing determined by permit officer. Certification night, weekend, and holidays.
GEORGIA	23,000	46,000	Night & weekend certification. Each load & route analyzed. Movement must have a permit.
HAWAII	(3)	(3)	
IDAHO	25,000(1)	50,000(1)	Routing restricted. Certification required for excessive loads or continuous travel.
ILLINOIS	24,000	48,000	Certification required for night or weekend.
INDIANA	28,000	48,000	Certification required for dimensions exceeding limits. (51,000 lbs tandem loads on A-tri).
IOWA	20,000	(*)	Escort required, approval of roads.
KANSAS	22,500(3)	45,000(3)	
KENTUCKY	22,000	45,000	Cases reviewed on an individual basis.

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	SINGLE AXLE	TANDEM AXLE	REMARKS
LOUISIANA	24,000	48,000	Only if made by govt. owned vehicle. Certification required.
MAINE	24,200	46,000	Prior approval for excesses in axle weight.
MARYLAND	30,000	30,000/axle	Nondivisible load only. Certification night, weekend and holidays. 30,000/axle for overweight moves only.
MASSACHUSETTS	22,400	36,000	Nonreducible load w/escort vehicle, and/or certification and/or additional restrictions.
MICHIGAN	60,000(4)	45,000/axle	
MINNESOTA	20,000	40,000	Each move considered separately. Special route check. Special certification required for over legal weight for spring time moves.
MISSISSIPPI	20,000	34,000	Cases reviewed on an individual basis.
MISSOURI	20,000	40,000	Extensive analysis of routing.
MONTANA	24,000	48,000	
NEBRASKA	20,000	40,000(14)	Permits issued for selected routes and times.
NEVADA	28,000	46,000	
NEW HAMPSHIRE	22,400	36,000	Beyond limits noted, route analysis.
NEW JERSEY	(9)	(9)	
NEW MEXICO	21,600	40,000	Military cargo exempt from regulations. Motor transportation to be notified.
NEW YORK	22,400	40,000	
NORTH CAROLINA	25,000	50,000	Special engineering study.
NORTH DAKOTA	24,000	42,000	(57,000 lbs triple; 68,000 lbs 4-axle).
OHIO	29,000	46,000	(53,000 lbs triple); Permit required.
OKLAHOMA	20,000	40,000	Night & weekend certification, Permit required for movement. Loads and route analysis.
OREGON	21,500	43,000	Nondivisible load only. 65' length designated highways.

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	SINGLE AXLE	TANDEM AXLE	REMARKS
PENNSYLVANIA	27,000	52,000	Military certification necessary for nighttime, weekend and holiday movements.
PUERTO RICO	--	--	Military owned and operated vehicles exempt.
RHODE ISLAND	22,400	(18)	Special permit and safety procedure.
SOUTH CAROLINA	20,000	40,000	(a) Under hardship cases. Height (10).
SOUTH DAKOTA	22,000	40,000	
TENNESSEE	20,000	40,000	See footnote #1.
TEXAS	25,000(7)	45,000	None.
UTAH	25,000	50,000	Nondivisible load only.
VERMONT	24,640	37,800	Cases reviewed on an individual basis. Special route check.
VIRGINIA	24,000	44,000	Length up to 150' depends upon routing & length of move. Military certification necessary.
WASHINGTON	22,000	43,000	In circumstances greater weight or dimension allowed, especially when certified as military necessary.
WEST VIRGINIA	28,000	45,000	Additional height possible depending upon route of travel.
WISCONSIN	30,000(19)	65,000	81,000 lbs triple; 90,000 lbs 4 axle. Up to 190,000 lbs gross and 16' height with special route.
WYOMING	25,000(10)	55,000(10)	Nondivisible load only.
GUAM	20,000	36,000	Route analysis and approval for overweight loads. Escorts for width greater than 14'.

* Up to and including 80,000 pounds, tandem gross weight is 34,000; 80,000 to 100,000 pounds, tandem gross weight 36,000; over 100,000 pounds, tandem gross weight is 40,000.

Footnotes:

(1) Dependent on number of axles and spacing.
(2) Two tires per axle.

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Footnotes: (Cont.)

(3) Engineering analysis required on all permits.
(4) Special Earth moving equipment based on 850 lb/inch of tire width. Limited moves only.
(5) 80,000 on interstate; 90,000 other roadways.
(6) Based on four or more load carrying axles.
(7) Based on 650 lb/inch of tire width.
(8) Tire loading maximum 750 lb/inch of tire width.
(9) Restricted to 800 lb/inch of tire tread.
(10) Controlled by route.
(11) Seven axles.
(12) Five axles, nine axles to 147,000 lbs.
(13) 75 feet over 8 feet wide, 100 feet less 8 feet wide.
(14) 50,000 triple.
(15) No limit.
(16) Unlimited.
(17) No regulation.
(18) Not specified.
(19) Dependent upon number of tires per axle.
(20) Typical 8 tire tandem is limited to approximately 46,000 pounds.
(21) As needed.
(22) Adequate axles.

NOTE: This chart was constructed from a questionnaire that MTMCTEA sent to each location listed. The questionnaire was authorized by the American Association of State Highway and Transportation Officials.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-52640B(ME)	Crane-Shovel, Crawler Mounted, 12-1/2-Ton at 12 Foot Radius, 3/4 Cu. Yd., DED, for Military Operations

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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